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Quantum Coherence Theory

A Constraint-Based Framework for Classical Emergence

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Abstract

Contemporary approaches to physical unification typically emphasize dynamical laws, interpretive frameworks, or information-theoretic reconstructions. Less attention is given to the structural constraints that determine which physical descriptions can remain coherent and stable under interaction, scale transition, and aggregation. This paper develops a constraint-first analysis of coherence, arguing that persistence and unification require restrictions on admissible physical continuations that are not reducible to dynamics alone.

Within the framework referred to here as Quantum Coherence Theory (QCT), coherence is treated not as a fragile quantum resource but as a structural property governed by interaction geometry and constraint density. On this view, classical behavior emerges when accumulated constraints sufficiently narrow admissible evolution, while quantum behavior re-emerges when constraints are relaxed. The account does not introduce new dynamics, collapse mechanisms, or modifications of quantum formalism, but instead clarifies the conditions under which existing formalisms yield stable, unified descriptions.

The resulting constraints distinguish QCT from observer-centric, intrinsic-collapse, and purely descriptive decoherence accounts, while remaining compatible with globally unitary dynamics. Formal analysis in later sections develops operational redundancy conditions, admissibility constraints on joint states, and rigidity-style arguments linking redundant record structure to the stability of classical descriptions. These results do not constitute a complete theory of unification, but they narrow the space of physically admissible descriptions and clarify the structural role of coherence in bridging quantum and classical regimes.

Scope, Commitments, and Structure

Quantum Coherence Theory (QCT) is presented as an explanatory framework concerned with the structural conditions under which coherence, collapse, and classical behavior emerge in physical systems. The analysis proceeds in two complementary stages.

Sections 1–10 develop the interpretive and conceptual foundation of the framework. These sections introduce the constraint-first perspective adopted throughout the work and clarify how coherence, admissibility, and persistence are to be understood within that explanatory posture. Sections 12–16 then begin the mathematical formalization of these ideas. In these sections, operational redundancy conditions, admissibility constraints on joint states, and structural restrictions on simultaneous record structures are developed in order to make the interpretive framework mathematically precise.

The formal development introduced here builds upon several established strands of research in quantum information theory and decoherence studies. The treatment of environmental fragments and redundant records follows the decoherence and quantum Darwinism program [1, 2, 3, 4]. Information-theoretic quantities such as entropy, mutual information, and accessible information follow the standard formalism of Shannon information and its quantum extensions[7, 8, 9].

The contribution of the present work is not to rederive these foundations, but to reformulate redundancy conditions as operational certification statements that act as structural constraints on admissible joint states. The present work therefore establishes the structural program and its initial formal development rather than a completed mathematical theory. Future work will focus on extending and refining the formalism introduced here and exploring its consequences across broader physical regimes.

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1 Introduction

This section establishes the central problem addressed by QCT: while quantum mechanics is empirically successful, it lacks an explanatory account of how classical definiteness emerges from quantum possibility. This section motivates a constraint-first framework that reframes coherence, collapse, and observation without invoking observer privilege, new dynamics, or metaphysical claims.

1.1 The Interpretive Gap in Quantum Mechanics

Quantum mechanics is among the most empirically successful frameworks in the history of science. Its mathematical formalism yields predictions of extraordinary precision across an immense range of physical phenomena, from atomic spectra to solid-state behavior to high-energy interactions. No competing theory has matched its predictive reach or experimental resilience. Yet this success coexists with a persistent and widely acknowledged interpretive gap: while the theory predicts outcomes with remarkable accuracy, it does not explain, in structural terms, why physical systems transition from quantum admissibility to classical definiteness.

This gap is not a failure of calculation, nor a lack of experimental confirmation. It is not resolved by improved measurement, larger datasets, or more refined instrumentation. It is a gap of explanatory linkage. The formalism specifies how probability amplitudes evolve, interfere, and decohere under interaction, but it remains largely silent on the conditions under which superposed possibilities cease to remain jointly admissible and instead manifest as stable, classical identities.

Quantum theory describes what is allowed. Classical physics describes what persists. The connection between these two descriptive regimes is operationally managed but not structurally explained.

Existing interpretations address this gap in different ways, but often by displacing rather than resolving it. Some elevate probabilistic postulates to foundational status, treating collapse or outcome selection as irreducible features of reality. Others invoke observer-relative updates, epistemic bookkeeping, or branching descriptions that preserve the formalism while shifting the burden of explanation onto interpretation or ontology. In still other approaches, decoherence is treated as sufficient explanation, despite its reliance on environmental assumptions that are themselves not structurally grounded.

In each case, the mathematical machinery remains intact. What remains unresolved is the underlying question: what constrains the universe such that only certain realizations persist, while others remain admissible in principle but unrealized in practice? The interpretive gap, as treated here, does not concern whether quantum mechanics is correct, complete, or empirically valid. Nor does it concern the introduction of new dynamics, hidden variables, preferred observers, or speculative ontologies. It concerns the absence of a constraint-based explanation connecting quantum permissiveness to classical persistence.

More precisely, the gap can be stated as follows. Quantum mechanics describes a space of admissible possibilities. Classical physics describes a space of persistent identities. The formalism alone does not specify why the former narrows into the latter, nor what structural pressures govern that narrowing. Collapse, decoherence, and measurement are described operationally, but their status as consequences of deeper limiting conditions is left implicit or undefined.

As a result, the transition from quantum coherence to classical behavior is often treated as either primitive, observer-dependent, or interpretation-relative. Yet the persistence of classical

structure across scale, interaction, and disturbance suggests that this transition is neither arbitrary nor incidental. Stable identities do not merely appear; they endure.

This work treats the interpretive gap as an explanatory deficiency rather than a metaphysical mystery. It assumes that the transition from quantum admissibility to classical persistence reflects the action of limiting conditions that govern stability itself. The task is therefore not to reinterpret quantum mechanics, revise its mathematics, or privilege particular observers, but to identify the structural constraints that must exist for the observed transition to occur.

Quantum Coherence Theory is introduced precisely to address this gap. It does so by shifting explanatory focus away from outcomes, observers, or postulated selection rules, and toward the constraints that determine which configurations remain admissible, stable, and persistent across scale and interaction. The aim is not to replace quantum theory, but to supply an explanatory layer that clarifies why its predictions resolve into the classical world that persists.

1.2 Limits of Probabilistic, Observer-Centric, and Instrumentalist Accounts

Many existing interpretations of quantum mechanics address the interpretive gap identified in the previous section by reframing the problem rather than resolving it. While these approaches differ in emphasis and philosophical posture, they share a common structural feature: each treats certain explanatory elements as primitive that, within a constraint-first framework, remain insufficiently grounded.

Probabilistic accounts typically elevate probability to foundational status. In such views, quantum outcomes are taken to be irreducibly stochastic, with no deeper explanation offered for why particular outcomes stabilize while others do not. Probability functions as an endpoint of explanation rather than a derived consequence. While this move preserves predictive adequacy, it leaves unanswered why probabilistic descriptions govern physical persistence at all, or why probabilistic behavior gives way to effectively deterministic classical behavior as systems scale, interact, and endure.

Put differently, probability describes distributions over admissible states, but it does not explain why admissibility narrows in practice. Treating probability as fundamental accounts for uncertainty, but not for durability.

Observer-centric accounts attempt to resolve the transition by assigning a privileged role to measurement, observation, or informational update. In these frameworks, definiteness arises not from structural properties of physical systems themselves, but from interactions involving observers or observer-like processes. While such accounts can be internally coherent, they implicitly relocate the explanatory burden. The question of why classical outcomes emerge is replaced by the question of why observers experience definiteness.

This displacement leaves the underlying physical transition unaddressed. Stable macroscopic structure persists regardless of whether it is observed, recorded, or interpreted. Any account that makes observation explanatorily central must still explain why unobserved systems obey the same stability constraints as observed ones. Without such an explanation, observer-centric approaches risk confusing epistemic update with physical persistence.

Instrumentalist approaches, by contrast, explicitly decline to offer an explanatory account. They treat the quantum formalism as a predictive tool whose role is exhausted by its success in forecasting experimental outcomes. From this perspective, questions about collapse, emergence, or persistence are regarded as either meaningless or unnecessary. While this stance avoids speculative

commitments, it also forgoes any attempt to explain why the formalism works as it does, or why its predictions align so consistently with a stable macroscopic world. Instrumentalism succeeds by narrowing the definition of explanation to prediction alone. In doing so, it renders the interpretive gap invisible rather than resolved.

From the perspective adopted in this work, these approaches are not rejected as incorrect. Each captures genuine aspects of quantum behavior and remains compatible with the formalism. Their limitation is instead explanatory completeness. Probabilistic accounts stop at chance. Observer-centric accounts stop at experience. Instrumentalist accounts stop at utility. None provide a structural account of the conditions under which physical possibilities cease to be jointly admissible and instead resolve into persistent configurations.

What these frameworks share is an under-articulation of constraint. In probabilistic and observer-centric accounts, constraints are present but implicit, folded into postulates or measurement rules. In instrumentalist accounts, constraints are bracketed entirely. As a result, the transition from quantum permissiveness to classical stability is managed operationally but not explained structurally.

This absence is not accidental. Constraint-based explanations occupy a conceptual layer that sits between formal dynamics and observed outcomes. Without explicitly naming this layer, interpretive frameworks are forced either to treat narrowing as primitive or to relocate it to observers, probability, or pragmatics.

Quantum Coherence Theory does not seek to replace these interpretations or to adjudicate between them. Instead, it treats their shared limitation as diagnostic. The persistent absence of an explicit account of constraint suggests not that such constraints are unnecessary, but that they have remained untheorized. The task, therefore, is not to dispute probability, observation, or prediction, but to articulate the limiting conditions under which these features emerge as secondary consequences of deeper structural restrictions governing admissibility and persistence.

1.3 Motivation for a Constraint-First Explanatory Framework

The limitations outlined in the preceding sections motivate a shift in explanatory posture rather than the introduction of new physical postulates. Quantum mechanics remains empirically unrivaled in its predictive accuracy, and nothing in this work challenges its formal validity. What remains unresolved is not whether the theory works, but why its permissive descriptions reliably give rise to stable, classical outcomes without invoking observer privilege, irreducible chance, or explanatory abstention.

At the center of this unresolved question lies a simple but underexamined fact: persistence is not automatic. Physical descriptions permit many possible configurations, but only a narrow subset of those configurations remain identifiable over time. For a system to persist as the same system, alternatives must be excluded in a structured and sustained way. Stability, in this sense, is not merely the continuation of state evolution, but the result of limiting conditions that restrict which possibilities remain admissible under interaction.

A constraint-first framework begins from this observation. Rather than asking why a particular outcome is selected from among many, it asks under what conditions multiple possibilities remain jointly admissible, and under what conditions they cease to be so. The focus shifts from outcomes to admissibility, and from observers to structure. Collapse, decoherence, and classical definiteness

are treated not as special events requiring interpretive intervention, but as consequences of how constraints accumulate, interact, and enforce exclusion across scale.

This reframing also alters the status of probability within explanation. Probability is not treated as a primitive governing principle, but as a descriptive artifact that arises when multiple admissible configurations coexist under insufficient constraint. Where admissibility remains broad, probabilistic descriptions are appropriate. Where constraints aggregate and admissibility narrows, probabilistic freedom diminishes and persistence emerges.

Probability, in this view, tracks permissiveness rather than causation. Similarly, measurement is not granted ontological privilege. Measurement is treated as one class of interaction among many that can introduce additional constraints by coupling a system to an environment with its own persistence requirements. Observation does not create definiteness; it participates in constraint aggregation. Predictive success, while indispensable, is therefore not taken as an endpoint of explanation, but as a surface manifestation of deeper regularities governing admissibility and stability.

The motivation for this approach is not metaphysical. It arises from a practical explanatory need. Without an explicit account of how constraints operate across scale and interaction, the transition from quantum coherence to classical determinism remains partitioned across interpretive boundaries. Decoherence explains suppression of interference, collapse postulates explain outcome definiteness, and classical physics explains stability, but no single structural account explains why these regimes connect as reliably as they do.

By contrast, a constraint-first framework aims to describe the quantum–classical transition as a continuous process governed by the aggregation and interaction of limiting conditions, rather than as a categorical jump requiring special intervention. It treats stability as an achievement rather than a default, and collapse as an enforced reduction of admissibility rather than an unexplained selection.

Importantly, this framework does not presume that constraints are externally imposed, intelligently selected, or teleologically ordered. Constraints are treated descriptively, as conditions that arise from interaction, coupling, boundary formation, symmetry restriction, and mutual compatibility requirements among physical systems. Their role is not to direct outcomes, but to delimit persistence.

The goal of Quantum Coherence Theory is therefore not to reinterpret quantum mechanics from first principles, nor to compete with existing interpretations on their own terms. Its aim is narrower and more foundational: to articulate how constraint structure governs admissibility, stability, and collapse across regimes, providing a unifying explanatory layer beneath probabilistic description, observer-centric narratives, and instrumental success.

1.4 Methodological Closure and Scope Discipline

Before proceeding to the foundational assumptions of Quantum Coherence Theory, it is necessary to state explicitly the methodological limits within which the framework operates. These limits are not provisional hedges or rhetorical safeguards. They are structural constraints imposed to preserve explanatory clarity, maintain internal consistency, and prevent category drift as the theory develops.

Quantum Coherence Theory adopts methodological closure in the following sense: the observable universe is treated as an informationally closed domain for purposes of explanation. This assumption is methodological rather than ontological. It does not assert that the observable domain is complete, unique, or exclusive. Nor does it deny the possibility of structures, domains, or

realities beyond empirical access. It reflects a disciplined refusal to rely on unobservable external inputs to explain persistence, stability, or collapse within the observable regime.

Methodological closure establishes an evidentiary boundary. All explanations advanced within QCT must be grounded in interactions, constraints, and structural relations that arise within the domain of empirical accessibility. No explanatory weight is assigned to hidden external agencies, privileged reference frames, transcendent selectors, or metaphysical interventions. Where explanation cannot proceed without such appeals, the limit is acknowledged rather than bypassed.

This posture is pragmatic rather than doctrinal. It does not claim that external influences do not exist; it claims only that they cannot be invoked as explanatory resources without dissolving the distinction between explanation and speculation. QCT therefore treats unobservables as epistemically out of scope unless their effects can be traced through constraint structure within the closed description. Scope discipline follows directly from this commitment. Quantum Coherence Theory does not attempt to:

- derive fundamental constants
- specify underlying microdynamical mechanisms
- adjudicate among competing ontologies beyond what constraint structure requires
- or provide prescriptive rules for intervention, engineering, or control

Instead, QCT positions itself as an explanatory layer concerned with how admissibility, coherence, and collapse behave under varying constraint conditions. It does not address why the universe exists, what ultimately grounds physical law, or how outcomes should be optimized or manipulated. These questions are neither answered nor denied; they are simply not required to accomplish the explanatory task at hand.

This discipline also governs the relationship between QCT and adjacent works. Foundations of Coherence Science supplies a diagnostic vocabulary for identifying coherence, collapse, and failure across domains.

Artificial Coherence Intelligence explores applied systems that operationalize constraint-based reasoning under sustained pressure. QCT neither depends upon nor extends these efforts. It occupies a distinct role: articulating how constraint-driven persistence connects quantum permissiveness to classical stability without instrumental ambition or cross-domain generalization.

Finally, methodological closure entails a commitment to explanatory sufficiency rather than explanatory completeness. QCT aims to explain as much as can be explained without exceeding the evidentiary reach of the framework. Where questions remain unresolved, they are preserved as open boundaries rather than prematurely resolved through speculative extension.

With these limits established, the framework can proceed to its foundational assumptions without ambiguity. What follows is not a claim about ultimate reality, but a disciplined attempt to describe the minimal structural conditions under which persistence, differentiation, and collapse become intelligible within the observable domain.

2 Orientating QCT

This section establishes the orientation language required to speak meaningfully about persistence in QCT. It defines coherence as a structural condition rather than a metaphor, treats constraint as primary, and frames identity and information as exclusionary processes. These assumptions do not explain reality, but bound what kinds of explanations can remain viable under interaction, history, and irreversibility.

2.1 The Meaning of Coherence in QCT

Quantum Coherence Theory employs the term coherence in a deliberately non-standard sense. This choice is neither accidental nor rhetorical. It reflects the explanatory level at which QCT operates and the specific problem it addresses: not the dynamics of particular physical variables, but the conditions under which any physical description can support persistence, identity, and stable structure under constraint.

In standard quantum mechanics, coherence refers to well-defined phase relations between components of a quantum state, typically expressed through off-diagonal terms in a density matrix. This usage is precise, domain-local, and indispensable within quantum formalism. Nothing in Quantum Coherence Theory disputes or modifies that definition within its proper scope. QCT does not attempt to reinterpret quantum-mechanical coherence, nor to replace its technical role in wave mechanics, interference phenomena, or decoherence theory.

Instead, QCT introduces a structurally prior notion of coherence, inherited from Foundations of Coherence Science (referenced in Related Works), that operates at a different explanatory level. In this framework, coherence refers to the satisfaction of invariants under transformation: the capacity of a system, description, or world-model to remain self-consistent, non-contradictory, and persistently identifiable as constraints accumulate. Coherence, in this sense, is not a wave property, a phase relation, or a superpositional feature. It is a structural condition for persistence.

This distinction in usage reflects a difference in explanatory task rather than a conceptual conflict. Quantum mechanics presupposes a stable descriptive framework in which states, operators, and dynamics are meaningfully defined. Quantum Coherence Theory asks a prior question: what must be true of a description for such a framework to remain viable at all under interaction, coupling, and constraint? The notion of coherence employed here is introduced to answer that question.

Within QCT, coherence is therefore not synonymous with maximal superposition, maximal symmetry, or unrestricted dynamical freedom. On the contrary, as explained in the framework, coherence increases as admissible possibilities are selectively restricted in ways that preserve identity and stability. An example explored more thoroughly in the following sections is superposition. Superposition, when viewed through this lens, represents a state of low structural commitment: many possibilities are simultaneously admissible, but none are yet stabilized as records. As constraints accumulate—through interaction, decoherence, boundary conditions, or environmental coupling—the space of admissible states narrows. Coherence, in the QCT sense, increases as contradiction is metabolized rather than eliminated and as persistence becomes possible.

This usage is intentionally invariant-centered. A system is coherent to the extent that it satisfies the invariants required for its continued existence across transformations. Loss of coherence, correspondingly, is not primarily the loss of phase information but the failure of invariant satisfaction: internal contradiction, uncontrolled proliferation of admissible states, or collapse of identity under

constraint. In this sense, coherence is not opposed to constraint; it is produced by the right kind of constraint.

Readers familiar with quantum-mechanical terminology may experience an initial friction when encountering this definition. That friction does not arise from ambiguity within the theory, but from a shift in explanatory level. Quantum Coherence Theory explicitly declares semantic precedence: coherence is defined structurally at the level of Coherence Science, and quantum coherence appears as one domain-specific manifestation evaluated against that broader criterion. The theory is internally consistent once this hierarchy is established.

This framing is essential for the arguments that follow. Claims about energy, vacuum structure, entropy, and unification depend on a notion of coherence that can apply across domains without being tied to a specific formalism. The task of QCT is not to optimize coherence within quantum mechanics, but to assess whether quantum descriptions—alongside classical, thermodynamic, or cosmological ones—can satisfy the invariant requirements for persistence in a constrained world.

Accordingly, no attempt is made to reconcile terminological differences at the level of usage. Quantum coherence and QCT coherence are not competing definitions of the same object; they are definitions serving different explanatory roles. Where the same term appears, the distinction is one of scope, not contradiction. This clarification allows the theory to proceed without semantic confusion while preserving full compatibility with established physics.

This section therefore establishes a foundational assumption that governs the remainder of Quantum Coherence Theory: coherence is treated as invariant satisfaction under constraint, not as a wave-mechanical property.

2.2 Constraint as a Prerequisite for Existence

Within this theory, constraint is treated as a prerequisite for existence in a minimal and strictly descriptive sense. Without constraint, no distinction can persist long enough to be identified, tracked, or meaningfully referred to. This claim is not metaphysical. It does not assert that constraint creates existence, nor that existence is reducible to constraint. Rather, it specifies the conditions under which existence becomes referable as persistent structure within a given explanatory boundary.

Constraint, as used here, refers to any condition that reduces, structures, or delimits the set of states or transitions that remain accessible under the admissible dynamics of a chosen description. Such conditions may arise from boundary relations, conservation laws, interaction couplings, symmetry restrictions, environmental entanglement, or coarse-graining choices.

They are not assumed to be forces, agents, goals, or causes by default. Constraint is a structural condition on admissibility, not an intention, directive, or dynamical command. Operationally, a constraint is present wherever the admissible evolution of a system leaves certain correlations, distinctions, or patterns intact across interaction and time. In these cases, accessibility is limited in a way that allows some structures to remain identifiable despite disturbance. Constraint functions by restricting what can vary freely, thereby permitting invariance or metastability at the descriptive level.

The claim advanced here is deliberately narrow. Constraint is necessary for persistence, identity, and information to be defined. It is not sufficient to guarantee coherence, stability, objecthood, or functional viability. Many constrained configurations fail to endure. They fragment, decohere, or collapse under interaction. Constraint makes persistence possible, but it does not ensure it. It is the entry condition for describability, not a guarantee of survival.

At any given descriptive boundary—specified by a choice of coarse-graining, admissible interactions, tolerance, and timescale—the absence of effective constraint means that the description provides no internal basis for preserving distinctions across time. The effective dynamics explore the accessible set without leaving long-lived structures at that resolution. Distinctions may arise momentarily, but they dissolve as interaction disperses correlations across the available state space. From the standpoint of the reduced description, such a regime is indistinguishable from noise: not because nothing occurs, but because nothing remains re-identifiable.

Existence in Quantum Coherence Theory is therefore defined operationally, in terms of persistence. A structure exists within a given description only insofar as it remains re-identifiable under the admissible transformations of that description, within an appropriate tolerance and over a relevant timescale. This framing avoids two common errors: treating existence as an intrinsic essence independent of conditions, and treating transient fluctuations as equivalent to stable structure. Only what persists under effective constraint qualifies as existent within the theory’s descriptive scope.

Importantly, constraint does not imply rigidity or determinism. A system may be highly constrained and brittle, or lightly constrained and flexible. What matters is not the presence of constraint alone, but its configuration and its compatibility with persistence. Constraint that is too weak fails to stabilize structure. Constraint that is not approximately preserved under the induced dynamics or system–environment coupling suppresses variability or collapses the system into triviality. Persistence occupies a constrained but non-degenerate regime in which structure is both protected and responsive.

This perspective also clarifies why constraint cannot be equated with determinism. Constraints delimit what is admissible, but they do not fix outcomes. Probabilistic behavior, indeterminacy, and stochastic dynamics remain fully compatible with a constraint-first framework, provided the constraints defining admissibility are respected. Constraint limits possibility; it does not exhaust it.

Constraint in Quantum Coherence Theory is always relative to a descriptive boundary. What appears weakly constrained at one scale may consolidate into robust invariants at another. A system may exhibit few stabilizing distinctions at one resolution and many at another. This scale dependence is not a defect or ambiguity. It is a structural feature that allows the same underlying system to display different persistence profiles across regimes without invoking categorical ontological shifts.

By treating constraint as prerequisite rather than cause, Quantum Coherence Theory avoids both reductionism and metaphysical inflation. Constraint does not explain why something exists rather than nothing. It explains why, given existence, stable structure is rare, fragile, and conditional. It establishes the stage on which coherence, identity, and information can later be meaningfully discussed.

In summary, constraint is introduced here not as an explanatory endpoint, but as the minimal condition that makes explanation possible at all. Without effective constraint, there is no persistence. Without persistence, there is no identity. Without identity, there is nothing for theory to refer to.

2.3 Identity as Exclusion and Persistence

For this framework, identity is not treated as a primitive feature of reality, nor as an intrinsic essence attached to objects. Identity is not assumed at the outset of explanation. Instead,

it is defined relationally and conditionally, as a property that emerges when a system remains distinguishable under constraint across time, interaction, and transformation.

At the most basic level, identity requires exclusion. A system is identifiable only insofar as it occupies a restricted subset of admissible states within a larger space of possibility. To be one configuration rather than another is necessarily to exclude alternative configurations that remain admissible but unrealized. This exclusion is not metaphysical negation; it is structural differentiation. Identity begins when “this” is not “that” within a defined descriptive frame.

However, exclusion alone is insufficient to establish identity. A momentary restriction that collapses immediately under interaction does not constitute an identifiable system. In QCT, identity requires persistence: the capacity of a configuration to remain distinguishable as constraints act upon it. A configuration that appears briefly but cannot withstand interaction does not form an identity in any meaningful sense. Identity is therefore not instantaneous. It is extended, conditional, and evaluated over trajectories rather than snapshots.

Persistence is always relative to a constraint set. A system’s identity is defined by the transformations it can undergo without losing distinguishability. These transformations form an equivalence class under the active constraints. Changes that remain within this class preserve identity; changes that exceed it degrade identity. Importantly, identity loss is typically gradual rather than binary. Systems do not usually “cease to exist” in a single step. They erode, fragment, or dissolve as constraints overwhelm their capacity to maintain distinction.

This gradual degradation is a critical feature of QCT’s treatment of identity. Identity is not a toggle. It is a tolerance band. As constraints intensify or change, a system may retain partial identity, ambiguous identity, or scale-limited identity. Only when exclusion can no longer be maintained does identity fully dissolve. This allows QCT to describe transitions without invoking sharp ontological breaks or arbitrary thresholds.

Identity is also inherently scale-dependent. A configuration that persists at one scale may fail to remain distinguishable at another. Quantum states, classical objects, and macroscopic structures differ not by categorical ontology, but by the degree to which their identities withstand aggregated constraint across interactions. Classical stability reflects high tolerance to constraint, not a fundamentally different mode of being. What appears “object-like” is simply what survives constraint thickening without losing distinguishability.

Because identity arises from constraint structure, it is contextual without being observer-dependent. The conditions that define persistence are imposed by interaction, coupling, and environmental embedding, not by observation or interpretation. Observers may describe identity, classify it, or track its degradation, but they do not instantiate it. Identity does not depend on measurement for its existence. It depends on whether distinguishability can be maintained under the relevant constraints.

This distinction is essential for avoiding observer-centered interpretations of persistence. Identity does not emerge because something is noticed. It emerges because something remains. The role of the observer is descriptive, not constitutive. QCT therefore treats identity as an emergent structural property of constrained systems, not as a projection of epistemic access.

Within this framework, identity is neither absolute nor illusory. It is conditional, durable when constraints permit, fragile when they do not, and always bounded by the structure of admissibility. Identity is what remains distinguishable when alternatives are excluded and that exclusion can be sustained across interaction.

In its most compact form, identity in Quantum Coherence Theory is defined as the persistence of distinguishability under constraint. Where exclusion fails, identity cannot arise. Where persistence fails, identity cannot endure. Everything else follows from that condition.

2.4 Information as Constrained Realization

In Quantum Coherence Theory, information refers to realized differentiation under constraint. It does not denote semantic meaning, representation, reference, or observer-relative interpretation. Information names a physical condition: that one admissible configuration has been realized rather than others, producing distinguishability within a constrained description.

Information is not treated as a primitive of reality. It does not exist independently of constraint, nor does it arise from unconstrained fluctuation alone. A system may exhibit transient patterning or statistical variation, but without stabilizing conditions that preserve distinctions across interaction and time, no realized differentiation can persist long enough to count as information. Where nothing remains distinguishable, nothing informative exists.

This framing separates the use of information in Quantum Coherence Theory from both semantic and purely probabilistic accounts. Semantic theories require interpretation or reference; no such requirement appears here. Probabilistic theories describe likelihoods over admissible states, but probability alone does not constitute information in this framework. Information appears only when constraints restrict admissibility in a way that fixes realizations relative to excluded alternatives. In this sense, information is an effect of constraint, not its cause.

Information is inseparable from exclusion. To realize one configuration is necessarily to exclude others that remain admissible but unrealized within the same descriptive frame. This exclusion is not absolute or global. It is always relative to a chosen partition, scale, and constraint set. Information therefore exists only within a bounded description, not as a universal quantity independent of modeling choices.

A distinction counts as information in Quantum Coherence Theory when it is realized, remains distinguishable under a specified coarse-graining, and persists across a relevant class of interactions and timescales due to stabilizing constraints. Transient differentiation that cannot be re-identified under interaction does not qualify.

Not all constraints that stabilize information are reducible to internal statistical interactions within a system. For realized differentiation to persist across disturbance and scale change, constraint relations must remain compatible under transformation. Such constraints may take the form of boundary conditions, conservation relations, symmetry restrictions, inter-scale couplings, or historical contingencies. They do not require external agents, purposes, or designs. They need only be stabilizing conditions not implied by treating the same descriptive boundary as unconstrained noise. This requirement distinguishes persistent information from momentary patterning.

Information in Quantum Coherence Theory is therefore relational. A realization becomes informative only insofar as it remains mutually compatible with other constrained realizations across a system. Information does not reside in isolated states but in the network of constraints that preserves differentiation across interaction. When this compatibility erodes, information degrades gradually rather than disappearing abruptly, reflecting loss of constraint alignment rather than annihilation of states.

This account distinguishes information from entropy while allowing the two to remain related. Entropy may characterize dispersion or uncertainty relative to a chosen coarse-graining. Information refers instead to the existence of stabilized differentiation itself. Entropy may increase while information persists, provided constraints continue to preserve distinguishability at the relevant scale. Conversely, entropy reduction does not guarantee informational persistence if constraints fail to maintain identity under interaction.

Information is also scale-dependent. A configuration may carry information at one scale while failing to remain informative at another. Quantum states, classical objects, and macroscopic structures differ not by categorical informational status, but by the robustness with which their differ-

entiations persist under aligned constraint structures. Informational persistence reflects tolerance to interaction and disturbance, not a distinct informational ontology.

Quantum Coherence Theory makes no claim that information must arise, increase, or optimize over time. The framework does not assume an informational arrow or teleological accumulation. It asserts only that wherever persistent differentiation is observed, constraint conditions sufficient to stabilize that differentiation must exist. Whether such conditions emerge, persist, or fail remains contingent rather than guaranteed.

Under this view, information is neither fundamental nor illusory. It is the observable trace of constraint acting on potential, marking where realization has occurred and endured. Information names the fact of stabilized difference, not its meaning, purpose, or value.

3 Possibility, Constraint, and Ontology

This section establishes the minimal ontological language used by QCT. It reframes energy, matter, and scale as expressions of constrained possibility rather than primitive entities. By treating scale as a proxy for constraint density, this section provides a neutral descriptive bridge between quantum admissibility and classical persistence without invoking hidden substances, teleology, or observer privilege.

3.1 The Zero-Sum Field

Any discussion of structure presupposes a contrast against which structure can be recognized. Before distinctions such as energy, matter, or identity can be meaningfully introduced, there must exist a reference condition in which none of these distinctions yet operate decisively. The zero-sum field is introduced here as that reference condition. The zero-sum field is not proposed as a physical field, nor as an ontological claim about the ultimate nature or origin of reality. It is a methodological construct used to represent a state of maximal admissibility: a condition in which no configuration is privileged, no outcome is stabilized, and no asymmetry yet enforces persistence. Its role is to provide a neutral baseline against which the effects of constraint can later be described.

“Zero-sum” in this context does not refer to competition, cancellation, or conservation in the economic or game-theoretic sense. It denotes the absence of net structural preference. All configurations are equally admissible; no pathway carries authority over another; no distinction yet carries the force required to endure. The field is “flat” with respect to structure, not empty of content.

This construct is deliberately minimal. It does not assume dynamics, causality, or time-directed evolution. Nor does it assume randomness as a productive principle. Possibilities may be enumerated, but none are yet selected. Variation may be conceivable, but no variation is retained. In this sense, the zero-sum field is not chaotic; it is non-committal.

The usefulness of this baseline lies in what it does not contain. By excluding implicit structure at the outset, it allows later sections to identify precisely where and how structure is introduced. Any appearance of persistence, localization, or durability must arise from additions to this baseline rather than being silently inherited from it.

It is important to emphasize that the zero-sum field is not intended to describe an actual historical or physical state of the universe. It is a limiting construct, analogous to idealized reference conditions used throughout physics and mathematics. Its purpose is comparative, not descriptive: to make the contribution of constraint legible by contrast.

In the sections that follow, departures from this baseline will be introduced gradually. Energy will first appear as a means of restricting admissible transitions without yet enforcing localization. Subsequent refinements will show how additional layers of constraint further narrow what can persist. The zero-sum field thus marks the point at which no such narrowing has yet occurred.

3.2 Constraint as the Structuring of Potential

In Quantum Coherence Theory, “constraint” is not a physical entity, force, or governing principle. It is a descriptive classifier: a way of characterizing how the space of admissible states is structured so that some configurations remain meaningful, stable, and predictively usable under

interaction, while others do not. “Potential” denotes the full set of configurations that are kinematically and dynamically allowed within a given description. “Constraint” characterizes how that space is shaped—by interaction, coupling, and modeling choice—into subsets that either sustain or fail to sustain persistence. Constraint does not produce outcomes. It defines the conditions under which a description continues to function without breakdown.

Admissibility in QCT is always evaluated relative to a descriptive partition. A partition specifies which variables count as the effective state of a system (macroscopic observables, coarse-grained degrees of freedom, pointer variables, or other operationally stable quantities). The partition induces an equivalence relation on microstates: different microstates are treated as equivalent when they realize the same effective state at the description’s resolution.

Persistence refers to predictive stability of those effective variables. A description persists when microstates that are equivalent under the partition continue to evolve into states that remain equivalent—within tolerance—over the timescale of interest. A microstate is admissible (relative to a partition) when its evolution preserves that predictive stability.

Constraint therefore functions as a viability condition for descriptions, not as a causal mechanism acting on physical states. Saying “constraint structures potential” does not mean constraint causes events. No new dynamics are introduced; underlying evolution remains standard physics (unitary dynamics on the combined system–environment, plus effective open-system dynamics derived from that unitary description). Constraint names the fact that, under interaction and a chosen partition, some configurations fail to remain jointly representable as stable descriptors.

These configurations are not eliminated by force or fundamental selection. They become inadmissible because they do not support persistence of the effective variables defined by the description. Multiple physical structures can implement the same constraint pattern. Conservation laws, symmetries, interaction Hamiltonians, boundary conditions, and environmental monitoring all instantiate constraints within their respective formalisms. QCT abstracts across these implementations to describe their shared effect: restriction of the admissible state space relative to a description.

Constraint accumulates through interaction. When systems couple, admissibility is no longer evaluated independently for each subsystem but jointly under the coupled dynamics. States admissible for isolated systems may become incompatible with persistence once correlations, entanglement, or environmental coupling are introduced. This narrowing does not require collapse, selection, or privileged observation; it follows from compatibility requirements under continued interaction. As interaction compounds, the admissible set may shrink gradually rather than discontinuously.

Constraint density refers to how strongly interaction reduces the measure of microstates that remain admissible under a chosen partition, tolerance, and timescale. The measure is defined over the microstates relevant to the description (e.g., induced by a preparation ensemble, a typicality assumption, or another sampling over microstates consistent with the description). At low constraint density, multiple alternatives may remain jointly admissible, allowing interference, superposition, and indeterminacy in the effective description. At higher constraint density, compatibility requirements render interference terms inaccessible to the effective algebra defined by the partition. Those terms become non-contributory to stable predictions, reducing the number of equivalence classes that remain jointly admissible.

Constraint is always partition-relative, but partitions are not arbitrary. Legitimate partitions are justified by operational accessibility, dynamical robustness, and predictive stability. Variables are selected because they remain well-defined under interaction and are redundantly supported by environmental correlations. Constraint density is therefore relational rather than absolute, but it is not unconstrained by modeling choice. Different partitions can assign different constraint densities to the same underlying system.

This is not a defect of the framework. It reflects the fact that persistence and admissibility are properties of descriptions, not of bare microstates in isolation. In standard open-system models, increasing constraint density often correlates with stronger environment-induced monitoring (faster decoherence), greater redundancy of environmental records, and increased closure of the effective observable algebra. QCT does not privilege any single formalism as uniquely definitive of these effects; it treats them as different implementations of the same descriptive role.

Constraint does not imply rigidity. A highly constrained system may be robust or brittle depending on how constraints are configured. Some constraint structures preserve compatibility under disturbance and stabilize persistence. Others overdetermine behavior and precipitate breakdown of the effective description. QCT therefore does not equate “more constraint” with “more stability.” Stability arises when constraints are mutually compatible and proportionally corrective rather than conflicting or excessive.

This framework clarifies decoherence and collapse without new postulates. Decoherence corresponds to progressive restriction of admissible phase relations under environmental coupling: interference terms become dynamically and operationally inaccessible relative to the chosen partition, and thus irrelevant to stable prediction within that description. Collapse, in QCT, refers only to effective collapse: the point at which alternatives are no longer jointly admissible within the effective description because aggregated constraints render them incompatible with stable representation. No fundamental non-unitary event is implied; underlying dynamics remain unitary.

Exclusivity emerges as exclusivity of joint representability and record-consistent prediction under the partition, not as selection among ontic outcomes. Classical behavior emerges when constraint density becomes sufficiently high that only a narrow equivalence class of configurations remains admissible under continued interaction. Determinism, in this sense, is emergent rather than fundamental. Quantum dynamics do not cease to apply; their expressive capacity collapses at the descriptive level because the admissible set has narrowed to configurations that behave classically.

In QCT, constraint is the means by which potential becomes structured into persistence. It neither creates reality nor selects outcomes. It delineates the conditions under which realizations endure, interactions remain compatible, and identity is preserved across scale. It is not an ontological addition to physics, but a unifying descriptive lens for how permissive quantum possibilities give rise to persistent structure.

3.3 Energy as Less Constrained not Unconstrained

Energy is traditionally treated in physics as a conserved quantity, formally defined within mathematical frameworks such as Hamiltonian and Lagrangian mechanics and operationalized through measurable exchanges between systems. Nothing in Quantum Coherence Theory disputes this role. The conservation of energy, its calculational utility, and its empirical indispensability remain fully intact. The distinction introduced here is not a rejection of energy’s formal function, but a clarification of its explanatory status within a constraint-first account of persistence and structure.

Within Quantum Coherence Theory, energy is not treated as an ontic substance or independently existing entity. Instead, it is understood as a representation of admissible potential for change: a parameter that tracks which state transitions are permitted relative to a specified dynamical description. Energy encodes which transitions are allowed given a Hamiltonian, its symmetries, and its boundary conditions; it persists relationally, through the constraint structure that references it, rather than as an independently localized form. It therefore does not itself constitute structure,

identity, or persistence in the sense required for record-bearing, locally referencable configurations. This reframing preserves all standard physical uses of energy while avoiding unnecessary ontological commitments.

This clarification matters because foundational confusion often arises when energy is implicitly asked to perform incompatible roles at once: as a conserved scalar, a causal agent, and a material substrate. In established physical practice, energy fulfills none of these roles in isolation. While energy constrains which transitions are dynamically accessible, it does not by itself determine which configurations persist, decohere, or stabilize as records. Those outcomes depend on constraint structure—interaction terms, coupling topology, spectra, symmetries, and boundary conditions—not on the magnitude of energy alone.

Within a constraint-first framework, potential refers to the space of admissible configurations available to a system relative to a chosen descriptive boundary. Energy is one powerful way of parameterizing that space, particularly in dynamical models where evolution is represented as motion through state space. However, potential is not exhausted by energetic description. Two systems may possess identical total energy and yet differ radically in the number, accessibility, and stability of the configurations they can sustain. What distinguishes such systems is not energy content, but constraint structure.

This perspective also clarifies the meaning of energy conservation. In closed descriptions exhibiting time-translation symmetry, a conserved energy functional serves as an invariant that restricts admissible trajectories through state space. Conservation does not imply that a “thing” called energy is preserved as a substance, nor that the overall space of possibilities is conserved in any ontological sense. Rather, conservation reflects the invariance of a scalar functional under a chosen description. Redistribution of energy corresponds to a redistribution of admissible transitions across degrees of freedom, not to the creation of new ontological content. In explicitly time-dependent drivings, non-stationary spacetimes, or certain general-relativistic settings, a globally conserved energy may be non-unique or undefined; the present argument concerns cases where a conserved energy functional is well-posed within the chosen model.

Energy should therefore not be described as unconstrained. It is constrained by global symmetries, conservation laws, and boundary conditions. What distinguishes energy from ontic structures is not the absence of constraint, but the kind of constraint it bears. Energy is globally constrained yet non-authoritative: it does not supply identity conditions, persistence criteria, or record authority. Where stability, decoherence, or classicality arise, they do so through the full Hamiltonian structure—its couplings, spectra, locality properties, and environmental interactions—not through the scalar value of energy alone. In standard formalisms the Hamiltonian generates dynamics; the claim here is that the scalar quantity “energy” is not ontologically generative. Explanatory work is done by the Hamiltonian’s structural content, with energy serving as one invariant parameter within that structure.

For this reason, energy is not itself a persisting form or configuration. Where energy appears localized or embodied, what persists are constrained field or matter degrees of freedom whose states admit energetic description relative to a chosen Hamiltonian. Energy is an attribute of such states, not the bearer of form. Form requires exclusion and persistence: to have form is to rule out alternatives and remain distinguishable under constraint. Energy, by contrast, indexes configurability without enforcing exclusion on its own.

This explains why energy alone cannot ground ontology. An unconstrained or weakly constrained distribution of potential would admit many configurations without stabilizing any. Ontological structure arises only when constraints channel potential into stable patterns that exclude alternatives. Energy participates in this process descriptively, but it does not drive it. Greater energy does not imply greater structure; absent appropriate constraints, it implies only a larger

space of admissible reconfiguration.

The representation-dependence of energy further reinforces this stance. Energy values depend on reference frames, coordinate choices, coarse-graining, and modeling assumptions. Changes in description can alter energetic representations without necessarily altering the underlying dynamics. This confirms that energy functions as a coordinate for describing potential within a model, not as the ground from which persistence arises.

Importantly, this reframing does not privilege any particular physical formalism. Whether energy is expressed through classical fields, quantum operators, or thermodynamic variables, its role remains consistent: to encode how constraints shape admissible change within a given description. Quantum Coherence Theory remains agnostic about which formalism is ultimately fundamental, focusing instead on the invariant feature they share—the tracking of potential as it is redistributed under constraint.

By treating energy as globally conserved but non-authoritative potential, Quantum Coherence Theory clears conceptual space for constraint to serve as the primary explanatory primitive of persistence. Energy does not cause coherence, collapse, or classicality as an independent ontic driver. It indexes how constraint pathways are implemented within existing formalisms, with energy scales entering as parameters of interaction structure rather than as autonomous agencies. Stabilization and collapse occur not because energy “acts,” but because admissible possibilities are progressively restricted as systems interact and aggregate constraints.

This framing also provides a natural interpretation of mass–energy relations in bound systems. In standard physics, the mass of a composite system is often less than the sum of the masses of its constituents, with the difference accounted for as binding energy. Within a constraint-first framework, this reflects a reduction in constraint redundancy. When systems bind, degrees of freedom that previously required independent stabilization become jointly governed under shared constraints. The resulting mass defect does not represent a “cost” paid to bind the system, but the release of energetic potential rendered unnecessary by the consolidation of constraint structure. Energy conservation remains intact; what changes is the density and redundancy of constraint required to sustain distinct configurations.

This section therefore establishes a disciplined modeling posture. Energy is retained fully within its empirical and mathematical roles, but it is not granted explanatory primacy. The explanation of why particular structures persist, decohere, or stabilize must be sought in how potential is constrained, not in the mere presence or magnitude of energy. This shift is methodological rather than metaphysical, conservative rather than speculative, and fully consistent with established physics.

3.4 Vacuum Energy as a Boundary Case

Vacuum energy provides a uniquely clear boundary case for any account that treats energetic presence as inherently sufficient to generate persistent structure. In modern physics, vacuum states are empirically real, mathematically indispensable, and in some contexts gravitationally consequential. Yet under fixed conditions they do not, by themselves, give rise to localized, record-bearing forms. This makes vacuum energy an ideal test of the central claim of Quantum Coherence Theory: energy can be real, conserved, and globally influential while still failing to determine persistence, identity, or form. What performs the explanatory work is not energy alone, but the architecture of constraints that restrict and stabilize admissible possibilities.

In quantum field theory, the vacuum is not “nothing,” but the ground state of interacting fields relative to a specified Hamiltonian and boundary conditions. As a state rather than a substance, it

can possess rich internal structure: nontrivial correlations, entanglement across regions, and well-defined symmetries. Observable effects associated with vacuum structure include forces that arise when boundary conditions are altered and vacuum-like contributions to the stress-energy tensor that, in general relativity, influence cosmic expansion. These facts establish that vacuum energy is physically consequential rather than a mere accounting convention.

At the same time, vacuum energy is not a locally actionable reservoir. In nongravitational quantum field theory, the absolute value of the vacuum energy can be shifted without changing the dynamics; what is physically meaningful are correlations and differences between configurations, especially when boundaries or interactions change. Even where vacuum-like energy has observable gravitational effects, its influence is uniform rather than selective. The presence of energy does not by itself specify where structure should appear, which configurations should persist, or how identities should stabilize.

This helps clarify a frequent misunderstanding. Vacuum fluctuations are real, but they do not consist of transient packets of freely extractable energy. They reflect nonzero variances and correlations characteristic of a stationary state. Under fixed constraints, such a state does not yield sustained work or stable structure. To obtain localized form, one must introduce additional conditions that alter the constraint environment: boundaries, gradients, symmetry breaking, non-equilibrium driving, or time-dependent backgrounds. Energy becomes actionable only when the constraints governing it change.

A homogeneous, translation-invariant vacuum also makes clear why vacuum structure does not yield persistent localized form. Such a state has no preferred location, orientation, or internal mechanism for selecting one configuration over another. Time evolution under the same governing conditions preserves this homogeneity. While the vacuum contains structure in the sense of correlations, this is not the same as stable, classically legible form. Persistent identities, records, and macroscopic structures require mechanisms that both select and stabilize particular configurations—mechanisms absent in a stationary vacuum. Gravity sharpens rather than weakens this lesson. Vacuum-like energy components in general relativity can influence global dynamics, such as cosmic expansion, through a uniform energy density with negative pressure. Yet gravitational influence does not imply local structure formation. A Lorentz-invariant vacuum component remains homogeneous and does not, by itself, clump into atoms, boundaries, or records. Structure arises only when additional dynamical processes—instabilities, dissipation, clustering matter, or symmetry breaking—restrict admissible outcomes and stabilize particular forms.

Apparent counterexamples reinforce this point rather than undermine it. Particle production associated with horizons, time-dependent backgrounds, or rapidly changing boundaries occurs only because the constraint environment has changed. Inflationary cosmology relies on accelerated expansion and horizon dynamics; particle production near black holes depends on horizon structure and observer-dependent conditions; boundary-driven effects require external driving. In every case, vacuum energy becomes form-relevant only because new constraints are introduced that select and stabilize specific outcomes.

Even situations involving degenerate or metastable vacua do not escape this logic. Where multiple minima exist, the appearance of domains, defects, or ordered phases depends on how constraints partition the space of possibilities and how dynamics selects among them. Cooling histories, boundary conditions, interactions, and environmental coupling determine which configurations persist. Energy alone does not explain the result; constraint architecture does.

Vacuum energy therefore illustrates a fundamental limitation of energetic explanation. Energetic accounting can be complete and even globally decisive while still leaving the emergence of form unexplained. Energy presence underdetermines which degrees of freedom become localized, symmetry-broken, and stabilized as persistent structures. The decisive factor is how constraints re-

strict admissible possibilities and exclude alternatives strongly enough to support enduring records.

Within Quantum Coherence Theory, vacuum energy thus functions as a boundary case rather than a foundational driver. It demonstrates that energy can be real, constrained, and consequential without being form-generating. Persistence arises not from energetic abundance, but from the organization of constraints that select, localize, and stabilize possibility.

3.5 Matter as Localized Identity Under Constraint

In Quantum Coherence Theory, matter is not treated as a primitive substance. It is understood as a stable, localized configuration of constrained potential—a pattern that constitutes a re-identifiable identity under interaction. What distinguishes matter from a transient fluctuation is not composition, nor longevity, but the fact that an identity structure exists at all. Persistence here does not mean indefinite duration; it means that identity is well-defined in the presence of interaction.

Re-identifiability is understood structurally rather than procedurally. A configuration constitutes an identity if its defining equivalence relation remains well-defined under interaction, rather than collapsing into indistinguishability. Identity is thus treated as a robustness property of patterns under interaction, not as an intrinsic label attached to a microstate. A configuration either admits re-identification under interaction, or it does not.

Here, constraint refers generically to structural restrictions—dynamical, relational, or boundary-based—that limit the admissible continuations of a configuration. Constraints are not limited to external boundaries; they include dynamical invariants, symmetry restrictions, selection rules, conservation-governed couplings, and interaction structures that restrict how a configuration may evolve. Many patterns may arise within a system, but only those supported by sufficient constraint density—that is, by constraints that significantly restrict admissible evolutions relative to an identity equivalence class—constitute coherent identities.

Equivalently, higher constraint density corresponds to a smaller set of admissible future trajectories consistent with the configuration remaining the same kind of thing under interaction. Matter is therefore not defined by stasis. It is defined by coherence through change, where transformation is permitted but bounded in ways that preserve identity for as long as that identity exists.

Identity in this framework is not identity of microstate. A material configuration may undergo continuous internal rearrangement while remaining the same entity in any operationally meaningful sense. What is preserved, while the identity exists, is an equivalence class of admissible configurations defined by stabilizing constraints. The equivalence relation is fixed by whatever features are preserved by those constraints—such as conserved quantum numbers, robust response signatures, topological invariants, or dynamically maintained boundary conditions. These constraints determine which variations preserve identity and which transitions constitute transformation into a different identity.

Crucially, identity is necessarily local and asymmetric. For a configuration to count as matter, it must distinguish itself from its surroundings in a non-arbitrary way. This requires constraints that close locally, meaning that the identity conditions primarily reference internal relations rather than being wholly specified by external bookkeeping. Local closure does not require complete isolation: identities may be stabilized either by internally closed constraints or by structured environmental support, provided that re-identification remains well-defined under the relevant interactions.

Such closure yields a bounded region—spatial, relational, or functional—whose admissible internal configurations differ from those outside it. Without local asymmetry, there is no principled

basis for re-identification, and therefore no identity. Matter is thus defined by the presence of a localized identity structure, not by separation from interaction.

This clarifies why isolation cannot serve as a criterion for material identity. Identity is not supplied by suppressing interaction; it is defined relative to the interaction structures that couple a configuration to its environment. A configuration whose identity is only meaningful in perfect isolation does not constitute material identity in this operational sense. Interaction does not test matter into existence; it is the context in which identity is defined at all.

Energy, as treated in Quantum Coherence Theory, is ontically constrained, but as a conserved quantity it does not by itself supply an individuation rule for carving the world into bounded, re-identifiable entities. Conserved quantities constrain transitions and enforce continuity across interactions, but individuation requires additional constraints—such as binding structures, symmetry breaking, gaps, or topological protections—that generate localized, stable sectors. Matter names those regimes in which such constraints produce localized identity.

At macroscopic scales, this localized identity manifests phenomenologically as discreteness and objecthood. High constraint density yields bounded, repeatable patterns that appear as distinct objects. An atom constitutes a re-identifiable identity under generic coupling and exchange; an interference pattern or thermal fluctuation typically does not, except under carefully maintained boundary conditions that supply additional stabilizing constraints. Where constraint density weakens, localization degrades: configurations spread, interfere, or lose distinguishability.

These regimes form a continuum, not a sharp dichotomy. Matter-likeness increases with the breadth of interaction classes under which re-identification remains stable. Calling this a continuum does not erase the usefulness of the term matter; it explains why objecthood can be sharp in some regimes and progressively ill-defined in others as constraint organization changes.

Matter, then, is not assumed at the outset of Quantum Coherence Theory. It emerges wherever constraint induces a localized, re-identifiable identity, regardless of how long that identity lasts. The distinction is structural rather than metaphysical: what is ontologically present as an identity versus what never constitutes one. Matter names the former—not as a fundamental substance, but as a stabilized mode of being.

3.6 Locality and Gravity from Constraint Aggregation

In Quantum Coherence Theory, locality is not treated as a primitive geometric fact, nor as a coordinate assignment imposed upon otherwise unconstrained entities. It is understood as a structural consequence of identity persistence under dense mutual constraint. Once a configuration constitutes a re-identifiable identity under interaction, the conditions required to preserve that identity necessarily restrict how it may relate to other identities. Locality arises as the regime in which admissible interactions are constrained by the requirement that identities remain mutually recognizable within a shared constraint environment.

An identity, in this framework, is not a fixed object or microstate, but an equivalence class of admissible configurations that remain re-identifiable across interaction. Re-identification fails not when change occurs, but when the constraints defining that equivalence class are violated. Constraint density refers to the number and strength of independent compatibility conditions that must be simultaneously satisfied for re-identification to remain possible. As constraint density increases, admissible continuations narrow, and identity persistence becomes increasingly sensitive to relational compatibility.

Locality, therefore, is not merely spatial proximity. It is a relational compatibility regime defined by how tightly the admissible evolutions of one identity depend upon the admissible evolutions of

others. A configuration is local when its continued identity depends primarily on maintaining compatibility with a bounded relational neighborhood of other identities, such that departures from shared constraints rapidly undermine re-identifiability. Localization is reinforced collectively: each localized identity participates in a network of mutual constraint that stabilizes the locality of others.

This framing explains why locality strengthens rather than weakens as constraint density increases. As constraints accumulate, transitions that would delocalize an identity do not merely become improbable; they become structurally inadmissible. Such transitions violate the equivalence relations required for re-identification and therefore cease to count as valid continuations of that identity at all. Locality is thus enforced internally, as a consequence of identity preservation under interaction, rather than imposed by an external background or limiting principle.

At sufficiently high aggregate constraint density, an additional stabilizing behavior emerges. When large numbers of localized identities participate in a shared constraint environment, configurations that preserve broad relational compatibility across the system become preferentially admissible. This large-scale stabilization response is identified phenomenologically as gravity. Quantum Coherence Theory does not introduce gravity as a fundamental interaction, nor does it seek to replace its established mathematical descriptions. Instead, gravity is treated structurally as the macroscopic expression of constraint aggregation: a global bias toward configurations that preserve compatibility among many densely constrained identities.

From this perspective, gravity is not anomalously weak at small scales and dominant at large ones. Its scaling behavior follows directly from its structural role. Where identities are weakly constrained or sparsely coupled, the stabilizing effect associated with constraint aggregation is negligible. Where identities are densely constrained and mutually reinforcing, that stabilizing effect becomes increasingly significant. The macroscopic prominence of gravity is therefore not an independent mystery, but an expected consequence of dense localization across large collections of identities.

Locality and gravity, though often treated as distinct phenomena, arise from the same underlying structural condition. Both reflect system behavior under high constraint density, where admissible configurations are restricted not only internally but relationally. Locality governs the near-scale restriction of admissible interactions by enforcing compatibility within bounded relational neighborhoods. Gravity governs the large-scale restriction of admissible configurations by favoring globally compatible arrangements of those neighborhoods. They are not separate mechanisms, but different expressions of the same constraint regime operating across scale.

The result is a unified account of classical persistence that avoids interpretive discontinuities. Localized matter remains localized because the constraints that define its identity are reinforced through mutual participation in a dense constraint environment. Apparent delocalization decoheres rapidly because it violates the equivalence relations required for re-identification under interaction. Large-scale structure remains stable because configurations that preserve global constraint compatibility remain dynamically viable.

This account neither replaces nor competes with existing physical theories. It functions as an organizing framework. It clarifies why classical behavior emerges continuously from quantum regimes as constraint density increases, why localization is reinforced rather than destroyed by interaction, and why gravity becomes dominant at scale without being fundamental in the same sense as microscopic interaction rules.

Locality and gravity, then, are not additional assumptions within Quantum Coherence Theory. They follow from the same structural requirement that defines matter itself: the persistence of re-identifiable identity under interaction. Where constraint closes locally, identity stabilizes. Where constraint aggregates globally, stabilization extends. The classical world is not imposed upon

quantum reality; it is the regime in which constraint has become sufficiently dense to enforce it.

Scope Note

This work is intentionally introductory rather than exhaustive. Its purpose is to present the organizing framework—identity persistence under interaction, locality as a compatibility regime, and gravity as the macroscopic expression of aggregated constraint stabilization—without attempting a full formal derivation or a complete mathematical instantiation.

Several natural extensions would reinforce and further discipline the account developed here. These include, for example, explicit definitions of identity classes and constraint density within a chosen formal substrate; derivations of emergent locality in terms of bounded dependency or propagation structure; and a more detailed treatment of how large-scale constraint aggregation reproduces familiar gravitational limits. These developments are not denied or set aside as unnecessary. They are deliberately deferred to later refinement, where they can be addressed with the specificity and technical care they require.

Accordingly, the present section should be read as a structural orientation rather than a completed formal theory. Its role is to clarify how locality and gravity can be understood as consequences of the same constraint regime, and why these phenomena emerge together as constraint density increases. Formalization, quantitative commitments, and derived limits are left to subsequent work, where their treatment would be both appropriate and meaningful.

3.7 Scale as a Proxy for Constraint Density

Just as material identity emerges when constraints close locally, classical behavior emerges when constraints accumulate globally across interaction. For QCT, scale is consequently not treated as a fundamental explanatory variable. Differences between quantum and classical behavior are not attributed to size, complexity, or aggregation in themselves, but to how constraint density changes as systems interact, compose, and persist. Scale functions only as a proxy: it correlates with the number, strength, and interdependence of limiting conditions acting on admissible configurations within a given description.

At small scales, systems are often weakly constrained in an operational sense. Degrees of freedom may remain loosely coupled, environmental interaction may be limited, and admissible configurations may remain broad relative to the chosen partition. Under these conditions, multiple alternatives can remain jointly admissible without immediate exclusion. Superposition, interference, and probabilistic manifestation arise not because small systems are ontologically special, but because constraint density is low. Low constraint density does not imply disorder or randomness; it implies permissiveness. Many configurations are allowed to coexist without forcing resolution.

As systems compose and endure, constraint density often increases. Composition introduces additional interaction pathways, coupling becomes more pervasive, and compatibility requirements multiply. Environmental entanglement, boundary formation, repeated interaction, and coarse-grained stabilization impose overlapping restrictions on admissible configurations. This accumulation does not require new principles, thresholds, or scale-dependent laws. It follows naturally from the composition of constraint relations across interacting parts.

Classical behavior emerges when constraint density becomes sufficiently high that only a narrow subset of configurations remains mutually compatible under continued interaction. Under such conditions, superposed alternatives are rapidly rendered non-contributory to stable prediction, recoverability of interference collapses, and persistent identities emerge across interaction.

Determinism, in this view, is not fundamental but effective. It reflects the scarcity of admissible alternatives relative to the prevailing constraint structure, not the absence of underlying possibility.

Scale, however, is an imperfect proxy. Large systems can remain weakly constrained with respect to particular degrees of freedom when interaction is limited, coupling channels are restricted, or constraint pathways are aligned to protect coherence rather than record alternatives. Empirical cases such as superconductivity, Bose–Einstein condensation, and macroscopic quantum interference demonstrate that coherence can persist at large scales when environmental monitoring is suppressed and constraint structures are coherence-preserving rather than exclusivity-enforcing. These cases do not contradict the framework; they confirm it. What matters is not size, but the organization and function of constraints.

Conversely, small systems can exhibit high effective constraint density when strongly coupled to their environment or subjected to restrictive boundary conditions. Rapid decoherence in microscopic systems illustrates that quantum behavior is not guaranteed by scale alone. Scale correlates with constraint density only insofar as it increases opportunities for interaction, redundancy, and incompatibility.

This framing resolves a persistent ambiguity in discussions of the quantum–classical transition. Rather than positing a sharp boundary or invoking scale-dependent regimes of law, QCT treats the transition as continuous. As constraint density increases, admissibility narrows. As admissibility narrows, persistence stabilizes. No categorical jump is required. Quantum and classical behaviors occupy different regions of the same constraint landscape.

Scale-dependence in Quantum Coherence Theory is therefore descriptive rather than ontological. It reflects how constraints aggregate under interaction and composition, not a division of reality into separate domains. A single system may exhibit quantum behavior under one partition and classical behavior under another, depending on how constraints are evaluated and at what resolution.

In summary, scale matters in Quantum Coherence Theory only because constraints tend to accumulate—or, in some cases, align—through interaction, aggregation, and coupling. Where constraint density is low, permissiveness persists. Where constraint density is high, stable identity and predictability emerge. The quantum–classical transition is not imposed by size, but revealed by the structure and function of constraints acting across scale.

4 Quantum Coherence as Low Constraint Persistence

This section characterizes quantum coherence as a regime of low effective constraint density, where multiple configurations remain jointly admissible under interaction. Superposition, interference, and entanglement are treated not as anomalies or primitives, but as natural consequences of permissive constraint structure. Quantum behavior reflects persistence without forced exclusion, rather than indeterminacy or observer dependence.

4.1 Stability of Coherence at Low Constraint Density

Quantum coherence, as treated within Quantum Coherence Theory, refers to the sustained mutual compatibility of multiple admissible configurations under a given constraint set. In this section, coherence is not introduced as a special quantum oddity, nor as a paradoxical coexistence of mutually exclusive realities. Instead, it names a structural condition: a regime in which the limiting conditions acting on a system are insufficient to enforce exclusion among alternative configurations. Coherence, in this sense, is not a mystery requiring resolution, but the expected behavior of systems operating under permissive constraint environments.

At low effective constraint density, limiting conditions are sparse, weakly coupled, or non-aggregating. Interactions do not yet impose incompatibility across the full space of admissible configurations, and prior structure does not strongly restrict future continuation. As a result, multiple possibilities can remain jointly admissible without generating contradiction or instability within the description. Coherence persists not because the system is indecisive or indeterminate, but because the structural pressure required to differentiate and exclude alternatives has not yet accumulated.

This framing applies across the quantum regime and underlies the phenomena examined throughout this section. What are often presented as distinct quantum effects—such as superposition, interference, tunneling, and entanglement—are treated here as specific manifestations of coherence under permissive constraint conditions rather than as independent primitives. Each reflects a situation in which exclusion has not yet been structurally enforced.

Seen in this light, superposition does not assert that reality is “many things at once” in any absolute sense. It reflects underdetermination rather than multiplicity. Superposition marks a condition in which constraints have not yet forced differentiation among admissible configurations. The system’s description remains open not because of ignorance or vagueness, but because exclusion has not yet been required by the constraint structure.

Admissibility is therefore central. A coherent system is one for which multiple state descriptions remain admissible under the same constraint set. These descriptions are never arbitrary and are never unconstrained. They are bounded by conservation relations, boundary conditions, coupling structure, and prior interaction history. Coherence is thus freedom within limits: permissiveness structured by constraints that have not yet become mutually incompatible.

This explains why coherence is most robust in regimes characterized by isolation, limited interaction history, and weak or carefully structured environmental coupling. Small systems, short-lived processes, and controlled experimental arrangements often exhibit coherence not because smallness confers special privilege, but because constraint accumulation remains minimal. Under such conditions, coherence is not fragile or exceptional. It is the default behavior of permissive dynamics.

Crucially, coherence stability is not equivalent to dynamical inactivity. Coherent systems may evolve, interfere, exchange energy, and undergo complex transformations while remaining coherent, provided such evolution does not introduce constraints that partition admissible alternatives into mutually incompatible outcomes. Change itself does not destabilize coherence. What destabilizes coherence is exclusion: the accumulation of constraints that render alternatives no longer jointly admissible.

In physical terms, constraint accumulation corresponds to the progressive creation of correlations that render alternative configurations distinguishable under interaction, thereby suppressing their joint admissibility as stable descriptions. When interactions produce no such distinguishing structure, coherence persists. When they do, coherence degrades—not because a choice is made, but because compatibility fails.

The stability of coherence is also resolution-dependent. At a coarse descriptive scale, a system may appear coherent even as fine-grained distinctions fluctuate beneath the modeling threshold. Conversely, increasing descriptive resolution can reveal latent constraint accumulation that renders coherence unstable. This dependence on partition and resolution reinforces that coherence is not an absolute property of systems, but a structural condition relative to how constraints are evaluated.

As interactions accumulate, constraint density increases. Coupling to additional degrees of freedom, entanglement with uncontrolled environments, and repeated perturbations introduce new restrictions on admissibility. These restrictions need not be strong individually. Their significance lies in their aggregation. Coherence degrades when the combined constraint structure no longer permits multiple configurations to remain mutually compatible as continuations of the same description.

Importantly, Quantum Coherence Theory does not treat coherence loss as a failure mode or anomaly. It is the expected outcome of constraint thickening. Coherence persists precisely as long as permissiveness remains structurally viable, and no longer. The transition from coherent to incoherent behavior is therefore continuous rather than categorical, reflecting gradual narrowing of admissibility rather than abrupt ontological change.

Coherence, in this framework, is neither metaphysical excess nor epistemic artifact. It is the predictable behavior of systems whose constraint environment has not yet enforced differentiation. Wherever limiting conditions remain insufficient to impose exclusion, coherence will persist. Wherever constraints accumulate to the point of incompatibility, coherence will dissolve. This principle governs the quantum regime examined in this section and prepares the ground for the transition mechanisms addressed in the following section.

4.2 Wave–Particle Duality as Constraint-Dependent Description

Wave–particle duality is among the most persistent ideas in quantum mechanics, yet it is also among the most misleading. The term survives primarily as a legacy label for the mismatch between classical categories and quantum formalism, not because it reflects a genuine duality in the nature of quantum systems. What it describes is not a system alternating between two physical modes, but a difference in how a single quantum state is manifested under different interaction contexts.

Quantum mechanics does not contain separate “wave” and “particle” dynamics. It contains a quantum state—represented by a state vector or density operator—that evolves unitarily when sufficiently isolated, together with rules for assigning probabilities to outcomes when the system undergoes a measurement interaction. The phenomena historically called “wave-like” arise from

coherent superposition with phase-sensitive effects, while those called “particle-like” arise from discrete, localized detection events associated with particular measurement records. Both behaviors emerge from the same formal structure.

The classical language of duality arose because early experiments violated the sharp distinction between waves and particles inherited from pre-quantum physics. Quantum systems produced interference patterns characteristic of waves while also yielding discrete interaction events characteristic of particles. In the absence of a framework capable of describing coherence under interaction and constraint, both classical categories were retained, and their coexistence was labeled a duality. While pedagogically convenient, this language obscures the underlying structure.

From a coherence-based perspective, the distinction reduces to whether the relevant alternatives remain coherent or become distinguishable through interaction. When a quantum system evolves without becoming correlated with degrees of freedom that record which alternative occurred, phase relations are preserved and interference can be observed. When the system interacts with an apparatus or environment in a way that correlates distinct alternatives with stable records, those phase relations become inaccessible, interference is suppressed, and outcomes appear as discrete, localized events. Nothing changes in the ontology of the system itself; what changes is the constraint structure imposed by the interaction.

A common source of confusion is the assumption that spatial localization is a property carried continuously through time. It is not. Between interactions, a quantum state need not be spatially localized at all. What is preserved in closed-system evolution is unitarity—norm, inner products, and the conservation laws associated with symmetries—not position. Apparent “loss of coherence” in realistic situations reflects the redistribution of coherence into entanglement with uncontrolled degrees of freedom, rather than a fundamental destruction of the quantum state.

The global consistency of quantum predictions already presupposes nonlocal correlation structure. Entangled states encode correlations that cannot be reduced to independent local descriptions, even across spacelike separation. At the same time, the theory remains no-signaling: local outcome statistics do not depend on distant measurement choices. The nonlocality involved is therefore not one of control or influence, but of correlation encoded in the joint quantum state. Descriptions invoking global admissibility or constraint should be understood in this sense—as reflecting the structure of joint states and their consistency conditions, not superluminal dynamics.

In this light, wave-particle duality should not be treated as an explanatory principle. It remains useful as historical shorthand, but it obscures the roles of unitary evolution, measurement interaction, and decoherence, and it encourages the mistaken impression that quantum systems alternate between incompatible forms of being. A more accurate description is therefore adopted here: quantum states evolve coherently under unitary dynamics, and discrete, localized outcomes arise when interactions generate stable records that render alternatives distinguishable. What differs between “wave-like” and “particle-like” phenomena is not ontology, but context—specifically, whether coherence relevant to interference is preserved or constrained by interaction.

4.3 Superposition as Unresolved Admissible Possibility

Here superposition is not treated as a paradoxical coexistence of mutually exclusive realities, nor as an ontological claim that all possibilities are equally real. It is treated instead as a structural condition: the simultaneous admissibility of multiple configurations under insufficient constraint. Superposition names a regime in which exclusion has not yet been enforced, rather than a claim about reality multiplying itself.

Within a constraint-first framework, a system occupies a superposed state when the limiting conditions acting upon it do not yet exclude alternative realizations. Superposition therefore reflects permissiveness, not multiplicity. It indicates that the system’s constraint environment allows several distinct configurations to remain viable descriptions at the chosen resolution, including their relational structure, rather than forcing differentiation into a single stabilized alternative.

This reframing removes the need to interpret superposition as a statement about reality “being many things at once” in any absolute sense. Instead, it is a statement about what has not yet been ruled out by interaction. The system is not indecisive; it is underdetermined. Its description remains open because the constraints required to enforce exclusion and stabilize identity have not yet accumulated.

Admissibility is central to this account. A superposed system is one for which multiple state descriptions remain jointly admissible within the same constraint set. These descriptions are not arbitrary, nor are they merely epistemic placeholders. They are restricted by conservation relations, boundary conditions, coupling structure, and prior interactions, and they remain mutually compatible because the active constraints have not yet rendered them exclusive.

Superposition is therefore not unconstrained freedom, but freedom within limits. This perspective clarifies why superposition is stable at small scales and fragile at larger ones without appealing to size as a primitive. At low effective constraint density, interactions are sparse, coupling is weak, and exclusion pressure is minimal.

Under these conditions, admissible alternatives can coexist without forcing resolution, and relational structure among them remains intact. As constraint density increases through interaction, entanglement, or environmental coupling, admissibility narrows. Superposition does not end because a choice is made, but because alternatives cease to be jointly permissible as stable descriptions.

Importantly, this account does not treat superposition as a purely epistemic artifact. The permissiveness it describes is physically instantiated through interaction structure and constraint distribution. At the same time, it does not require superposition to be ontologically fundamental. It is a regime-dependent description that applies wherever constraint aggregation has not yet enforced exclusion, and it fades precisely as exclusion becomes structurally unavoidable.

Superposition, in Quantum Coherence Theory, is therefore neither mystery nor metaphysics. It is the expected behavior of systems whose constraint environment has not yet imposed sufficient differentiation. It persists exactly as long as admissibility remains unresolved under interaction—and no longer.

4.4 Tunneling as Constraint-Compatible Persistence

Tunneling, in regards to this framework, is not treated as an exception to constraint or as a temporary violation of physical law. It is a canonical case of persistence evaluated at the level of the full constraint structure rather than by isolated classical pointwise criteria. What appears classically as an impenetrable barrier corresponds, in QCT terms, to a region of elevated local constraint density: the space of admissible configurations is narrowed and strongly suppressed, but not necessarily eliminated.

In standard quantum mechanics, tunneling arises because physical admissibility is determined by globally consistent solutions to the governing dynamical equations, subject to continuity, normalization, and boundary conditions. A system must satisfy these constraints across the entirety of configuration space, including regions in which classical motion would be forbidden. In such

regions, admissible solutions change character rather than disappearing: oscillatory behavior gives way to evanescent structure. From a QCT perspective, this reflects the fact that admissibility is evaluated structurally, not by the existence of a locally classical trajectory at each point.

This framing avoids the familiar but misleading narrative that tunneling involves “borrowing energy” or suspending conservation laws. No law is violated, and no special dynamics are introduced. Energy conservation holds globally, and the system satisfies the same constraints everywhere. In QCT terms, the system does not locally evade constraint; rather, a globally admissible configuration exists whose support spans both sides of the barrier. Persistence here refers to the continuity of admissible configurations under unitary evolution and boundary constraints, not to traversal along a classical path through a forbidden region.

Constraint density in tunneling admits a direct operational interpretation. Increasing barrier height, increasing the width of the classically forbidden region, increasing mass, or strengthening coupling all suppress admissibility continuously rather than abruptly. The admissible weight associated with persistence across the barrier decreases smoothly as constraints intensify. In the high-constraint limit, this weight becomes so small that persistence is effectively unobservable for all practical purposes, and classical impenetrability emerges as a limiting behavior rather than as a new postulate.

Two distinct mechanisms contribute to this suppression. The first is intrinsic, arising from the constraint structure of the barrier itself, which reduces the admissible weight of configurations that span it. The second arises from environmental coupling.

Interaction with uncontrolled degrees of freedom increases effective constraint density by entangling the system with its environment, suppressing coherent phase relations between alternatives. This does not remove tunneling-compatible solutions from the closed-system description; instead, it limits the stability and observability of interference-based records in the reduced description. QCT treats these contributions as structurally distinct but complementary: one suppresses admissibility weight, the other suppresses coherence.

Tunneling does not presuppose observation, measurement, or collapse. It occurs entirely within the regime of coherent persistence, prior to record formation. In this respect, tunneling anticipates later discussions of decoherence and classicality by illustrating how exclusion sharpens gradually as constraint density increases, rather than appearing suddenly through the introduction of new explanatory primitives.

Interpreted as constraint-compatible persistence, tunneling ceases to appear anomalous. It is a direct consequence of evaluating admissibility at the level of the global constraint structure rather than at locally classical slices. The phenomenon demonstrates how global compatibility governs persistence in low-constraint regimes and how classical exclusion emerges smoothly as constraints accumulate, providing a continuous bridge between quantum coherence and classical stability within a single explanatory framework.

4.5 Entanglement as Shared Constraint Structure

In QCT, entanglement is not treated as a mysterious linkage between distant objects, nor as a signal-carrying mechanism. It is understood structurally, as the emergence and persistence of shared constraint relations across multiple subsystems such that admissibility can no longer be evaluated independently for each part.

When two or more systems interact in a manner that correlates their admissible states, the constraint sets governing their evolution become partially shared. From that point onward, the

admissible configurations of each subsystem are no longer separable from those of the others. What remains admissible for one is conditional on what remains admissible for the rest. Entanglement is the persistence of this joint admissibility structure under subsequent evolution.

In QCT, a composite system is entangled iff its joint admissibility structure cannot be represented as any convex mixture of independent local admissibility structures—i.e., there is no decomposition of the form “sometimes (A) is admissible in way (i) while (B) is admissible in way (i), with only classical mixing between (i).” When no such decomposition exists, the admissible state space itself does not factor into local parts, and entanglement is present.

This framing removes the need to treat entanglement as a special kind of influence. No information is exchanged at the moment correlations are observed, and no dynamical coupling is required at the time outcomes are recorded. The correlations arise because the shared constraint structure was established during prior interaction and has remained intact. Observation does not generate this structure; it reveals which joint realization within an already-restricted admissible space has occurred (including the familiar distinction between unconditional local statistics—unchanged at a distance—and conditional updates once outcomes are compared).

Entanglement therefore reflects relational persistence, not intrinsic connection. The systems involved need not be spatially adjacent, dynamically coupled at the time of observation, or continuously interacting. What matters is that their admissible configurations were constrained together in the past and that no subsequent interactions have introduced sufficient constraint incompatibility to dissolve that shared structure.

Within this view, the non-classical features of entanglement follow directly. Joint outcome statistics fail to decompose into independent local descriptions because the admissible state space itself does not decompose into independent parts. Local descriptions become incomplete not because of hidden mechanisms or instantaneous influence, but because the relevant constraints are distributed across the composite system rather than localized within its components.

QCT’s structural account is fully compatible with no superluminal signaling, but it does not reinstate Bell-local factorization. In Bell terms, the joint admissibility constraints for entangled systems are irreducibly global, so the theory does not assume that outcomes can be generated by purely local specifications plus shared randomness while preserving the observed quantum correlations. The “no influence at measurement” claim is therefore a no-signaling claim, not a claim of Bell-local causality.

The stability of entanglement is governed by the same principles as coherence more generally. At low effective constraint density, shared constraint structures can persist across extended interaction histories. As additional interactions accumulate—particularly those that couple subsystems to largely independent environments—the shared constraints fragment. When this occurs, admissibility separates and entanglement degrades continuously, without requiring any discontinuous event or interpretive intervention.

QCT does not treat entanglement as fundamental or universal. It is contingent on how constraints arise, align, and persist. Nor does entanglement imply global coherence or universal connectedness. It is a local structural condition, emergent from specific interaction histories and vulnerable to constraint incompatibility.

Under this framework, entanglement is neither paradoxical nor exceptional. It is the expected manifestation of coherence when constraint relations span more than one subsystem. Where constraints are shared, admissibility is shared. Where admissibility is shared, entanglement appears.

4.6 Nonlocal Correlation Without Causal Violation

Nonlocal correlation, under QCT, is understood as a structural consequence of shared admissibility rather than as evidence of superluminal influence, hidden communication, or action at a distance. Correlations spanning spatial separation arise because the constraints governing admissible outcomes were established jointly through prior interaction and persist thereafter. Nothing propagates between subsystems at the moment of observation; what persists is a global constraint structure within which outcomes are realized.

Once systems share such a structure, admissibility can no longer be evaluated independently for each part. The relevant space of admissible configurations becomes composite: what remains admissible for one subsystem is conditional on what remains admissible for the others. Spatial separation does not restore independence, because distance alone does not alter admissibility.

Constraints are not carriers of influence localized in space; they are relations defining which joint configurations can persist coherently under the governing dynamics. Interactions and measurements remain local, but the admissibility relations they probe are defined at the level of the composite system.

This framing preserves relativistic causality in its operational sense. Local outcome statistics remain invariant under distant measurement choices. No controllable signal is transmitted, no information is sent across space, and no update propagates between subsystems at measurement. Observation does not generate correlation; it conditions description on one realized element of an already-restricted joint admissibility structure. The familiar distinction between unconditional local statistics and conditional updates once outcomes are compared is preserved without invoking superluminal causation.

From a QCT perspective, Bell-type violations do not indicate faster-than-light dynamics. They indicate the failure of a separability assumption: namely, that admissibility for spatially separated systems can always be decomposed into independent local components supplemented by shared randomness. QCT explicitly rejects this form of Bell-local factorization. When constraints are shared, no admissibility-complete specification exists that restores independent local descriptions while preserving observed correlations. The violation is therefore structural rather than dynamical, reflecting the irreducibly global character of admissibility once systems have interacted.

This account does not rely on hidden variables, retrocausal influence, or observer-dependent collapse. The underlying dynamics may remain globally unitary. What changes is the adequacy of purely local state assignments once constraints are shared. Local realism fails not because reality involves nonlocal causal influence, but because locality alone is insufficient to characterize admissibility when the relevant structures span multiple subsystems.

Nonlocal correlation thus marks the limits of description based on isolated systems. It signals that the appropriate unit of persistence is the composite constraint structure rather than its individual components. Where admissibility is globally defined, explanation must also be global. Attempts to recover locality by insisting on independent subsystem descriptions misidentify the source of correlation and reintroduce paradoxes that the structural account avoids.

As with coherence and entanglement, nonlocal correlation is contingent rather than universal. It persists only so long as shared constraints remain intact. Environmental interaction, incompatible coupling, or accumulated constraint conflict can fragment the joint structure, yielding effective separability and classical locality at the level of accessible description. This transition does not require any special dynamical postulate; it reflects the loss of shared admissibility rather than the imposition of a new law.

In QCT, nonlocal correlation is therefore neither paradox nor anomaly. It is the expected signature of admissibility structures that are relational rather than separable. Where admissibility

is shared, correlation follows. Where admissibility separates, locality reasserts itself.

5 Collapse and the Quantum-Classical Transition

This Section explains collapse and classical emergence as effective transitions produced by constraint aggregation rather than as primitive events, observer effects, or fundamental discontinuities. As interactions accumulate, joint admissibility narrows until multiple configurations can no longer persist together. Classical behavior emerges as a high-constraint limit of the same underlying dynamics that permit quantum coherence.

5.1 Constraint Aggregation and Collapse Thresholds

Importantly, collapse is not treated as a primitive event, a stochastic interruption, or a consequence of observation in QCT. It is understood as an effective transition arising from constraint aggregation: the accumulation and interaction of limiting conditions that progressively reduce the joint admissibility of multiple configurations until their simultaneous persistence becomes structurally untenable within the description in use.

At low effective constraint density, multiple admissible configurations can coexist without mutual exclusion. In this regime, distinctions remain underdetermined, environmental records are sparse or non-redundant, and no single realization is structurally forced to dominate. This is the domain in which quantum coherence and superposition persist. Crucially, the absence of collapse here does not imply that reality is “waiting on observation,” but that insufficient structural pressure has accumulated to enforce exclusion among alternatives.

As interactions accumulate, constraints do not merely add linearly; they couple, reinforce, and overlap. Environmental entanglement, repeated interaction, boundary formation, and coarse-grained stabilization introduce correlated restrictions on admissible states. When these restrictions remain mutually compatible—selecting aligned structures or preserving recoverability—joint admissibility can remain open and superposition may persist, even across increasing scale or duration. When compatibility fails, admissibility narrows, not by decision, but by structural consequence.

Collapse occurs when the aggregated constraint structure becomes incompatible with the simultaneous persistence of multiple configurations. At this threshold, exclusion is no longer partial, local, or practically reversible. One realization stabilizes not because it is selected, observed, or favored, but because alternative configurations can no longer coexist without violating the combined constraints imposed by interaction and coupling. Collapse, in this sense, is the exhaustion of joint admissibility, rather than the invocation of a new rule or a discontinuity in the underlying dynamics.

This framing also clarifies why superposition can, under certain conditions, be extended beyond minimal scales. Scale alone does not force collapse. What matters is whether increasing interaction introduces incompatible or redundant constraints. Systems engineered or situated such that interaction pathways remain sparse, aligned, or correctable can support extended coherence, while systems exposed to heterogeneous, proliferating, or record-forming interactions transition rapidly into classical stability. Collapse thresholds therefore depend on constraint geometry and coupling structure, not on size, distance, or duration in isolation.

Constraint aggregation further explains why collapse is typically abrupt in appearance but gradual in origin. Long before a classical outcome becomes manifest, admissibility may already be

narrowing. Redundancy accumulates, recoverability erodes, and alternative configurations become increasingly incompatible under the total constraint load. The visible transition marks the point at which further coexistence becomes structurally unavailable at the level of description being maintained, not the moment at which change began.

Within this framework, collapse is neither an anomaly nor a failure of quantum theory. It is the natural consequence of persistence under pressure. Where constraints remain relaxed and compatible, coherence endures. Where they accumulate beyond tolerance, resolution follows. This account requires no appeal to observer privilege, hidden selection mechanisms, or fundamental non-unitary dynamics, remaining compatible with globally unitary evolution and effective irreversibility at the descriptive level.

Collapse, then, is not the negation of quantum behavior but its boundary condition. It marks the limit at which permissive admissibility gives way to enforced persistence, governed by the structure and interaction of constraints rather than by chance, intention, or external intervention.

Scope Note: This section’s purpose is explanatory: it states how QCT interprets the quantum–classical transition in constraint-first terms. Questions of what would count as a QCT-specific empirical commitment—novel predictions, distinguishing criteria, or potential falsifiers—are deferred to the QCT Explanatory Closure section, where the framework’s evaluative burden is addressed directly.

5.2 Decoherence as Enforced Admissibility Reduction

In QCT, decoherence is not treated as a mysterious loss of quantum behavior, nor as an observer-driven update, nor as a fundamental dynamical rupture. It is understood as a structural process by which admissible configurations are progressively restricted through interaction, coupling, and record formation. Decoherence enforces admissibility reduction: it narrows the space of configurations that can remain mutually compatible under sustained constraint.

In low-constraint regimes, quantum systems may support multiple admissible configurations simultaneously. Interference persists because no interaction imposes sufficient restriction to privilege one configuration over others. As systems interact with their environments, however, admissibility is no longer evaluated in isolation. Coupling introduces external degrees of freedom that participate in the system’s state space, expanding the effective description while simultaneously restricting which configurations can persist without contradiction.

From an operational standpoint, this process corresponds to increasing entanglement between a system and its environment. Degrees of freedom that were previously internal become correlated with uncontrolled external states. As a result, interference between alternative configurations becomes inaccessible to subsystems of the whole. In standard quantum formalism, this appears as the suppression of off-diagonal terms in the reduced density matrix relative to a particular basis. In QCT terms, it is the enforcement of incompatibility between alternatives under aggregated constraint.

Crucially, decoherence does not eliminate states. It does not destroy possibilities, collapse wavefunctions by fiat, or introduce non-unitary dynamics. Instead, it restructures admissibility. Configurations that were once jointly admissible can no longer coexist because their persistence would require mutually incompatible constraint satisfaction across interacting degrees of freedom. Decoherence therefore marks a transition in what can remain coherently realized, not a physical annihilation of unrealized alternatives.

This framing clarifies why decoherence is both ubiquitous and context-dependent. The rate and strength of admissibility reduction increase with coupling density, environmental mode availability, temperature, scattering probability, and record redundancy. Conversely, coherence can persist when constraint pathways are minimized or aligned—through isolation, cooling, dynamical decoupling, or carefully structured environments. These are not exceptions to decoherence, but confirmations of its constraint-based character.

Decoherence also explains the apparent basis-dependence of classical outcomes. The configurations that remain admissible under continued interaction are those that are stable with respect to the dominant coupling structure. These stable configurations—often termed pointer states in standard treatments—are not selected arbitrarily. They are the configurations least disrupted by constraint aggregation, and therefore the only ones capable of persistent realization. In QCT terms, admissibility reduction is structured, not uniform.

Importantly, enforced admissibility reduction should not be conflated with determinism. Decoherence restricts coexistence, not necessarily outcome statistics. Probabilistic behavior remains compatible with this framework, provided probabilities are understood as descriptions over admissible configurations prior to enforced exclusion. What decoherence removes is not uncertainty, but the simultaneous persistence of mutually incompatible alternatives.

This account also distinguishes decoherence from collapse. Decoherence explains why interference disappears and why classical behavior emerges as stable, predictable structure. Whether one interprets the remaining admissible configuration as a single realized outcome, a branch among many, or an effective classical trajectory depends on interpretive commitments external to QCT. Quantum Coherence Theory brackets those commitments. Its claim is narrower: enforced admissibility reduction is sufficient to explain the emergence of classical stability from quantum permissiveness.

In this sense, decoherence is not a failure of quantum mechanics, but its structural completion under constraint. It is the mechanism by which permissive descriptions become restrictive enough to support persistent identity, classical predictability, and stable records. Where admissibility remains broad, coherence persists. Where admissibility is enforced, classical behavior follows.

Decoherence, then, is not the negation of quantum theory but the condition under which quantum descriptions cease to admit multiple coexisting realizations. It marks the point at which constraint structure overwhelms permissiveness, transforming quantum possibility into classical persistence without invoking observers, hidden variables, or additional physical postulates.

5.3 Classical Determinism as High-Constraint Behavior

Classical determinism is not treated as a fundamental mode of physical law, nor as a separate ontological regime, in this framework. It is understood as a limiting behavioral condition that emerges when constraint density becomes sufficiently high to suppress admissibility variation across all relevant degrees of freedom. Classical behavior is therefore not imposed by new rules, but produced by the saturation of constraints that enforce persistent identity and predictable evolution.

At high constraint density, admissible configurations are so narrowly restricted that effective freedom collapses to a single viable trajectory. Alternative possibilities are not merely unlikely; they are structurally inaccessible. Under these conditions, system evolution appears deterministic, not because the universe has become rigid or fully specified, but because deviation would violate constraints required for continued persistence. Determinism, in this sense, is the visible trace of constraint exhaustion.

This framing resolves a long-standing tension between quantum indeterminacy and classical predictability. Classical determinism does not override quantum permissiveness; it emerges where permissiveness has been progressively constrained by interaction, coupling, redundancy, and environmental embedding. The same underlying dynamics persist across regimes. What changes is not the law governing evolution, but the space of admissible evolution available to the system.

Importantly, high constraint density does not imply absolute rigidity. Classical systems may still exhibit stochastic behavior, noise, and sensitivity to initial conditions. What distinguishes them is not the absence of variation, but the boundedness of variation. Deviations remain confined within narrow stability regions that preserve identity and function. Deterministic descriptions succeed not because they capture all microdynamics, but because the constraint structure renders those details irrelevant at the descriptive scale.

This perspective clarifies why classical determinism is scale-dependent and context-sensitive. Systems appear more deterministic as constraint aggregation increases through mass, interaction rate, coupling strength, or environmental redundancy. Conversely, when constraints are relaxed—through isolation, cooling, shielding, or engineered decoupling—classical predictability degrades and quantum behavior reasserts itself. Determinism is therefore not a permanent property of systems, but a contingent feature of their constraint environment.

Classical laws of motion, conservation, and causality can be understood within this framework as effective summaries of behavior under sustained high constraint density. They describe what happens when admissibility has been narrowed so thoroughly that alternative histories are no longer structurally viable. Causality, in this regime, reflects enforced sequence rather than fundamental necessity. Events occur in a determinate order because constraint compatibility requires them to do so.

This also explains why classical descriptions fail under extreme conditions. When constraint structures are disrupted—through high energy density, rapid interaction, or strong coupling—determinism erodes. Classical trajectories fragment, probabilistic descriptions regain relevance, and admissibility expands. These failures do not indicate breakdowns of physical law, but transitions between constraint regimes.

Within QCT, classical determinism is therefore treated as an emergent regularity, not a foundational principle. It is the behavioral signature of systems that have traded configurational freedom for persistence. Determinism appears where sustainability demands it, and disappears where it cannot be maintained.

In summary, classical determinism is the consequence of high constraint density acting on potential. It is not a separate ontology, but a boundary condition of quantum permissiveness. Where constraints are strong and compatible, outcomes are enforced and trajectories stabilize. Where constraints relax, determinism dissolves. Classical behavior, in this view, is not the opposite of quantum behavior, but its constrained limit.

5.4 Continuity Across Regimes

The quantum–classical transition is not a categorical divide, but a continuous shift in constraint conditions. Quantum and classical behaviors are not governed by separate principles or ontologies; they occupy different regions within a single explanatory landscape defined by admissibility and persistence under constraint.

Across this landscape, no sharp boundary separates coherence from collapse or indeterminacy from determinism. Instead, systems move gradually along a spectrum as constraint density increases or decreases. What appears experimentally as a sudden transition reflects the exhaustion

of admissible configurations at the descriptive scale, not the introduction of a new physical rule. The underlying dynamics remain continuous even when observable behavior changes abruptly.

This continuity explains why intermediate regimes are both common and difficult to classify. Mesoscopic systems, weakly measured states, and partially decohered configurations often exhibit mixed characteristics: residual interference alongside emergent stability, probabilistic behavior constrained by robust patterns, or classical trajectories punctuated by quantum sensitivity. These hybrid behaviors are not anomalies; they are natural expressions of systems operating near constraint thresholds.

Within QCT, regime classification is therefore contextual rather than absolute. A system may behave quantum mechanically with respect to some degrees of freedom while appearing classical with respect to others. Constraint density need not be uniform across a system or its environment. Different partitions, interaction pathways, or observational resolutions may reveal different effective regimes without contradiction. Continuity is preserved because all such descriptions are anchored in the same admissibility structure.

This perspective also clarifies why classical descriptions remain extraordinarily successful despite their known limitations. Classical models function as effective approximations precisely because high constraint density suppresses sensitivity to micro-level variation. When admissibility is narrow, many distinct microhistories converge on the same macroscopic behavior. Deterministic laws succeed not by capturing fundamental dynamics, but by exploiting constraint-induced redundancy.

Conversely, the re-emergence of quantum behavior under isolation or constraint relaxation does not represent a violation of classical principles, but a return to a broader admissibility space. As constraints loosen, distinctions once suppressed become relevant again. Interference, indeterminacy, and superposition reappear because they were never eliminated—only rendered inaccessible under prior conditions.

Continuity across regimes therefore supports a unified explanatory account. Quantum mechanics describes behavior in regions of low effective constraint density. Classical physics describes behavior in regions of high constraint density. Neither description is privileged absolutely. Each is valid within the domain where its assumptions about admissibility and persistence hold.

By framing the quantum–classical relationship in this way, QCT avoids both reductionism and fragmentation. It does not seek to derive one regime entirely from the other, nor to isolate them as fundamentally distinct. Instead, it explains their coexistence as expressions of the same constraint-governed structure operating under different conditions.

5.5 Experimental Evidence for Quantum-Classical Transition

Experimental results over the past several decades demonstrate that the quantum–classical transition is not governed by intrinsic limits on spatial scale, mass, material composition, or duration. Instead, the persistence or loss of observable quantum coherence correlates most strongly with how entangling couplings to uncontrolled degrees of freedom are structured, suppressed, or rendered effectively correctable. These findings motivate a reframing of quantum scalability away from absolute thresholds and toward the accumulation and organization of decohering interaction channels—that is, the number, strength, and independence of the pathways through which which-alternative information becomes redundantly recorded in the environment.

Matter-wave interferometry provides an early and continuing illustration of this pattern. Interference has been observed not only for elementary particles, but for increasingly massive and

structurally complex systems, including large molecules. In these experiments, the degradation of interference tracks environmental coupling—such as scattering, thermal excitation, internal vibrational and rotational modes, blackbody emission and absorption, and uncontrolled phase interactions—rather than mass or compositional complexity in isolation. When such couplings are sufficiently constrained or filtered, coherence persists well beyond naïve expectations based on scale alone.

Related behavior appears in superconducting circuits and mesoscopic quantum devices. Coherence times increase dramatically when systems are cooled, shielded, filtered, dynamically decoupled, or otherwise isolated from uncontrolled modes. Conversely, coherence degrades rapidly when interaction pathways multiply, diversify, or generate redundant environmental records. These observations indicate that the stability of coherent quantum behavior depends on the number, strength, and effective independence of decohering interactions, not on intrinsic limits tied to size, material composition, or architectural complexity.

Optomechanical systems and macroscopic resonators further reinforce this conclusion. Experimental efforts to place increasingly massive objects into quantum states show that coherence can be maintained provided that vibrational modes, thermal noise, excess phase noise, and unwanted environmental entanglement are sufficiently restricted. Loss of coherence in these regimes correlates with exposure to heterogeneous, information-leaking, and effectively irreversible interactions, rather than with any categorical transition from “quantum” to “classical” scale.

Across these domains, a consistent empirical regularity emerges: coherent quantum behavior remains observable when decohering couplings are sparse, structured, or (in limited regimes) correctable, and degrades as independent decoherence channels accumulate and reinforce one another through redundant record formation. No confirmed experiment to date requires the postulation of a universal size cutoff, a privileged observer role, or an intrinsic collapse trigger to account for the observed quantum–classical transition, even as experimental programs continue to place increasingly strong bounds on such possibilities.

These results do not imply that coherence can be extended arbitrarily. They demonstrate instead that scalability is conditional. Constraint management can delay the loss of observable coherence, but it cannot eliminate the structural pressures that arise as interactions proliferate and environmental records become redundant. Coherent behavior is therefore not preserved indefinitely, but maintained only so long as relevant couplings remain jointly suppressible or effectively correctable as systems scale.

The empirical record thus supports a constraint-first interpretation of the quantum–classical transition. The loss of observable coherence does not track scale, complexity, or observation as such, but the proliferation of decohering couplings and redundant environmental records. While these experiments do not resolve the ultimate ontological status of collapse, they sharply constrain any viable explanatory framework: whatever mechanism accounts for classicality must arise from the same structural pressures that govern interaction-channel accumulation and information leakage.

6 Observation Without Ontological Privilege

This section reframes observation as a physical interaction that imposes constraints and forms records, not as an ontologically privileged act. Stabilization and collapse-like behavior arise from interaction structure and record formation, independent of awareness. Classical persistence therefore reflects universal constraint dynamics, operating with or without observers.

6.1 Observation as Constraint Interaction

Observation is not treated as a primitive epistemic act or an ontologically privileged event. It is understood instead as a particular class of physical interaction—one that couples a system to structures capable of stabilizing, preserving, and propagating correlations over time. In this sense, observation is a subset of interaction defined by its constraint consequences, not by the presence of awareness, intent, or interpretation.

This framing distinguishes three concepts that are often conflated in discussions of quantum foundations. Measurement refers to the establishment of correlations between degrees of freedom through interaction or coupling. Observation refers to those measurement interactions that produce durable, record-bearing constraints capable of persisting beyond the immediate interaction. Collapse refers to the subsequent condition in which aggregated constraints render the simultaneous persistence of multiple alternatives physically unavailable. These distinctions are structural, not semantic, and they allow QCT to treat observation without invoking observer privilege.

An interaction qualifies as observational in the QCT sense only if it produces constraints that endure. A transient correlation that can be fully reversed, erased, or reabsorbed without consequence does not constitute observation. By contrast, an interaction that imprints correlations into degrees of freedom that are robust to local perturbation, redundantly accessible, or environmentally amplified imposes lasting constraints on admissibility. The defining feature is not that information is acquired, but that constraint structure is altered in a way that persists.

This distinction resolves common ambiguities surrounding unread detectors and unattended measurements. A detector that irreversibly records which-path information constrains admissibility regardless of whether its output is ever examined. Conversely, in carefully engineered reversible interactions—such as weak measurements or quantum eraser arrangements—observation does not occur in the QCT sense precisely because constraint pathways remain compatible and recoverable. Observation is therefore neither synonymous with interaction nor guaranteed by correlation alone.

Observation, however, is neither necessary nor sufficient for collapse. Collapse does not occur simply because a record exists, nor does it require an observational interaction at all. Collapse occurs only when the accumulated constraint structure becomes incompatible with the joint persistence of alternatives. An observational interaction may contribute to this accumulation, but collapse depends on whether constraints conflict, reinforce, and irreversibly narrow admissibility—not on whether a system has been “observed.”

This framing situates observation within the same physical arrow that governs irreversibility more generally. Observational interactions tend to align with the thermodynamic arrow because they disperse correlations into many degrees of freedom, increasing redundancy and reducing recoverability. The apparent directionality of observation is therefore not imposed by consciousness or agency, but by the asymmetry introduced when constraints propagate beyond local control.

Importantly, QCT does not treat observation as a causal trigger. Observation does not select outcomes, resolve superposition, or terminate quantum evolution. It modifies the constraint landscape within which admissibility is evaluated. Whether that modification contributes to collapse depends on how it composes with other constraints already acting on the system. Observation is thus one contributor among many to constraint aggregation, not a privileged switch.

By reframing observation as constraint interaction, QCT preserves continuity between laboratory measurement, natural processes, and unobserved dynamics. The same principles govern whether a quantum system is monitored by an experimental apparatus, embedded in an environment, or evolving in isolation. No appeal to epistemic acts, experiential states, or subjective access is required to explain why observational interactions matter.

This treatment prepares the ground for the following section. If observation is nothing more—and nothing less—than a physical interaction that produces persistent constraints, then no additional explanatory role remains for consciousness in the account of collapse. Collapse, if it occurs, must be explainable entirely in terms of constraint structure and compatibility.

6.2 Collapse Without Consciousness

Accounts that assign a privileged causal role to consciousness, observation, or subjective awareness in collapse fail to meet the requirements of an observer-independent explanation. The transition from quantum permissiveness to classical persistence must be treated as a physical process governed by constraint structure, not as an event triggered by experience, cognition, or informational uptake by an agent.

This is not a matter of philosophical taste but of explanatory necessity. Physical systems stabilize, decohere, and produce persistent outcomes in environments entirely devoid of observers. Astrophysical processes, geological formation, radioactive decay, and cosmological structure formation all exhibit durable classical structure without any plausible appeal to conscious monitoring. Any account that ties definiteness to observation must therefore explain why unobserved systems obey the same stability constraints as observed ones. No such explanation has been successfully provided.

Observation, properly understood, is a class of physical interaction, not a distinct ontological category. Detectors, sensors, and measurement devices impose constraints through coupling, record formation, and environmental embedding. Whether a resulting record is later read, interpreted, or consciously registered plays no role in the constraint structure imposed. A detector that produces a stable, irreversibly encoded record constrains future admissibility regardless of human involvement.

This distinction is essential. Measurement as interaction has physical consequence; measurement as experience does not. Treating awareness as explanatorily active introduces boundary ambiguities—about which systems qualify as observers, when consciousness arises, or how subjective states could exert causal influence on physical admissibility—that do no explanatory work. Constraint-based accounts avoid these ambiguities by grounding persistence entirely in interaction structure.

Collapse, in this structural sense, does not denote a fundamental non-unitary discontinuity unless such dynamics are independently established by experiment. It denotes an effective physical transition in which interference between alternatives becomes inaccessible because constraint-driven entanglement, record formation, and irreversibility render joint persistence incompatible. Decoherence describes the suppression of interference; collapse names the point at which incompatible alternatives can no longer coexist within the active constraint network.

This framing preserves compatibility with unitary evolution at the global level while accounting for effective irreversibility at the descriptive level. Alternatives need not be ontologically annihilated, nor must a privileged moment of selection be postulated. What matters is that incompatibility between constraint-stabilized records makes continued coexistence physically unavailable. Persistence becomes exclusive not because something chooses, but because compatibility fails.

Crucially, this narrowing of admissibility is not epistemic. It does not depend on knowledge, belief, or informational access. An unread detector that irreversibly records which-path information constrains future interaction just as effectively as one whose data is examined. Conversely, in carefully engineered reversible interactions—such as quantum eraser scenarios—collapse does not occur precisely because constraints have not yet become incompatible or irrecoverable. Irreversibility, not awareness, marks the boundary.

The framework does not prescribe a specific microdynamical rule for collapse here. The claim is architectural rather than algorithmic. Collapse follows when the constraint structure governing a system no longer admits the simultaneous persistence of alternatives. How that structure is represented mathematically—through reduced density matrices, decoherence functionals, or other formalisms—is secondary to the requirement that incompatibility be physically enforced.

By removing consciousness from the explanatory role, the account gains both generality and parsimony. The same principles apply uniformly across observed and unobserved systems, laboratory experiments and natural processes, microscopic interactions and macroscopic stability. Collapse is not an act of observation, nor a mystery awaiting mental intervention. It is the structural consequence of persistence under constraint.

In this sense, collapse without consciousness is not a radical proposal. It is the minimal claim required for a coherent, observer-independent account of physical stability. The universe does not wait to be noticed in order to persist.

6.3 Observerless Quantum Systems

Any account of stabilization, decoherence, or collapse must apply universally, including in the absence of observers. Physical systems evolve, interact, and stabilize whether or not they are monitored, recorded, or experienced. Chemical reactions, stellar fusion, geological processes, and cosmological structure formation all proceed without observation, yet display the same transition from quantum permissiveness to stable classical outcomes. An explanatory framework that assigns a necessary causal role to observation must therefore account for why identical stabilizing behavior occurs where observation is entirely absent.

Within QCT, observers are not privileged entities but physical subsystems embedded within the same constraint landscape as all others. An observer differs only in its capacity to form stable internal correlations with existing environmental records. That capacity does not introduce new dynamics. It extends an already-established network of constraints by adding further physical correlations. Observation, in this sense, tracks stabilization; it does not create it.

The mechanisms that govern admissibility narrowing operate independently of awareness. Constraints arise from interaction structure: coupling Hamiltonians, boundary conditions, environmental entanglement, and coarse-graining. Admissibility refers to the subset of state histories or configurations that remain dynamically stable and mutually compatible under those constraints at a given descriptive resolution. As interactions accumulate, admissible alternatives may be reduced, but this reduction depends solely on physical coupling and compatibility, not on whether any subsystem registers the result.

This principle applies equally to unattended measurement devices. A detector that interacts with a system and produces a stable, redundantly encoded correlation imposes the same constraints whether or not its output is later examined. What matters is not the act of reading, but whether the interaction produces durable records—correlations that are robust against local perturbation and effectively irreversible due to dispersion into many environmental degrees of freedom. Observation adds no special causal step beyond this physical process.

Delayed-choice and quantum eraser experiments do not challenge this account. Later operations alter which correlations are accessible and which conditional statistics are selected, but they do not retroactively modify the constraint structure established by earlier interactions.

Where records remain non-redundant or reversible, admissibility may remain broad. Where records become amplified and dispersed, admissibility narrows regardless of when or whether results are observed. Changes in inference do not imply changes in past physical constraint pathways.

In QCT, collapse is therefore not defined by observation but by constraint structure. Collapse refers to an effective transition in which interference between alternatives becomes practically inaccessible due to entanglement with uncontrolled degrees of freedom and the formation of stable records. This usage does not require fundamental non-unitary dynamics, nor does it invoke consciousness. It identifies the point at which admissible alternatives can no longer coexist under the accumulated constraints imposed by interaction.

Any theory that assigns collapse, stabilization, or classicality to conscious registration fails a basic universality requirement: it cannot account for observerless persistence. QCT instead adopts observer-independence as a structural criterion. The same constraint-driven processes that yield classical stability in laboratory measurements must also govern the unattended evolution of the physical world.

This section, therefore establishes a necessary condition for any viable account of the quantum–classical transition: the mechanisms responsible for stabilization must operate without reference to observation. Observation may extend constraint networks by adding new correlations, but it cannot serve as their source. The next section makes this conclusion explicit by addressing collapse directly and demonstrating why consciousness plays no constitutive role in its occurrence.

6.4 Removal of Observer Exceptionalism

Quantum Coherence Theory rejects observer exceptionalism as an explanatory principle. The mechanisms that govern the stabilization of physical systems, the loss of interference, and the emergence of classical behavior must operate universally, including in regimes where no observers are present and no measurements are read or recorded by conscious agents. Any account that assigns a privileged causal role to observation or awareness therefore inherits an explanatory burden: it must explain why the same stabilizing behavior occurs reliably in the absence of observers across cosmological, geological, chemical, and microscopic domains.

Within QCT, observation is treated as a particular class of physical interaction, not as a distinct ontological category. An observer is simply a subsystem capable of forming stable internal correlations with existing records. This capacity introduces no new physical law. It extends the network of constraints already in place by coupling an additional system to structures that have already stabilized through interaction. Observation, in this sense, participates in constraint propagation; it does not initiate it.

This framing dissolves the distinction between “observed” and “unobserved” systems at the level of physical evolution. A detector left unread imposes the same constraints on a system as

one whose output is later accessed. What matters is whether an interaction produces durable, redundantly encoded correlations—records that persist under subsequent perturbation—not whether those records are later consulted. Subsequent access changes which correlations are physically retrieved and which conditional statistics are selected; it does not retroactively alter the interaction history that generated the records.

The removal of observer exceptionalism also clarifies the status of delayed-choice and quantum eraser experiments. In such cases, later interactions modify which correlations remain accessible or which interference patterns can be reconstructed, but they do not change earlier constraint-imposing interactions. The apparent tension arises from conditioning on different subsets of correlations, not from retrocausal influence or observer-induced revision of past states. Constraint pathways include all relevant interactions, including those occurring later in time; no special role is played by awareness or choice.

Collapse-like behavior, within this framework, is not triggered by observation. It arises when aggregated constraints render interference between alternatives operationally inaccessible and prevent multiple configurations from remaining mutually coherent at the relevant coarse-graining. This typically occurs when environmental interactions distinguish alternatives in non-commuting or competing stable bases, dispersing correlations into many degrees of freedom and producing robust pointer-like structures. Formally, this corresponds to the suppression of interference terms in the reduced density matrix as correlations spread irreversibly into the environment. No appeal to conscious registration is required to account for this transition.

By treating observation as constraint interaction rather than ontological intervention, QCT maintains explanatory continuity across scales and contexts. The same principles that govern unattended nuclear reactions in stars, crystallization in cooling materials, or molecular stabilization in chemistry also govern laboratory measurements and detector clicks. Any theory that requires observation as a necessary physical trigger fails this universality requirement.

The removal of observer exceptionalism is therefore not a metaphysical assertion but a methodological constraint. Explanations of persistence, decoherence, and classical stability must remain valid whether or not observers are present. QCT satisfies this requirement by locating the source of stabilization in the structure and compatibility of constraints themselves, rather than in acts of observation or awareness.

7 Informational Closure and Curl Equilibrium

This section establishes informational closure as explanatory sufficiency rather than metaphysical finality. It shows how shared constraints determine admissibility across part–whole descriptions, how unresolved incompatibilities redistribute rather than vanish under closure, and how stability can emerge without rigidity. The section frames closure as a disciplined boundary on explanation, not a new dynamics or ontology.

7.1 Informational Closure as Explanatory Exhaustion

Informational closure, as used in Quantum Coherence Theory, refers to explanatory exhaustion rather than metaphysical finality. A description is informationally closed when no additional ontic primitives are required to account for transitions between behavioral regimes—specifically, the persistence of coherence under low constraint density and the emergence of classical stability under high constraint density. Closure marks the point at which explanatory power no longer increases by introducing new entities, forces, or postulates, and instead rests entirely on the structure, compatibility, and interaction of constraints governing admissible realizations.

This notion of closure is explicitly methodological. It does not assert that reality itself is complete, finite, or fully captured by the present framework. Nor does it deny the possible existence of hidden variables, deeper dynamics, or future theoretical extensions. Informational closure claims only that, within the explanatory scope defined here, invoking such additions is unnecessary to account for the observed quantum–classical transition. If explanation can be carried to completion using constraint structure and admissibility alone, then explanatory closure has been achieved, regardless of what deeper descriptions may later emerge.

Importantly, informational closure does not imply that information is a conserved substance, a causal agent, or a fundamental physical field. In QCT, information is descriptive of constrained realization: it characterizes how possibilities are restricted, correlated, and stabilized under interaction. Closure therefore does not rest on information as an ontological primitive, but on the sufficiency of constraint-based descriptions to account for persistence, decoherence, and stabilization without remainder.

For shared constraints to function as shared, constraint-relevant information must be able to circulate across the system within a timescale compatible with identity persistence. Informational closure therefore presupposes neither instantaneous global coordination nor static isolation, but finite, tolerance-bounded propagation of constraint correlations. If propagation were effectively unbounded, local differentiation could not stabilize before global coupling dissolved structure. If propagation were effectively negligible, parts could not maintain joint admissibility before decoherence fragmented identity. Closure thus depends not on the speed of propagation as a parameter, but on the existence of finite circulation that enables coordinated persistence without rigid lockstep.

The criterion for informational closure is operational and falsifiable. If explaining the transition from coherent multiplicity to classical definiteness requires the introduction of an independent dynamical variable not expressible in terms of constraint structure, admissibility, and their circulation—such as a primitive collapse trigger, observer-dependent intervention, or irreducible stochastic law—then closure fails. Conversely, if observed behavior can be fully accounted for by how interac-

tions impose compatibility conditions, restrict joint admissibility, propagate constraint correlations, and render interference operationally inaccessible, then closure holds at the explanatory level.

Closure should therefore not be confused with reductionism or finality. It does not claim that all phenomena reduce to a single mechanism, nor that future theory is foreclosed. Instead, it delineates a boundary of sufficiency: beyond this boundary, additional assumptions do not increase explanatory resolution for the phenomena under consideration. Within that boundary, constraint structure and its finite circulation provide a complete account of why certain configurations persist while others fail to remain jointly admissible.

By framing closure in this way, QCT avoids both instrumental minimalism and metaphysical overreach. It neither dismisses unanswered questions nor inflates its scope to address them prematurely. Informational closure marks the point at which explanation has done all the work it can do using the resources already introduced. What lies beyond that point may be real, significant, or even necessary for other domains—but it is not required to explain the quantum–classical transition as a problem of constrained persistence.

In this sense, informational closure functions as a disciplinary safeguard rather than a proclamation. It prevents the unnecessary multiplication of primitives while remaining open to future refinement. The framework is considered closed not because inquiry must stop, but because explanation, at the level addressed here, has reached sufficiency without remainder.

7.2 Part–Whole Equivalence Under Shared Constraints

Part–whole equivalence, within Quantum Coherence Theory, is not a claim about ontological identity, reducibility, or the elimination of subsystem distinctions. It is a conditional statement about explanatory sufficiency. When a system and its subsystems are governed by the same constraint structure, descriptions at the part level and at the whole level are equivalent with respect to explaining persistence, exclusion, and stability under those constraints.

This equivalence applies only within a defined domain: systems that share a common constraint boundary and whose admissible realizations are restricted by the same interaction conditions. Under such circumstances, shifting between part-level and whole-level descriptions does not introduce new explanatory primitives. The same constraints delimit admissibility regardless of how the system is partitioned, and no additional assumptions are required to account for which configurations persist and which are excluded.

Equivalence here is therefore methodological rather than ontological. It concerns what must be assumed to explain behavior, not which properties belong to which components. Part–whole equivalence does not imply that subsystems possess all the properties of the whole, nor that collective phenomena reduce to local attributes. Emergent properties may arise at the composite level that are not attributable to any single part. The claim is instead that, where constraints are fully shared, such properties do not require additional explanatory resources beyond those already present in the constraint structure.

This principle becomes most visible in systems exhibiting strong correlation or entanglement. In such cases, admissibility is determined at the joint level rather than independently for each component. Local descriptions remain operationally useful, but they do not fully specify which global configurations are stable or excluded. The need for a joint description does not arise from metaphysical holism, but from the fact that the relevant constraints span the partition. Admissibility, under these conditions, is a property of the constrained composite, not of its parts in isolation.

Crucially, shared constraints also imply shared incompatibilities. When a configuration is locally admissible in one part but incompatible with the joint constraint structure, the resulting tension

cannot be resolved by appeal to any single subsystem. Instead, incompatibility must be evaluated at the level where constraints are shared. This establishes the conditions under which unresolved contradiction cannot be eliminated locally and must instead persist at the composite level.

Part-whole equivalence therefore sets the stage for understanding how systems maintain stability in the presence of incompatibility. When admissibility is evaluated jointly and constraints are shared, local resolution of contradiction may be impossible without violating global persistence. Under such conditions, stability is preserved not by eliminating contradiction, but by allowing it to circulate within the constraint structure without collapsing identity.

Importantly, shared constraints do not guarantee equivalence across all descriptions. When constraints differ across partitions—when subsystems are subject to distinct boundary conditions, interaction regimes, or coupling strengths—part-whole equivalence breaks down. Subsystem behavior may then diverge from whole-system behavior in ways that require separate explanatory treatment. Equivalence is therefore conditional, not universal.

By framing part-whole relations in this way, QCT avoids both reductionism and monism. It neither insists that wholes are nothing more than their parts, nor that parts dissolve into an undifferentiated totality. Instead, it identifies the circumstances under which explanatory distinctions between part and whole become unnecessary because constraint satisfaction is evaluated at the same structural level in both descriptions.

Part-whole equivalence under shared constraints thus functions as a guide for explanatory economy and as a boundary condition for stability. Where the same constraint structure governs admissibility, multiplying descriptive layers does not deepen explanation. Where constraints diverge, distinctions reassert themselves. This principle does not erase structure; it clarifies when structure must be described jointly and prepares the ground for understanding how unresolved incompatibilities are sustained without collapse.

7.3 Curl Equilibrium

Curl equilibrium, within Quantum Coherence Theory, is a diagnostic label for a specific structural pattern that emerges once informational closure has been reached. It denotes the persistence of residual incompatibilities that cannot be resolved through further local constraint satisfaction and therefore remain distributed across correlated degrees of freedom. The concept is descriptive rather than dynamical: it introduces no new forces, operators, or laws of evolution, but characterizes how unresolved constraint demands are organized after all reducible inconsistencies have been exhausted.

Constraint closure, as used here, denotes the condition in which all propagatable compatibility conditions have been enforced. At closure, no further local reconfiguration can reduce incompatibility without generating it elsewhere. What remains is not a failure of explanation, but a remainder supported by the global structure of constraints themselves. Curl equilibrium names this condition of globally consistent closure accompanied by irreducible, non-localizable residual incompatibility.

The term curl is used analogically, not as a vector calculus operator. It indicates that residual incompatibility is supported on cycles or loops within the constraint structure rather than at any single site. In such cases, incompatibility cannot be eliminated outright; it can only be redistributed across the cycle. Any local adjustment that resolves it at one point necessarily reintroduces it elsewhere. The equilibrium consists in the containment and circulation of this residual within the closed structure of admissibility.

Metaphorically, curl equilibrium may be understood as unresolved informational incompatibility that can no longer propagate outward into resolution. Once constraint closure has been achieved,

no additional degrees of freedom remain available to absorb or discharge residual incompatibility. What persists is therefore not an outward flow of unresolved information, but its confinement within the internal topology of the constraint structure itself. In this sense, information may be said to “curl inward,” not spatially or causally, but structurally—remaining trapped on cycles of admissibility where it can neither be eliminated nor localized.

A simple illustrative case is a triangular constraint structure in which three variables are pairwise constrained in mutually incompatible ways. Any two constraints can be satisfied simultaneously, but not all three. After closure, one incompatibility must remain. Importantly, which constraint appears violated depends on representation: through reparameterization or redistribution, the residual incompatibility can be shifted from one edge of the triangle to another, but never eliminated entirely. The incompatibility is cycle-supported rather than point-localized. This pattern is what curl equilibrium denotes.

Residual incompatibility in this sense should not be interpreted as logical inconsistency or paradox. It refers specifically to non-co-satisfiable admissibility demands: sets of constraints that cannot all be simultaneously realized within a single configuration, even though each is individually valid. Curl equilibrium therefore describes a bookkeeping property of admissibility under closure, not a breakdown of consistency, causality, or physical law.

This pattern has analogues in familiar physical and mathematical contexts. In constrained systems containing cycles—such as frustrated networks or contextual correlation structures—global consistency can coexist with irreducible local tradeoffs. Similarly, in correlated quantum systems, admissibility structures may exist in which no single global assignment satisfies all local contexts, even though each context is internally consistent. Curl equilibrium is intended to remain compatible with such phenomena without importing their formal machinery or committing to a specific interpretation.

Curl equilibrium arises only under identifiable conditions: (i) a constraint structure containing cycles or equivalent non-tree-like dependencies, (ii) mutually incompatible constraints on at least one such cycle, and (iii) closure of all reducible incompatibilities. When these conditions are met, residual incompatibility cannot permanently localize; it remains supported on the cycle space of the constraint structure.

The concept carries limited but definite diagnostic content. If, after closure, residual incompatibilities were observed to vanish entirely in systems whose constraint geometry contains irreducible cycles, curl equilibrium would not apply. Likewise, if residual strain consistently localized to a single site without redistribution under reparameterization, the notion would be inappropriate. Curl equilibrium therefore does not claim universality; it applies only where constraint geometry supports it.

Finally, curl equilibrium should not be mistaken for a mechanism of stabilization, a principle of optimization, or a source of dynamics. It does not explain how systems evolve. Instead, it clarifies how unresolved incompatibility is organized once evolution and interaction have enforced all possible local consistency. In this sense, curl equilibrium functions as a descriptive invariant of closed admissibility structures rather than an additional explanatory engine.

7.4 Closure Without Rigidity

Informational closure does not imply causal isolation, ontic completeness, or predictive finality. It does not assert that a system contains all information required to determine its future, nor that external influences are absent. Closure, as used here, is a property of an explanatory account at

a given resolution: the transition from quantum coherence to classical stability can be explained without introducing additional ontic primitives beyond constraint structure, admissibility, and standard physical dynamics.

This distinction is essential. Informational closure constrains explanatory vocabulary rather than physical behavior. It specifies that no supplementary collapse triggers, observer-dependent interventions, teleological principles, or ad hoc classicality postulates are required to account for persistence, decoherence, and stabilization. It does not deny unitary evolution, Hamiltonian dynamics, or established physical laws. Instead, it asserts that such dynamics, when combined with interaction-induced constraints and admissibility reduction, are sufficient for explanation at the level addressed by QCT.

Closure therefore does not entail rigidity or strict determinism. It does not fix unique trajectories or eliminate variability. Deterministic behavior, where it appears, is an effective property of high constraint density—regimes in which environmental coupling, coarse-graining, and rapid decoherence narrow admissibility so severely that alternative configurations cease to persist. This determinism concerns predictability and stability, not ontic inevitability. Where constraint density is low or weakly coupled, admissibility remains broad, and behavior may remain indeterminate, probabilistic, or interference-capable.

Probabilistic descriptions remain fully legitimate within a closed explanatory framework. Two roles of probability are preserved. First, probability may reflect incomplete constraint specification, coarse-graining, or limited access to environmental correlations—an operational or epistemic role. Second, probability arises formally within quantum theory itself, as in Born-weighted statistics. Quantum Coherence Theory does not take a position on the metaphysical status of quantum probability. Its claim is narrower: probabilistic models remain valid effective descriptions under closure, provided they respect the same constraint and admissibility structure.

Informational closure also excludes teleology and optimization as explanatory primitives. Stability does not arise because systems minimize, maximize, or pursue preferred outcomes. It arises because incompatible configurations fail to remain jointly admissible under interaction. Persistence is therefore explained through exclusion rather than selection: outcomes endure because alternatives are structurally disallowed, not because they are chosen or favored.

Nor does closure collapse all descriptions into a single scale or perspective. Multiple effective models—quantum, classical, probabilistic—may coexist so long as they agree on shared invariants, respect the same constraint structure, and yield consistent predictions in overlapping regimes.

Descriptive plurality is constrained but not eliminated. Closure restricts explanatory inflation, not legitimate modeling diversity. Finally, informational closure remains conditional and revisable. If future empirical findings require the introduction of a genuinely new explanatory primitive—such as an irreducible, context-dependent collapse dynamics not expressible in terms of constraint aggregation or admissibility reduction—then closure, as defined here, would fail for those phenomena. Until such necessity is demonstrated, closure functions as a discipline rather than a doctrine: it prevents unnecessary multiplication of primitives while remaining open to refinement.

Closure without rigidity thus preserves explanatory sufficiency alongside physical openness. It explains persistence without overdetermining behavior, accommodates variability without mystification, and maintains continuity with established physics while setting clear boundaries on what additional assumptions are warranted.

8 Non-Sequential Informational Consistency

This section argues that quantum correlations and collapse-like updates need not be explained through time-ordered causal influence. Instead, they arise from global constraint consistency across an entire experimental context. Temporal order remains valid for interventions and signaling, but fundamental admissibility is evaluated non-sequentially, preserving no-signaling, locality of control, and compatibility with established physics.

8.1 Non-Sequential Informational Consistency

This section addresses how quantum correlations and collapse-like transitions can be explained without treating temporal sequence as a fundamental explanatory primitive. In Quantum Coherence Theory, temporal ordering remains indispensable for describing interventions, control, and experience, but it is not required as the primary explanatory basis for admissibility, persistence, or correlation.

QCT treats admissibility as prior to temporal narration. Admissibility denotes the set of realizations—understood minimally as complete event-record assignments for a given experimental context—that are compatible with the full constraint configuration of that context. This notion is distinct from probability. Admissibility specifies which configurations can persist without contradiction; probability weights are supplied by standard quantum formalism and are not derived within QCT.

Non-sequential consistency does not imply that physical events occur outside time, nor that time is illusory. It asserts only that the consistency conditions governing joint outcomes need not be expressed as a step-by-step causal chain ordered in time. In particular, correlations observed in entangled systems need not be explained by signals propagating between subsystems, nor by a privileged temporal slicing of the experimental arrangement. Instead, joint outcomes satisfy a shared constraint structure defined by preparation conditions, interaction topology, and record-forming couplings.

Global constraint satisfaction must be distinguished sharply from signaling or control. While admissibility is evaluated relative to the complete constraint configuration of an experiment, local interventions remain free boundary conditions within that configuration. Parameter independence and no-signaling are preserved: no agent can exploit global consistency to transmit information or influence outcomes at a distance. The global aspect constrains joint realizations, not transmittable causal influence.

Collapse-like transitions, in this framework, correspond to admissibility reduction under constraint aggregation. As interactions with an environment produce stable, redundant records, the set of admissible realizations compatible with those records contracts. This reduction is effectively irreversible at macroscopic scales. QCT does not fix whether this reduction should be interpreted as a physical pruning of realized configurations or as an effective update of physically grounded descriptions; it characterizes only the structural conditions under which collapse-like behavior appears unavoidable.

Temporal asymmetry emerges when constraint density becomes sufficiently high that admissibility reduction acquires an effectively one-way character. Under coarse-graining, this manifests as

entropy increase, decoherence, and classical stability. The arrow of time is thus treated as a derived feature of high-constraint regimes rather than as a fundamental ordering principle imposed at the outset.

Finally, QCT’s non-sequential account is intended to remain compatible with established physical laws at the level of observable predictions. No-signaling constraints are preserved, relativistic causal structure remains intact for controllable interventions, and standard decoherence theory is inherited rather than replaced. The explanatory role of this section is to clarify why temporal sequence and observer reference are not required as foundational explanatory primitives, even though they remain indispensable for description, practice, and experience.

8.2 Global Constraint Satisfaction Versus Local Causation

Global constraint satisfaction is introduced to address a specific explanatory gap: how joint outcome structures arise in correlated systems without appeal to superluminal influence, observer privilege, or hidden sequential mechanisms. It is not proposed as a competing causal process, nor as a replacement for local dynamical laws. Local causation remains sufficient and indispensable for describing controllable interventions, signal propagation, and ordinary physical interactions. Global constraint satisfaction operates at a different explanatory level, accounting for the compatibility relations that govern which joint configurations are admissible once all relevant constraints are in place.

In this framework, a constraint refers to any physical or formal condition that restricts the set of admissible joint realizations of a system. These include, but are not limited to, the prepared quantum state, interaction structure, boundary conditions imposed by experimental settings, conservation laws, and the formation of stable records through environmental coupling. A realization denotes a complete assignment of outcome records for a given experimental run that is consistent with these conditions. Admissibility is therefore a support concept: it distinguishes configurations that can persist under the full constraint structure from those that cannot, without itself assigning probabilities or weights.

Quantum correlations present cases in which admissibility is determined at the composite level rather than by independent local assignments. In such systems, joint outcome distributions cannot be factorized into products of locally causal contributions without loss of consistency. This is not merely an inconvenience of representation, but a structural fact: the relevant constraints act across the partition, restricting which combinations of local outcomes can jointly occur. The failure here is not of operational locality, but of Bell-style outcome independence. Parameter independence and no-signaling conditions are preserved; what fails is the assumption that joint outcomes can be decomposed into independent local result assignments conditioned only on shared past variables.

Global constraint satisfaction therefore does not imply global signaling or controllable nonlocal influence. Constraints restrict the set of jointly admissible outcomes, but they do not provide agents with leverage to manipulate distant results beyond what is already encoded in the shared constraint structure. Local interventions alter local boundary conditions—such as measurement settings or coupling regimes—but they do not enable the transmission of information outside locally causal channels. The global character of admissibility concerns correlation structure, not causal control.

Local causation retains its full operational role within this account. Where agents intervene, vary experimental parameters, or propagate signals, causal descriptions remain appropriate and necessary. QCT does not weaken or replace such descriptions. Instead, it situates them within

a broader explanatory context: local causal narratives describe how boundary conditions are implemented and propagated, while global constraint satisfaction describes why only certain joint configurations remain admissible once those boundaries are fixed. The two levels address different questions and are not in competition.

This distinction clarifies why purely sequential causal stories often fail to account for quantum correlations. Attempts to force a time-ordered micro-causal explanation typically introduce additional assumptions—hidden variables, privileged foliations, or implicit signaling mechanisms—that exceed what is empirically required. QCT avoids these additions by recognizing that admissibility is evaluated relative to the complete constraint configuration of the experimental context, not generated step-by-step through local influence. The absence of a sequential causal microstory is therefore not a deficiency, but a reflection of the explanatory level at which the phenomenon is constrained.

Importantly, global constraint satisfaction does not introduce teleology, optimization, or goal-directed behavior. Systems do not select outcomes in order to satisfy constraints. Rather, configurations incompatible with the constraint structure are assigned zero admissibility: they do not correspond to any realization consistent with the full set of conditions. Stability arises through exclusion, not choice. The language of satisfaction refers to compatibility within a constraint system, not to evaluation, preference, or purpose.

By distinguishing global constraint satisfaction from local causation, this framework preserves the integrity of causal explanation while accounting for phenomena that exceed its scope. Local causation governs how changes propagate and how interventions take effect; global constraints govern which joint configurations are admissible once all relevant conditions are specified. This separation allows QCT to explain nonlocal correlation without invoking nonlocal causation, to maintain no-signaling and intervention locality, and to remain continuous with established physical practice while extending explanatory reach where purely sequential accounts prove insufficient.

8.3 Emergence of Temporal Asymmetry Under Entropy

Temporal asymmetry is not treated as a fundamental property of microscopic law, but as an emergent feature of constraint aggregation and admissibility reduction in high-constraint regimes. The framework does not posit an intrinsic arrow of time at the level of unitary dynamics. Instead, it explains why time-directed behavior reliably appears once interaction, record formation, and environmental coupling narrow the space of admissible realizations in a manner that is effectively irreversible.

Entropy, in this account, functions as a descriptive proxy rather than a causal driver. Rising entropy reflects the proliferation and entrenchment of constraints imposed by interaction, coarse-graining, and record redundancy. As systems interact with increasingly complex environments, admissible configurations compatible with prior records contract rapidly. This contraction is asymmetric in practice, not because the underlying dynamics forbid reversal, but because reversing the accumulated constraints would require coordinated control over an astronomically large number of correlated degrees of freedom. QCT therefore presupposes low initial constraint density as a boundary condition for temporal asymmetry, without attempting to explain why such a condition obtained.

Irreversibility, as used here, is operational rather than fundamental. Microscopic laws may remain time-reversal invariant, but admissibility reduction becomes effectively one-directional once constraint density passes a threshold at which recoherence or global reversal is no longer practically

attainable. Configurations compatible with existing records dominate the admissible space, while alternatives shrink to measure-zero sets relative to realistic control and access. The arrow of time thus emerges from constraint geometry and record stability, not from an added dynamical asymmetry.

This perspective clarifies why temporal asymmetry is graded rather than absolute. Different subsystems may exhibit different degrees of reversibility depending on their coupling to the environment and the density of constraints they experience. Weakly coupled or carefully isolated subsystems may retain low constraint density and support reversible or interference-capable behavior, even while the larger environment exhibits robust classical irreversibility. Temporal directionality is therefore not a universal switch but a spectrum tied to constraint aggregation.

Importantly, this account avoids teleology. Systems do not evolve toward preferred futures, minimize entropy, or optimize outcomes. Stability arises through exclusion: configurations incompatible with accumulated constraints fail to remain jointly admissible, while compatible configurations persist. Temporal asymmetry reflects the asymmetry of admissibility under record-conditioned interaction, not the pursuit of an end state.

Finally, the emergence of temporal asymmetry under entropy is compatible with established physical descriptions. It aligns with decoherence-based accounts, redundancy of environmental records, and thermodynamic irreversibility, while reframing them in constraint-based terms. QCT does not replace these frameworks; it explains why they jointly produce a stable arrow of time without requiring fundamental temporal bias. Time’s direction, in this view, is not imposed by law but arises from the structural consequences of constrained persistence.

8.4 Compatibility With Established Physical Laws

This framework is explicitly constrained by compatibility requirements rather than motivated by revisionary ambition. Its purpose is explanatory clarification, not modification of the operational content of quantum theory or its relativistic and thermodynamic extensions. Compatibility here means that for any preparation and measurement protocol within the empirically validated domain of quantum mechanics and quantum field theory, the framework yields the same operational predictions and statistical outcomes as standard formulations. It does not alter probabilities, signal structures, or experimentally accessible observables.

In particular, no-signaling and operational locality are preserved. Although admissibility is evaluated jointly across correlated systems, global admissibility is defined over—and therefore inherits—the joint probability structure supplied by quantum theory itself. It does not introduce an independent outcome-selection rule. Local interventions affect only local marginals, and spacelike-separated choices do not permit controllable influence. Any apparent nonlocality arises at the level of joint correlation structure rather than causal propagation, fully consistent with standard quantum constraints.

Relativistic compatibility is maintained by treating global constraint satisfaction as non-propagating and non-sequential. Constraints are not transmitted through spacetime, nor do they require a preferred foliation or temporal ordering. Where time asymmetry appears, it arises from boundary conditions and record formation rather than from violations of Lorentz invariance or microcausality. Nothing in the account conflicts with relativistic covariance at the level of observable predictions.

Unitary dynamics remain intact. Schrödinger evolution, Hamiltonian structure, and quantum field dynamics are not modified. Admissibility reduction does not function as a collapse law or an additional dynamical process. Configurations deemed inadmissible are not eliminated by force

or suppression; rather, they fail to form stable, record-bearing realizations under environmental interaction and constraint closure. In this sense, the framework reinterprets the emergence of classical definiteness without introducing new physical operators or stochastic triggers.

Thermodynamic compatibility is likewise preserved. Macroscopic irreversibility, entropy increase, and time-directed behavior are treated as effective consequences of constraint aggregation and record redundancy under standard low-entropy boundary conditions and coarse-graining. The framework neither derives the second law from first principles nor denies the time symmetry of microscopic dynamics. Instead, it explains why irreversibility becomes robust once constraint density exceeds a threshold beyond which admissibility reduction is effectively irreversible for practical systems.

Decoherence theory and related accounts of environmental monitoring are neither displaced nor duplicated. The framework remains compatible with decoherence, environment-induced superselection, and quantum Darwinism, while reframing their significance in terms of admissibility and constraint closure. Decoherence explains how interference becomes inaccessible; the present account explains why only a restricted subset of configurations remains structurally capable of persistence and record formation once closure is achieved.

Compatibility, then, is not a claim of novelty at the level of prediction. It is a constraint on explanation. Explanatory additions that violate no-signaling, introduce observer-dependent ontology, impose teleological selection principles, or require new dynamical primitives beyond established physics are explicitly ruled out. At the same time, the framework remains provisional: should future empirical findings require genuinely new explanatory elements not expressible in terms of constraint structure and admissibility, revision would be required.

This section therefore functions as a boundary condition on interpretation rather than a doctrinal endpoint. It affirms continuity with established physical laws while clarifying the level at which the framework operates—not as an alternative formalism, but as a constraint-first explanatory account of persistence, correlation, and classical stability that remains fully consistent with the physics it seeks to explain.

9 Shared Invariants and Mutual Constraint Participation

This Section formalizes a central requirement of interaction and persistence: systems can only participate in a shared world if they are reducible to at least one shared invariant. Interaction is understood as mutual constraint participation, and shared invariants are shown to be the minimal structures that make coherence, compatibility, and viability across systems possible.

9.1 Shared Invariants and Reducibility

The preceding sections have established that persistence, identity, and information all require constraint, and that constraint enforces persistence only through invariance. A structure persists only insofar as something about it remains fixed across admissible transformation. Information exists only where realized differentiation remains stable relative to excluded alternatives, and record authority arises only when such stability constrains future interaction. These results apply locally, at the level of individual systems and interactions, and do not yet presuppose any global ontological commitment.

A further structural consequence now follows. If multiple physical structures interact in a way that produces persistent joint effects—durable correlations, records, or stabilized mutual restrictions—then their interaction must be expressible in terms of at least one shared invariant. Without a shared invariant, no joint constraint can be enforced, and no persistent interaction can occur. Apparent coordination may arise momentarily, but it cannot stabilize as an authoritative restriction on admissible evolution.

In Quantum Coherence Theory, interaction is defined as mutual participation in a constraint relation that restricts admissible evolution for all participants. For such a restriction to be jointly binding, it must be referenced to something that remains invariant across the coupled evolution of the interacting structures. This invariant need not be fundamental, scalar, or unique; it may be relational, approximate, or emergent. It need only be sufficient to support re-identification and exclusion across interaction. Where no such invariant exists, interaction collapses into transient coincidence rather than enforceable constraint.

This requirement is already implicit in the earlier treatment of persistence and identity. Persistence requires invariance, and when persistence extends across interacting systems, invariance must be shared. Otherwise, admissibility fragments into incompatible descriptions that cannot jointly constrain future evolution. In such cases, each system may exhibit local stability, but no globally authoritative interaction can be sustained.

If the ontology under consideration is closed—that is, if all interaction, persistence, and record formation occur entirely within a single self-contained descriptive boundary—then a stronger requirement follows. In a closed ontology, all shared invariants that support persistent interaction must themselves be mutually reducible. If they were not, the ontology would decompose into disjoint constraint domains incapable of enforcing global admissibility. Records stabilized in one invariant regime would lack authority in another, and persistence would become local but not transferable.

This reducibility requirement does not imply that all systems are identical, nor that all dynamics collapse to a single variable. It is not eliminative reduction. It is a condition on comparability

and mutual constraint participation. For two structures to jointly constrain one another within a closed ontology, their invariants must be mappable—directly or indirectly—into a common invariant framework. Where no such mapping exists, interaction cannot be globally authoritative.

The same condition applies to information. Stabilized information can propagate and remain mutually accessible only if the exclusion relations that define it are preserved under interaction. That preservation is impossible unless the interacting systems share a reducible invariant structure. Information that cannot be reduced to a shared invariant cannot survive interaction; it decoheres into incompatibility rather than persisting as constraint.

Importantly, this claim remains structural rather than metaphysical. It does not assert what the shared invariant is, nor that it is unique in any ultimate sense, nor that it is directly observable. It asserts only that viability within a closed constraint system requires invariance closure: all local invariants that participate in persistent interaction must ultimately be mutually reducible. Without such reducibility, persistence cannot be globally enforced, and a coherent world-description cannot be sustained.

With this condition made explicit, the internal architecture of Quantum Coherence Theory is complete. Constraint enforces admissibility. Invariance enables persistence. Shared invariance enables interaction. Reducibility of invariants enables global enforceability within a closed descriptive boundary. What remains is not a further refinement of these principles, but an evaluation of which ontological descriptions can satisfy them without contradiction.

9.2 Modes of Shared Invariance and Reducibility

The information throughout this work established a necessary structural condition of Quantum Coherence Theory: persistent interaction within a closed constraint system requires shared invariance, and global enforceability requires that such invariants be mutually reducible. That condition, however, does not specify a single form that invariance or reducibility must take. Shared invariance may be exact or approximate, local or global, fundamental or emergent. Reducibility may be direct, indirect, structural, or statistical. The requirement imposed by QCT is not uniformity of form, but preservation of constraint authority across interaction.

In QCT, an invariant is defined operationally by its role rather than its representation. An invariant is whatever stabilizes exclusion and re-identification under admissible evolution. It is that which allows a realized differentiation to remain authoritative across interaction, disturbance, and time. Shared invariance therefore means that interacting structures preserve a common basis for exclusion and identity, even if that basis is encoded differently within each system. Reducibility, in turn, requires that these bases be translatable in a way that preserves their constraint-enforcing role.

In practice, invariants are often approximate rather than exact. Persistence is therefore always relative to a tolerance and a timescale. A shared invariant may stabilize interaction only within a bounded regime, degrading gradually as drift, noise, or coupling erodes its reliability. QCT treats persistence as graded rather than absolute: record authority weakens continuously as invariance degrades, rather than disappearing abruptly. This does not weaken the framework; it reflects the operational conditions under which constraint enforcement occurs in real systems.

Shared invariants may also be emergent rather than fundamental. Many interaction-relevant invariants—thermodynamic variables, functional states, computational configurations—do not reduce to unique micro-level descriptors. QCT permits such emergence provided that the invariant continues to support exclusion and re-identification under interaction. Reducibility in this context

does not require one-to-one correspondence. It requires only that there exist a mapping, possibly many-to-one, that preserves the constraint-relevant structure. What matters is not preservation of detail, but preservation of enforceability.

Reducibility likewise need not take the form of literal identity. Distinct invariant frameworks may encode the same constraint regime through different structures that are nonetheless inter-translatable. Such equivalence may be indirect or mediated by intermediate descriptions, and it may operate at the level of relations rather than variables. QCT treats these cases as admissible so long as the translations preserve exclusion relations and re-identification conditions. Dual or complementary descriptions therefore do not fragment the ontology unless their invariants fail to compose under interaction.

Stochastic reducibility is also compatible with closure. In many systems, constraint authority is preserved only in expectation or within bounded variance. Statistical mechanics, measurement processes, and adaptive systems often operate in this regime. QCT allows such cases provided that the stochastic translation preserves admissibility and record authority at the level governing interaction. Reducibility need not be deterministic to be effective; it must only be sufficient to maintain constraint enforcement across coupling.

Finally, QCT distinguishes genuine shared invariants from representational artifacts. Gauge freedoms, coordinate conventions, and descriptive redundancies do not count as invariants unless they participate in interaction constraints. Reducibility is imposed only on invariants defined over the physically admissible configuration space—after quotienting out purely representational degrees of freedom. This ensures that closure tracks real interaction capacity rather than notational overlap.

Taken together, these modes clarify the scope of the reducibility requirement without weakening it. Invariance may be approximate, emergent, dual, or statistical; reducibility may be homomorphic, indirect, or probabilistic. Across all cases, the criterion is the same: whatever carries constraint authority across interaction must remain translatable without breaking persistence. This preserves the internal completeness of QCT while preparing the ground for a subsequent evaluation of which ontological descriptions can satisfy this requirement without contradiction.

9.3 Invariants as the Basis of Mutual Constraint

Persistence requires invariance and that interaction, in Quantum Coherence Theory, is defined as mutual participation in a constraint that restricts admissible evolution for all participants. This definition already excludes purely causal influence or coincidental correlation. A constraint that is genuinely mutual must bind each participant under the same admissibility conditions across time. The question this section addresses is what makes such mutual binding possible at all.

For a constraint to be mutual rather than merely asymmetric, it must possess identity across the interaction history. A restriction that applies at one moment but is not the same restriction at the next cannot function as an enforceable boundary on admissibility. Without continuity, exclusions cannot accumulate, and no structure can stabilize. Invariance supplies the identity conditions of the constraint itself, fixing what counts as the same restriction across coupled evolution.

When two or more systems interact in a way that produces durable joint effects, their interaction must therefore be organized around at least one invariant that all participants remain answerable to. This invariant need not dictate the internal dynamics of each system, nor require similarity of structure or mechanism. It functions instead as a shared criterion of compatibility: each system may vary internally, but only variations that preserve the invariant remain admissible within the interaction. In this sense, the invariant is not merely conserved through interaction; it is what interaction coheres around.

If no such shared invariant exists, apparent mutual constraint collapses upon inspection. One system may influence another, but the influence cannot be reciprocally binding. Correlations may appear momentarily, but nothing enforces their continuation. Exclusions imposed at one moment cannot be re-imposed coherently at the next, because there is no preserved reference that fixes what exclusion means. What remains is mutual effect without mutual authority.

This distinction clarifies why invariance precedes records and information in QCT. A record constrains future interaction only if later admissibility remains answerable to the same exclusion relations that defined the record. That answerability presupposes an invariant structure that persists across time and coupling. Records therefore do not create mutual constraint; they are products of an interaction regime already stabilized by shared invariance.

In Quantum Coherence Theory, invariants are thus not passive quantities that happen to survive interaction. They are the structural basis of mutual accountability. They fix the identity of constraints across time, make exclusions enforceable, and allow interaction to rise above transient coincidence. Without shared invariance, there is no mutual constraint; without mutual constraint, there is no persistent interaction to be explained.

9.4 Shared Invariants and Record Authority

If interaction requires a shared invariant, then durable interaction requires more than momentary constraint satisfaction. It requires the ability to preserve interaction-state across time such that participants can re-identify what counts as “the same interaction” despite temporal separation, partial observation, or independent reconstruction. This persistence cannot be achieved by a shared invariant alone. It requires an invariant-preserving record.

Here, “record” does not mean a ledger, log, or centralized store. It refers to the minimal commitment state sufficient for participants to distinguish admissible continuations of interaction from noise. In other words, a record is the minimal invariant-reducible structure that allows interaction history to be reconstructed up to equivalence. Full historical detail is neither required nor generally possible.

This distinction matters because a shared invariant can constrain valid moves in the present, but it cannot by itself support temporal properties such as:

- distinguishing novelty from repetition,
- detecting omission, duplication, or reordering,
- preserving commitments across interaction boundaries.

These are diachronic constraints. A record is the mechanism by which the invariant is applied across time.

Record authority arises not from control, consensus, or enforcement, but from invariance. A record is authoritative to the extent that any participating system can independently verify that it belongs to the invariant-defined equivalence class of admissible histories. Authority here is not permission to write, but admissibility under shared constraint. Consensus mechanisms, when they exist, do not create authority; they merely help systems remain within invariant bounds.

Crucially, authoritative records are not located inside systems. They are maintained between systems as invariant-preserving relations. Individual systems may store internal representations, but those representations derive their authority only insofar as they remain reducible to the shared

invariant. When representations diverge, the invariant adjudicates—not by force, but by viability. What cannot be reduced together cannot be preserved together.

This means that purely private records cannot ground interaction. A private memory may feel authoritative locally, but unless it can be reconstructed and validated by others under the shared invariant, it cannot participate in mutual constraint. The moment interaction resumes, invariant compatibility—not belief, confidence, or recollection—determines what persists.

Record authority therefore precedes semantics. Participants may disagree about meaning, interpretation, or intent while still maintaining a shared record, so long as the invariant-relevant structure is preserved. Structural agreement enables stable disagreement. Meaning operates on top of record authority; it does not establish it.

There is also a hard limit: no record can exceed the expressive capacity of the invariant that sustains it. Distinctions not constrained by the shared invariant cannot be jointly preserved. They are not false, but non-shareable. Attempts to encode excess structure either decay into noise or fracture interaction viability. Record sufficiency is therefore a constraint, not a deficiency.

Finally, the authority of records depends on the openness or closure of the underlying ontology. In open ontologies, records remain provisional and revisable as new shared invariants emerge. In closed ontologies, record authority hardens: once invariant sufficiency is saturated, no further mutually admissible reduction is possible. This closure is structural, not procedural, and marks the transition from revisable interaction to finalized history.

In sum, shared invariants do not merely permit interaction. They determine what can be jointly remembered. Record authority is not granted, stored, or enforced; it is earned through invariance. What cannot be reconstructed together cannot function as part of a shared reality.

9.5 Closed Ontology and Invariant Sufficiency

The preceding sections have established that persistent interaction requires shared invariants, and that record authority depends on the preservation of exclusion relations under interaction. These results, however, do not yet specify what is required for an ontology to function as a single world-description rather than a collection of locally coherent domains. That distinction becomes decisive once closure is considered.

An ontology is closed if all interaction, persistence, and record formation occur within a single self-contained domain of admissibility. No appeal is made to external enforcement, privileged observers, or auxiliary structures outside the ontology itself. In such a setting, any constraint that acquires authority must do so entirely through relations internal to the ontology.

Closure therefore raises a stricter question: not merely whether invariants exist, but whether they are sufficient to coordinate constraint participation across the whole. Invariant sufficiency concerns whether the set of shared invariants operative within an ontology is adequate to support globally authoritative interaction. Local interactions may be stabilized by distinct invariants—emergent, approximate, or regime-specific—without difficulty. However, if those invariants cannot be mutually related, then constraint authority becomes fragmented. Records formed under one invariant regime cease to bind admissibility under another. Interaction remains locally meaningful but globally incoherent.

In a closed ontology, such fragmentation is not a benign feature. If invariant regimes are mutually incommensurable, then the ontology decomposes into effectively independent constraint domains. Each may sustain persistence internally, but none can claim authority across the whole. From within any one domain, other domains appear either opaque or irrelevant; from the standpoint

of the ontology as a whole, there is no unified admissibility structure. Worldhood dissolves into partition.

Invariant sufficiency therefore requires more than the mere presence of shared invariants. It requires that all invariants which participate in persistent interaction be jointly admissible—that is, capable of being situated within a single invariant framework that preserves constraint authority across domains. This does not demand a single primitive invariant, nor a reduction to a unique variable. It requires only that invariant regimes not be mutually exclusive in their constraint implications.

This condition is structural rather than metaphysical. It does not specify what the sufficient invariant framework must be, how it is realized, or at what scale it operates. It specifies only what failure looks like: an ontology in which constraint authority cannot compose globally cannot sustain a coherent account of persistence, identity, or record authority across its full descriptive scope.

Quantum Coherence Theory stops at this boundary. It identifies invariant sufficiency as a necessary condition for closure, but does not attempt to determine whether any particular ontology satisfies it. The task of assessing sufficiency—of determining whether an ontology’s invariant structure genuinely supports global admissibility without fragmentation—lies beyond QCT’s explanatory role.

What QCT establishes is the limit: without invariant sufficiency, closure fails. And where closure fails, no description can claim to represent a world rather than a collection of compatible but disconnected structures.

9.6 Viability as Reducibility to Shared Invariants

Progressively throughout this work, the structural requirements for persistence, interaction, and record authority have been established. Constraint enforces admissibility. Invariance enables persistence. Shared invariance enables interaction. Where interaction produces durable records or stabilized exclusions, those effects are authoritative only insofar as the participating structures remain jointly answerable to invariant constraints. These results have been established without appeal to interpretation, probability, or external selection.

A further consequence now becomes unavoidable. If a system of interacting structures is to remain viable as a single world-description—capable of sustaining persistence, interaction, and record authority across its full domain—then all constraint-relevant invariants that participate in those processes must be mutually reducible. Reducibility here does not mean identity or eliminative reduction.

It means that invariants must be translatable in a way that preserves admissibility, exclusion, and re-identification across interaction. Without such reducibility, persistence fragments into locally coherent but globally incompatible domains.

Viability, in this sense, is not an additional physical property layered onto an ontology. It is a structural condition on whether constraint can remain authoritative across the entirety of what the ontology claims to describe. An ontology may permit local persistence, localized records, or sector-relative stability while still failing to be viable as a unified world-description. Global viability requires that no interaction-authoritative invariant exist in isolation from the others that govern the same descriptive boundary.

This condition follows directly from the definition of record authority. A record constrains future interaction only if its exclusions remain meaningful across subsequent interactions. That meaningfulness depends on the ability of later systems to remain answerable to the same invariant

structure, either directly or through invariant-preserving translation. Where invariants are not mutually reducible, record authority cannot propagate. History becomes parochial, persistence becomes compartmentalized, and interaction loses global enforceability.

Quantum Coherence Theory therefore yields a precise criterion for viability: a system is viable as a closed world-description if and only if all invariants that participate in persistent interaction are reducible within a single constraint framework sufficient to enforce admissibility globally. This criterion is structural rather than metaphysical. It does not specify the nature of the invariant, its origin, or its ultimate form. It specifies only what must hold if persistence, interaction, and record authority are to coexist without contradiction.

At this point, the internal work of Quantum Coherence Theory is complete. QCT has identified the minimal structural conditions required for persistence, interaction, and information to function as physically authoritative features of a world. It has also identified the point at which those conditions exceed the scope of a theory concerned solely with constraint and invariance. Determining whether a given ontology satisfies the criterion of global viability—whether its invariants are in fact reducible in the required sense—is no longer a question of coherence theory alone.

What emerges from QCT, then, is not a unification, but a boundary. On one side lies the internal logic of constraint, invariance, and admissibility. On the other lies the evaluation of entire world-descriptions for their capacity to sustain those structures without fragmentation. That evaluation requires a framework whose primary task is not to describe interaction, but to assess whether a world can continue to exist as a world.

9.7 What This Claim Does Not Assert

The viability criterion established in the preceding section is structural in scope and limited in intent. It specifies necessary conditions for global persistence within an interacting ontology. It does not, by itself, resolve questions of ontology selection, metaphysical priority, or ultimate explanation. To avoid misinterpretation, several clarifications are required.

First, this claim does not assert that all invariants reduce to a single law, quantity, or equation. Reducibility to shared invariants is not eliminative unification. It does not demand that diverse constraint structures collapse into one description, nor that higher-level invariants disappear into lower-level formalisms. The requirement is only that invariant frameworks admit mappings that preserve exclusion relations and re-identification conditions. Structural compatibility, not descriptive uniformity, is the criterion.

Second, this claim does not assert that local descriptions lacking such reducibility are false or meaningless. Many frameworks may remain valid within restricted domains, under limited interaction, or for finite durations. The viability criterion distinguishes global persistence from local adequacy. A description may succeed empirically while failing to function as a world-description. This distinction concerns scope, not correctness.

Third, this claim does not assert that a viable ontology must be complete, closed under all future discovery, or immune to revision. Reducibility to shared invariants concerns the preservation of constraint authority under interaction, not the exhaustion of explanatory content. A world may remain viable while its descriptions evolve, provided that new constraints remain translatable into the shared invariant framework.

Fourth, this claim does not assert that shared invariants are directly observable, simple, or human-interpretable. The existence of a shared invariant framework is a structural necessity inferred from persistence, not a guarantee of epistemic accessibility. The theory does not presume

that such invariants are discoverable in final form, nor that they can be expressed without approximation or mediation.

Fifth, this claim does not assert that quantum coherence alone determines world viability. Quantum Coherence Theory establishes conditions under which constraint accumulation, interaction, and record authority are possible. It does not specify how these conditions are realized across scales, nor how competing invariant regimes are reconciled. The theory identifies a boundary, not a construction.

Finally, this claim does not assert inevitability. It does not guarantee that any given ontology satisfies the viability condition, nor that unification succeeds. It provides a diagnostic, not a promise. Where reducibility to shared invariants fails, fragmentation, parochial persistence, or ontological drift may result. Where it succeeds, global persistence becomes possible, but not assured.

These clarifications delimit the scope of Quantum Coherence Theory. The theory establishes what must be true if a world is to persist as a world. It does not declare which worlds do so, nor how such persistence is achieved in practice. Those questions require criteria beyond local coherence and constraint behavior.

With this boundary in place, Quantum Coherence Theory reaches explanatory closure. It has identified the structural conditions under which interaction, record authority, and persistence can coexist. What remains is not further explanation of quantum behavior, but evaluation of world-descriptions against the viability criterion now established.

10 Explanatory Closure

Section X closes Quantum Coherence Theory on its own terms by clarifying what QCT explains and, just as importantly, what it deliberately does not. QCT is shown to be explanatory rather than instrumental: a structural test for whether any account of reality can support stable records, persistence through interaction, and shared classical structure without appealing to selection, observation, or chance as primitives.

10.1 What QCT Explains

This section addresses a specific explanatory gap in quantum foundations: how classical definiteness, stable records, and shared macroscopic structure arise without appealing to observer-dependent collapse, global selection rules, or unreduced outcome proliferation. The scope is intentionally narrow. The aim is not to explain why the universe exists, why its laws take the form they do, or why particular initial conditions obtain. The aim is to explain how, given physical interaction and constraint aggregation, a single macroscopic continuation becomes authoritative.

Within this scope, collapse is treated as a structural phenomenon rather than a primitive event. It is not a stochastic jump, a mental act, or a global selection rule. It is the physical narrowing of admissibility that occurs when interactions generate sufficiently dense, redundant, and stabilizing constraints. When incompatible configurations can no longer jointly persist under the accumulated constraint network, admissibility is reduced. What remains is not selected by preference or probability, but enforced by physical incompatibility.

This framing clarifies why decoherence alone is insufficient for definiteness. Decoherence explains the dynamical autonomy of alternatives: why they cease to interfere. It does not, by itself, explain why only one macroscopic configuration governs future interaction. That additional step requires recognizing that once records are physically stabilized, they function as constraints on subsequent dynamics. Incompatible macroscopic continuations are not merely non-interfering; they are rendered inadmissible. Collapse, in this sense, is the lawful consequence of constraint aggregation, not an interpretive supplement.

The same structure explains record authority. A record is not merely a correlation or a narrative label for what happened. It is a physical structure whose presence alters what can happen next. Downstream interactions depend on it counterfactually. That dependence is objective and observer-independent. Record authority is therefore explained by constraint enforcement rather than by epistemic access, self-location, or perspective.

Classical robustness follows naturally. Classical behavior persists not because it is favored, optimal, or typical, but because it occupies regions of state space that remain admissible under dense interaction. As constraint density increases, incompatible configurations are progressively excluded. Classical regularities emerge as stable survivors of constraint pruning, not as outcomes selected by global rules or anthropic filtering.

Crucially, this account does not commit to a full ontology of worlds. It does not claim that only one possible universe exists, nor does it deny the mathematical existence of alternative solutions or histories. Its claim is conditional and physical: once constraint conditions sufficient for record stabilization obtain, admissibility collapses to a single macroscopic continuation. This claim is local, interaction-based, and in principle testable through its implications for stability, redundancy, and exclusion.

Equally important are the questions this account does not attempt to answer. It does not explain why constraint structures exist in the first place. It does not explain why the universe supports stable matter, long-lived records, or observers at all. It does not offer a cosmological selection principle or a metaphysical foundation. Those questions lie outside its intended domain.

For this reason, the framework should be read as a test case rather than a world theory. It demonstrates that a constraint-first explanatory discipline can resolve a long-standing problem at the quantum–classical boundary without importing observer dependence, teleology, or unreduced multiplicity. It shows that persistence, definiteness, and shared classical structure can be explained as consequences of physical interaction alone.

That result is sufficient for its role here. It establishes that constraint-based explanation is viable in a demanding domain. Whether the same discipline applies more broadly is a separate question—one that can only be asked once this narrower case has been allowed to stand or fail on its own terms.

10.2 Constraint Based Collapse

In Quantum Coherence Theory, collapse is not defined as a fundamental dynamical interruption, nor as the literal elimination of representable alternatives. It is defined as a transition in admissibility: the point at which accumulated physical constraints render macroscopic continuation (up to a specified coarse-graining C) conditionally singular with respect to future interactions. At this point, all but one candidate macroscopic continuation lack any physically realizable robust extension that preserves the stabilized records under the instantiated constraint network within tolerance.

Decoherence explains the suppression of interference between macroscopically distinct configurations. Quantum Darwinism explains why certain states become redundantly recorded and effectively objective within dynamically autonomous sectors. Neither, by itself, explains why a particular macroscopic record comes to function as an authoritative constraint on what can happen next. Quantum Coherence Theory addresses this gap by defining collapse in terms of constraint-based admissibility rather than representational exclusivity, epistemic update, or branch deletion.

A macroscopic continuation is admissible relative to a given coarse-graining, tolerance, and constraint network if there exists a physically realizable extension of the present state in which all stabilized records are preserved under the actual couplings, boundary conditions, and noise environment the system admits. Admissibility is therefore a matter of dynamical accessibility and robustness, not of description, awareness, or perspective. A continuation that cannot be extended without destabilizing existing records or amplifying error beyond tolerance is inadmissible, regardless of whether it remains mathematically representable.

For clarity, admissibility may be treated schematically as a context-relative set. Given a coarse-graining C , a stability tolerance ϵ , and a constraint network D , one may consider the set $A(C, \epsilon, D)$ of coarse-grained macroscopic continuations that admit physically realizable robust extensions preserving the stabilized record structure under D within ϵ . Collapse, in this sense, occurs when $A(C, \epsilon, D)$ becomes singleton up to tolerance. This singularity is approximate and scale-relative. It is defined by robustness criteria, not by exact microstate uniqueness.

A record, in this context, is not an arbitrary correlation. It is a macroscopic correlation pattern redundantly instantiated across degrees of freedom that are sufficiently decoupled to persist under expected perturbations. A record is stabilized when it remains robust under the actual couplings and noise the system admits. Stabilized records function as constraints because subsequent interactions must remain consistent with them in a counterfactual sense.

The notion of authority is central. A record has authority when it exerts intervention-relative counterfactual control over future macroscopic structure: varying that record through physically permitted interventions systematically alters downstream constraints and records, while attempts to instantiate incompatible alternatives fail to propagate stably under the same constraint network. Authority is therefore a physical property of constraint integration, not a function of observation, awareness, or representational status.

Within this framework, thermodynamic irreversibility aligns naturally with increasing constraint density over time. As interactions accumulate, stabilized records proliferate, redundancies grow, and admissible macroscopic continuations narrow. What is conventionally described as entropy increase corresponds, at the macroscopic level, to the progressive restriction of future admissibility imposed by accumulated constraints. This does not redefine entropy or reduce thermodynamics to QCT. It clarifies why entropy increase and irreversibility track the same structural phenomenon: once constraints are instantiated and redundantly enforced, restoring earlier admissibility would require unrealizable coordination across the constraint network.

Crucially, Quantum Coherence Theory does not assert that incompatible decohered configurations are dynamically erased, nor does it require that a global mathematical state reduce to a single term. The claim is narrower and structural. Once collapse occurs, only one macroscopic record retains counterfactual authority conditional on the realized constraint network. Other representable alternatives lack the capacity to govern future interactions within that context.

This definition is neutral with respect to deeper ontology. Under Everett-style interpretations, collapse is sector-relative: conditional singular admissibility holds within each dynamically autonomous sector. Under single-world realism, an additional ontic selector is required to determine which admissible continuation is realized. Quantum Coherence Theory does not supply that selector. It specifies what collapse must amount to structurally in order to explain persistence, record authority, and shared classical structure.

Collapse is not probabilistic selection. Probability may characterize distributions over microconditions or over which constraint networks become instantiated, but probability does not itself enforce admissibility. In Quantum Coherence Theory, narrowing is achieved through constraint enforcement. How probabilistic regularities arise from this process is a separate question and is not presupposed by the definition of collapse.

There is no universal collapse threshold or fixed timescale. The onset, scale, and sharpness of collapse depend on constraint density, redundancy, coupling strength, and error tolerance in the physical context. Collapse is therefore emergent, context-sensitive, and progressive rather than instantaneous or fundamental.

In summary, collapse in Quantum Coherence Theory is the transition to conditional singular admissibility: the point at which, given the stabilized record structure actually instantiated, only one macroscopic continuation remains capable of exerting counterfactual control over future interactions. It is not decoherence, not epistemic update, not branch deletion, and not a primitive postulate. It is the moment at which constraint aggregation renders macroscopic authority unique within a realized constraint network, thereby grounding persistence and shared classical structure.

10.3 Shared Classicality Without Selection

The persistence of a shared classical world is often treated as a mystery that demands resolution by selection. Why this outcome rather than another. Why one world rather than many. Why

agreement rather than fragmentation. Across interpretations, this pressure repeatedly reappears as an appeal to chance, preference, observation, optimization, or purpose. Each of these attempts seeks to explain shared reality by choosing among alternatives.

Within a constraint-first framework, no such choice is required. What must be explained is not why a particular macroscopic outcome is selected, but why macroscopic structure exhibits stable, intersubjective continuity across interactions. That continuity does not arise because a world is chosen. It arises because constraint-enforced records possess unique authority over what can happen next.

Once collapse is understood as conditional singular admissibility, shared classicality follows directly. When a macroscopic record becomes stabilized, it does not merely describe what has occurred. It becomes a physical constraint on subsequent interactions. Any future process that remains dynamically admissible must preserve consistency with that record within tolerance. Interactions that fail to do so do not propagate as robust extensions of the world. They are not forbidden by fiat. They are rendered dynamically inadmissible by the constraint network itself.

Because constraint authority is physical rather than epistemic, it applies uniformly across subsystems. Observers do not coordinate their perceptions in order to agree on reality. They encounter the same constraint structure. Agreement is enforced not by communication or consensus, but by the fact that all admissible interactions are already constrained to be mutually consistent with the same stabilized records.

This is why shared classicality does not depend on observation. An observer is simply a subsystem interacting with an already constrained environment. Whether or not a record is observed, it continues to exert authority. Whether or not it is represented, it continues to shape what can persist. The world does not wait for acknowledgment in order to remain coherent. Disagreement, illusion, and error are not excluded by this account. Local subsystems may encode false or incomplete correlations. But such deviations lack authority. They do not propagate stably across the broader constraint network. They are overwritten, damped, or confined to contexts where they cannot govern future macroscopic structure. What persists at scale is not belief, but constraint compatibility.

This also explains why shared classicality does not require the elimination of alternatives. Representationally distinct configurations may remain mathematically describable. Decohered sectors may remain dynamically autonomous. What matters is not representability, but authority. Only those macroscopic records that are compatible with the realized constraint network can function as sources of future constraints. Alternatives that lack this capacity may exist as descriptions, but they cannot act.

In this sense, the classical world is singular not because it is uniquely selected, but because constraint authority is uniquely instantiated. There is no global preference. There is no ranking of outcomes. There is only the progressive narrowing of what remains dynamically viable as records accumulate and stabilize.

This account also avoids teleology. Persistence does not occur because a configuration is optimal, successful, or favored. It occurs because it remains admissible under the constraints that exist. Survival is not achievement. It is compatibility. Apparent regularity reflects constraint satisfaction, not goal fulfillment.

Nor does shared classicality require ontological commitment beyond the structure already described. Under sector-relative interpretations, constraint authority is local to each dynamically autonomous context. Under single-world realism, an additional fact must determine which context is realized. In either case, the mechanism by which shared classicality arises is the same. QCT specifies the structure that any such account must respect, without supplying a selector of its own. The world appears unified because it is constrained to be so.

Once stabilized records possess authority, divergence cannot scale. Branching may occur representationally. Possibilities may proliferate mathematically. But macroscopic structure remains shared because only one continuation remains capable of governing what follows within a realized constraint network. The classical world is therefore not imposed, selected, or declared. It is enforced. Not by observers. Not by purpose. Not by probability. But by the accumulation of constraints that render all but one macroscopic continuation dynamically inadmissible as a robust extension. This is what it means for a world to persist.

10.4 What QCT Leaves Open by Design

The preceding sections have not proposed a mechanism for choosing outcomes. They have instead identified a structural condition that any account of persistence, record authority, and shared classical structure must satisfy. Collapse, as defined here, is not an event appended to dynamics, nor a metaphysical declaration about what exists. It is a diagnostic boundary: the point at which accumulated constraints render macroscopic continuation conditionally singular with respect to future interaction.

This reframing shifts the explanatory task. Traditional debates ask how outcomes are selected from among alternatives. A constraint-first account shows that selection is not the operative macroscopic phenomenon. What matters is whether stabilized records acquire the capacity to function as authoritative constraints on what can happen next. Where they do, persistence follows. Where they do not, classical structure fails to form—regardless of how many alternative outcomes remain representable in principle.

Seen this way, collapse is not a solution to the measurement problem. It is a criterion for ruling out explanations that cannot, even in principle, support persistence under interaction. Any framework that leaves incompatible macroscopic continuations equally admissible after record stabilization fails this criterion, because it leaves record authority physically ungrounded. Frameworks that appeal instead to epistemic update, global optimization, or irreducible selection also fail—not because they are incoherent, but because they relocate authority into something other than constraint: belief, preference, luck, or purpose.

The consequence is subtle but decisive. Once constraint authority is established, the question “why this outcome rather than another?” loses its force. The correct question becomes: which structures are capable of governing future admissibility? Classical history is not chosen; it is enforced. Agreement is not coordinated; it is compelled. Error is not eliminated; it is rendered non-authoritative.

Nothing in this account privileges a particular microphysical ontology. The constraint-based criterion applies equally to unitary, stochastic, hidden-variable, or sector-relative interpretations. What it excludes are not theories as such, but explanatory moves: those that attempt to recover persistence without providing a way for records to become binding constraints on subsequent interaction.

This boundary also clarifies the status of operational frameworks that treat coherence as a quantifiable resource within controlled regimes. Resource-theoretic approaches, for example, presuppose environments in which coherence persists long enough to be counted, converted, or expended. QCT neither competes with nor subsumes such frameworks. It addresses a prior condition they assume but do not explain: the structural circumstances under which coherence can acquire durability and authority under interaction, rather than evaporating immediately upon coupling.

This diagnostic role is intentional. The aim is not to replace existing formalisms, but to clarify what any viable world-description must already accomplish if it is to explain why anything like

a shared, stable world persists at all. Collapse, in this sense, is not the answer to the story of classicality. It is the point at which non-viable stories can no longer continue.

- **The Viability Test:** A framework passes the viability test only if, given its own admissible dynamics, stabilized records automatically acquire authority as constraints on future interaction. Where authority requires supplementation—observer, chance, optimization, or decree—the framework fails. Viability demands that persistence be self-enforcing through constraint, not externally justified by selection.
- **Contrast Case:** Record stabilization without authority yields worlds that do not cohere. Decoherence occurs, records form, but nothing binds them. Each stabilization event becomes a local correlation lacking power to govern subsequent admissibility. Classical structure fragments into isolated pockets, each internally stable but globally non-persistent. A world of that kind can be written, but not lived in.

What remains is not to decide between outcomes, but to determine which theoretical structures can sustain constraint authority without supplementation. That question does not belong to collapse itself. It belongs to viability.

10.5 Deferred Role of Mathematical Formalism

Quantum Coherence Theory (Sections 1-10) intentionally refrains from introducing a fixed mathematical formalism in this work. This decision is neither a denial of formalizability nor a claim that QCT operates outside mathematical description. It is a scope choice. The present framework is concerned with identifying structural relations that govern persistence, admissibility, and stabilization across regimes, prior to committing to domain-specific representations.

Mathematical formalism is not excluded by QCT. On the contrary, multiple formalisms may be appropriate depending on the system under consideration, including Hilbert-space dynamics, information-theoretic measures, dynamical systems models, or other constraint-based representations. Premature commitment to a single formal language would risk conflating explanatory structure with a particular implementation.

QCT therefore treats mathematical constructs as downstream instruments rather than foundational primitives. Formal models are expected to arise where explanatory pressure demands precision, prediction, or system-specific instantiation. Their absence here reflects the aim of establishing cross-domain explanatory coherence rather than delivering a closed technical formalism.

This stance preserves both rigor and flexibility. It allows QCT to remain compatible with existing mathematical frameworks while avoiding artificial restriction at the level of principle. Formalization is not rejected; it is deferred until necessity, rather than convention, determines its form.

10.6 Empirical Commitments, Distinguishers, and Falsifiers

This work is not introduced as a modification of quantum dynamics, nor as a competing collapse model. It does not posit new forces, additional variables, or departures from unitary evolution at

the global level. Its empirical commitments are therefore not expressed as novel equations of motion, but as structural claims about how known physical mechanisms scale, interact, and constrain admissible behavior across regimes. This section clarifies those commitments, identifies where QCT makes distinguishable claims relative to other explanatory frameworks, and specifies the conditions under which the theory would be undermined.

Empirical Commitments

QCT commits to the following empirically grounded claims:

- **Decoherence and Classical Emergence Track Interaction Structure, not Intrinsic Scale:** Loss of observable quantum coherence correlates with the accumulation, alignment, and redundancy of system–environment couplings, rather than with mass, size, duration, or complexity in isolation. Experiments in interferometry, mesoscopic systems, superconducting circuits, and optomechanics are interpreted as confirming this structural dependence.
- **Classical Determinism Emerges as a Limiting Behavioral Regime:** Deterministic classical descriptions become effective when constraint density is sufficiently high to restrict admissible evolution to narrow, stable manifolds. Determinism is therefore contingent and context-sensitive, not fundamental.
- **Pointer States and Classical Stability Reflect Constraint Compatibility:** The basis in which classical behavior appears is determined by the dominant coupling structure. Configurations that persist are those least disrupted by aggregated constraints, not those selected by observation or imposed collapse rules.
- **Quantum Behavior Can Re-Emerge when Constraints Are Relaxed:** Isolation, cooling, shielding, dynamical decoupling, and engineered environments can broaden admissibility and restore quantum coherence. Such reversals are not anomalies, but direct consequences of reduced constraint density.

These commitments align QCT with the empirical record while assigning explanatory priority to constraint geometry and interaction structure.

Distinguishers

QCT is distinguishable from other approaches in how it organizes and interprets these facts:

- **Against Observer-Centric Accounts:** QCT attributes neither decoherence nor classical emergence to observation, measurement updates, or epistemic intervention. Observers are treated as physical subsystems subject to the same constraints as any other.
- **Against Intrinsic Collapse Models:** QCT does not posit stochastic collapse triggers, mass thresholds, or gravity-induced reductions. Apparent collapse is understood as the exhaustion of joint admissibility under constraint aggregation, compatible with globally unitary dynamics.
- **Beyond Descriptive Decoherence:** While standard decoherence theory describes how interference is suppressed, QCT advances a unifying structural principle: admissibility reduction under accumulated constraint explains why decoherence, pointer selection, classical determinism, and scale dependence exhibit the patterns they do.

- Interpretation-Neutral: QCT brackets questions of single-outcome realization, branching ontology, or subjective probability. Its claims concern the emergence of stable, classical behavior, not the metaphysical status of unrealized alternatives.

Falsifiers and Limits

Because QCT does not introduce new dynamics, its falsification conditions concern structural failures, not parameter mismatches. The framework would be undermined if any of the following were reliably observed:

- Sustained Macroscopic Coherence under High, Heterogeneous Constraint Density: If systems exhibiting dense, redundant, and incompatible environmental coupling nonetheless maintained long-lived, recoverable quantum interference, the central claim that constraint aggregation enforces admissibility reduction would fail.
- Classical Determinism Emerging without Constraint Saturation: If deterministic classical behavior were shown to arise robustly in regimes lacking strong coupling, redundancy, or environmental embedding, QCT's account of determinism as a constrained limit would be incorrect.
- Decoherence Rates or Pointer Bases Independent of Coupling Structure: If experimental evidence demonstrated that decoherence and basis selection depend primarily on intrinsic system properties rather than interaction geometry, the constraint-first interpretation would be invalidated.
- A Confirmed Need for Intrinsic Collapse Dynamics: Empirical validation of non-unitary collapse mechanisms operating independently of interaction structure would place QCT outside the correct explanatory class.

Scope Clarification

QCT does not claim to derive the Born rule, resolve the ontology of outcomes, or replace existing quantum formalism. Its claim is narrower and more structural: constraint aggregation is sufficient to explain the emergence of classical stability, determinism, and predictability from quantum permissiveness. Any viable interpretation or extension of quantum mechanics must be compatible with this constraint-based structure.

In this sense, QCT is not an alternative to quantum theory, but a refinement of its explanatory architecture. It identifies where permissiveness ends, where stability begins, and why classical behavior appears not as a separate law, but as the constrained limit of quantum possibility.

10.7 Why QCT Is Explanatory, Not Instrumental

This framework is not introduced as a rival predictive formalism, nor as an auxiliary rule for generating outcomes. It does not modify the quantum calculus, introduce new probabilities, or specify procedures for selecting results. Its role is explanatory rather than instrumental. The question it addresses is prior to prediction: what must be structurally true of a world for stable records, persistence through interaction, and a shared classical domain to be possible at all.

Instrumental approaches focus on prediction. They ask how to compute probabilities, expectation values, or outcome distributions. The concern here is different. It asks what makes such computations interpretable as describing a single, persisting world rather than merely correlating reports. The target is not operational success, but explanatory adequacy at the level of world-conditions.

This distinction matters because predictive accuracy alone does not secure ontological coherence. A formalism may reproduce observed statistics while leaving unanswered why macroscopic records are binding rather than ephemeral, why facts persist rather than fragment, or why intersubjective agreement does not dissolve under continued interaction. The gap appears wherever outcome talk is possible without an account of why stabilized structure constrains what can happen next.

Many interpretive strategies attempt to bridge this gap by adding selection rules: stochastic collapses, observer-triggered updates, optimization principles, or decision-theoretic surrogates. Even when such strategies reproduce empirical frequencies, they often relocate the explanatory burden rather than resolve it. They explain which outcome is registered, but not why stabilized macroscopic structure enforces persistence rather than remaining a passive correlation.

The constraint-first approach reframes the burden entirely. Definiteness is not treated as an outcome-selection mechanism layered onto dynamics, but as a condition on admissibility. Once stabilized records exist, admissibility narrows. Incompatible macroscopic continuations are excluded not by preference, probability, or observation, but because they cannot be extended as robust continuations under the realized constraint network. Definiteness, on this view, is not injected into dynamics; it emerges when record-stabilized structure acquires authority over future interaction.

Because this analysis operates at the level of structural preconditions, it is deliberately silent on several questions often treated as foundational. It does not derive probabilities, specify sampling rules, or choose between ontological interpretations. Those tasks are instrumental. Probability, observation, and outcome talk presuppose persistence and record authority; they cannot replace them as foundations.

For this reason, the framework should not be read as an interpretation in the usual sense. It does not propose a picture of what ultimately exists. It specifies what any viable world-description must already satisfy if it is to explain why a shared, stable classical world persists at all. Its contribution is not a mechanism, but a criterion: an account of the structural conditions under which facts can become binding rather than merely reported.

In this sense, the contribution is explanatory rather than instrumental. It clarifies the dependency order between persistence, record authority, and the calculational machinery of physical theory. What remains—how different ontologies realize or supplement these conditions—is not a question of collapse or selection. It is a question of viability.

11 Summary and Synthesis of QCT

(Sections 1-10)

Quantum Coherence Theory (QCT) does not introduce new dynamical mechanisms, nor does it propose revisions to quantum formalism. Its contribution is explanatory and diagnostic. It clarifies what coherence must mean if quantum theory is to account for persistence, record authority, and shared classical structure in an interacting world.

At the experimental level, QCT aligns naturally with results across matter-wave interferometry, engineered qubit platforms, and decoherence studies by reframing coherence as a property that scales with environmental interaction rather than intrinsic isolation. The persistence of superposition is shown not to depend primarily on size or energy, but on the structure and accumulation of constraints through which which-alternative information is suppressed, redirected, or rendered non-authoritative. Coherence fails not because systems grow large, but because unconstrained interaction channels overwhelm stabilizing structure.

This reframing extends the definition of coherence beyond internal phase consistency to include interactive survivability. A state is coherent not merely when its internal relations are well-defined, but when those relations persist under coupling and acquire authority as constraints on future interaction. From this perspective, the quantum–classical transition is no longer a regression away from quantum behavior, but a forward progression toward higher constraint density and stabilized admissibility. Classicality is not imposed by measurement or selection; it is enforced by constraint.

Within this framework, energy is properly situated as a constraint regime rather than an explanatory primitive. Energy budgets, dissipation rates, and thermal effects remain indispensable to practice, but they function as bookkeeping measures of the cost of maintaining structure, not as the source of coherence or persistence themselves. Structural organization determines whether coherence survives; energy tracks what it takes to sustain that organization.

QCT also demystifies non-locality without invoking action at a distance. Locality appears as a high-constraint satisfaction regime in which admissible interactions are tightly restricted. When constraint structures span extended systems, correlations can remain binding without localized mediation. What appears non-local is, in this sense, globally constrained.

Central to this account is the explicit identification of record authority as a necessary structural feature of persistence. Standard treatments often assume that records, once formed, persist simply because dynamics continue. QCT shows this assumption to be insufficient. In purely dynamical systems—particularly at the quantum level—stabilized correlations would remain locally descriptive but globally violable under subsequent interaction. That this does not occur indicates that record authority is not an emergent convenience but a requirement. Once records stabilize, they constrain future admissibility. The past is not merely remembered; it is enforced.

This enforcement implies the necessity of shared invariant structure. Persistence across interaction requires global consistency, and global consistency requires invariant constraints respected across domains. QCT does not posit the nature of such invariants as primitives; it demonstrates their necessity diagnostically. Worlds lacking shared invariant structure cannot support authoritative records and therefore cannot sustain shared classical history. The implications of this requirement are explored more fully in Viable Worlds Theory.

From these considerations follow QCT’s closure conditions. The theory does not privilege unitary, stochastic, hidden-variable, or sector-relative interpretations. What it excludes are not theories as such, but explanatory moves: those that attempt to recover persistence without providing

a way for stabilized records to become binding constraints on future interaction. Where authority must be supplied by observer choice, chance, optimization, or decree, the account fails—not because it is inconsistent, but because it cannot explain persistence.

Accordingly, QCT proposes no mechanisms of collapse, outcome selection, or world branching. Instead, it sharply constrains what any proposed mechanism must satisfy in order to be operational. It narrows the space of viable explanations without isolating a single solution, identifying instead where current approaches remain incomplete and where future theoretical development is required.

In this sense, Quantum Coherence Theory does not close inquiry. It closes a class of non-viable stories. What remains is a clearer understanding of what coherence, persistence, and classical structure must mean if a shared, stable world is to exist at all.

12 Operational and Mathematical Consolidation of QCT

We now transition from the interpretive layer of QCT to a disciplined operational formulation. The aim is not to introduce new physical postulates, but to state QCT’s structural claims within a minimal open-systems setting where admissibility, redundancy, regime separation, and boundary conditions can be evaluated explicitly.

Section 12 fixes the mathematical setting and operational predicates used throughout the remainder of the manuscript. In particular, it separates (i) disjoint simultaneous redundancy (DSR), a fixed-state predicate applied to a single joint state $\rho_{SE}(t)$, from (ii) tested-channel redundancy, a controlled-run notion evaluated under fresh initialization. With these conventions in place, we state the Structural Admissibility Hypothesis in a form suitable for parameter-space hardening and adversarial stress testing in the next section.

12.1 Why Operational Formalization is Necessary

Earlier sections articulated the central structural tension of QCT at a conceptual level: whether disjoint environment fragments can carry high-quality records of two quantitatively incompatible PVMs on the same system, and under what conditions such redundancy is admissible. Conceptually, the tension is straightforward. Incompatible observables cannot be sharply recorded simultaneously on a single system instance. Yet redundancy arguments in decoherence theory are typically framed in terms of fragment-level record quality rather than joint sharp measurability. This leaves open a structural question:

For observables that remain quantitatively incompatible ($c \leq 1 - \eta$), what constraints must the joint $S:E$ state satisfy for $\text{DSR}_\delta(X, Z; t)$ to be logically attainable, and when does such redundancy force the state toward a restricted structural class?

Answering this question requires more than informal disturbance arguments or isolated entropy bounds. It demands an explicitly operational formulation in which logical attainability, structural restriction, and regime dependence are separated cleanly.

Three distinctions are essential:

1. **Logical attainability:** whether $\text{DSR}_\delta(X, Z; t)$ is compatible with quantum mechanics in a controlled setting.
2. **Structural restriction:** whether, if realized, it forces the global state toward a highly constrained configuration.
3. **Regime qualification:** whether compatibility and restriction coincide generically or only under fine-tuned structural conditions.

Conflating these issues obscures the distinction between logical possibility and structural restriction, and can lead either to premature rigidity claims or to underestimation of the constraints involved. Accordingly, redundancy will be evaluated in two logically distinct ways throughout this

framework. First, the *state-level* disjoint simultaneous redundancy (DSR) predicate $\text{DSR}_\delta(X, Z; t)$ is applied to a fixed joint state $\rho_{SE}(t)$. Second, we consider a *tested-channel* redundancy notion verified operationally under controlled initialization, where one observable is probed per task using fresh input preparation. No implication between these predicates is assumed unless explicitly established.

The objective of the present work is not to assert generic simultaneity of incompatible records nor to derive a universal rigidity theorem. Rather, this introductory analysis determines when $\text{DSR}_\delta(X, Z; t)$ is attainable, quantifies its non-generic character, and identifies the structural constraints that accompany it.

The analysis proceeds in five stages. Section 12 formalizes the operational setting and the state-level redundancy predicate $\text{DSR}_\delta(X, Z; t)$. Section 13 introduces a structural admissibility hypothesis and a falsification criterion. Section 14 develops an entropic obstruction and, under the rigidity strengthening of Hypothesis 14.1, shows that simultaneous high-quality fragment records of incompatible observables imply that the induced system state lies close to the compatibility set $\mathcal{C}_{XZ}$. Section 15 then presents an explicit engineered construction demonstrating logical attainability of tested-channel redundancy under controlled initialization. Finally, Section 16 analyzes perturbative stability, showing that the engineered regime is structurally fragile and nongeneric.

With these distinctions fixed, we now formalize the minimal operational setting.

12.2 Minimal Mathematical Setting

This section specifies the finite-dimensional open quantum system framework within which the structural claims of Quantum Coherence Theory (QCT) are formulated.

- We assume standard quantum mechanics: tensor-product subsystem structure, global unitary evolution, the Born rule, and the operational description of measurement interventions via completely positive, trace-nonincreasing instruments whose aggregate map is completely positive and trace-preserving (CPTP).
- All Hilbert spaces in this chapter are assumed finite-dimensional.
- No interpretive commitments beyond these conventional structures are introduced.

Subsystem Structure

We assume a fixed tensor-product decomposition

$$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E \tag{12.1}$$

representing a laboratory-defined decomposition into a system S and an environment E .

To analyze correlation distribution within the environment, we further assume a fixed finite tensor decomposition

$$\mathcal{H}_E = \bigotimes_{k=1}^N \mathcal{H}_{E_k}. \tag{12.2}$$

Each factor $\mathcal{H}_{E_k}$ is treated as an operationally addressable subsystem in the following limited sense: single-copy POVMs acting solely on $\mathcal{H}_{E_k}$ are permitted in principle. No assumption is made

regarding dynamical independence, spacelike separation, or simultaneous non-disturbing access to multiple fragments. This decomposition is treated as part of the physical modeling input and is not claimed to be unique or fundamental.

Global Dynamics

The composite state evolves unitarily:

$$\rho_{SE}(t) = U(t)\rho_{SE}(0)U^\dagger(t) \quad (12.3)$$

with

$$U(t) = e^{-iHt},$$

where H is self-adjoint and $\hbar = 1$.

Time-dependent Hamiltonians may be incorporated via time-ordered exponentials. No Markovian or weak-coupling assumptions are imposed at the kinematic level. In particular, no product-state restriction is assumed in the formulation of the global unitary dynamics. Operational restrictions to product initial states are introduced later only for redundancy analysis.

Reduced subsystem states are defined by partial trace:

$$\begin{aligned} \rho_S(t) &= \text{Tr}_E \rho_{SE}(t), \\ \rho_{E_k}(t) &= \text{Tr}_{S, E \setminus E_k} \rho_{SE}(t). \end{aligned}$$

Trace Norm and Trace Distance

For any operator A define the Schatten 1-norm

$$\|A\|_1 := \text{Tr}(\sqrt{A^\dagger A}).$$

For density operators ρ, σ define the trace distance

$$\text{dist}_{\text{tr}}(\rho, \sigma) := \frac{1}{2} \|\rho - \sigma\|_1. \quad (12.4)$$

For any nonempty $\mathcal{C} \subseteq \mathcal{D}(\mathcal{H}_S)$ define the distance to a set by

$$\text{dist}_{\text{tr}}(\rho, \mathcal{C}) := \inf_{\sigma \in \mathcal{C}} \text{dist}_{\text{tr}}(\rho, \sigma). \quad (12.5)$$

All trace distances in this section use the 1/2-normalized convention. In finite dimension the infimum is attained when $\mathcal{C}$ is compact.

Decoherence Criterion

Given an orthonormal basis $\{|x\rangle\}$ on $\mathcal{H}_S$, define

$$q_x(t) = \langle x | \rho_S(t) | x \rangle.$$

We say that $\rho_S(t)$ is ϵ -decohered in that basis at time t if

$$\frac{1}{2} \left\| \rho_S(t) - \sum_x q_x(t) |x\rangle\langle x| \right\|_1 \leq \epsilon. \quad (12.6)$$

This definition is basis-relative and time-local.

Remark: This trace-distance formulation of decoherence is consistent with the broader dynamical decoherence and environment-induced superselection (einselection) framework reviewed in [1]. The present treatment isolates only the finite-dimensional, operational aspects required for the redundancy analysis.

When the basis $\{|x\rangle\}$ corresponds to a projective measurement

$$X = \{P_x = |x\rangle\langle x|\},$$

the weights $q_x(t)$ coincide with the Born-rule probabilities of that measurement.

No claim of asymptotic behavior or pointer-basis uniqueness is made at this stage.

Information-Theoretic Quantities

For a classical-quantum ensemble $\{p_x, \rho_x\}$ define the Holevo quantity

$$\chi(\{p_x, \rho_x\}) = S\left(\sum_x p_x \rho_x\right) - \sum_x p_x S(\rho_x), \quad (12.7)$$

where $S(\rho) = -\text{Tr}(\rho \log \rho)$ denotes the von Neumann entropy.

The accessible information of the ensemble is defined as

$$I_{\text{acc}}(\{p_x, \rho_x\}) = \sup_{\{M_y\}} I(X : Y), \quad (12.8)$$

where the supremum is taken over all POVMs $\{M_y\}$ and $I(X : Y)$ denotes the classical mutual information between preparation label X and measurement outcome Y [6].

Observables and Incompatibility

A system observable X is represented by a POVM $\{N_x\}$ on $\mathcal{H}_S$.

Two POVMs $X = \{N_x\}$ and $Z = \{M_z\}$ are incompatible if there exists no joint POVM $\{G_{xz}\}$ satisfying

$$\sum_z G_{xz} = N_x, \quad \sum_x G_{xz} = M_z.$$

Non-commuting projective measurements provide a sufficient condition for incompatibility.

Measurement Instruments

Given a POVM $X = \{N_x\}$, a measurement instrument is a collection of completely positive, trace-nonincreasing maps $\{\mathcal{I}_x\}$ such that the aggregate map

$$\mathcal{I} = \sum_x \mathcal{I}_x$$

is completely positive and trace-preserving (CPTP), and

$$\text{Tr}[\mathcal{I}_x(\rho)] = \text{Tr}(N_x \rho)$$

for all density operators ρ .

Each $\mathcal{I}_x$ returns an unnormalized post-measurement state.

For projective measurements $N_x = P_x$ with

$$P_x^2 = P_x = P_x^\dagger, \quad \sum_x P_x = I,$$

we adopt the Lüders instrument:

$$\mathcal{I}_x(\rho) = P_x \rho P_x.$$

In §§12.3–14.4 we restrict attention to rank-1 PVMs and fix the Lüders measurement instrument throughout. Under this standing Lüders convention, the Lüders-conditioned cq state is uniquely determined by $\rho_{SE}(t)$ and the specified PVM. Alternative instruments realizing the same PVM are not considered here.

12.3 Operational Definitions: Records, Redundancy, and Structural Admissibility

This subsection fixes the operational predicates that carry the remainder of the argument: what it means for a fragment to contain a high-quality record of a system PVM, how redundancy is counted across fragments, and the disjoint simultaneous redundancy predicate $\text{DSR}_\delta(X, Z; t)$ for two observables. It also defines the system-level classical compatibility set $\mathcal{C}_{XZ}$ and concludes by stating the Structural Admissibility Hypothesis in time-local form.

We restrict to rank-1 PVMs $X = \{|x\rangle\langle x|\}_x$ and $Z = \{|z\rangle\langle z|\}_z$ interpreted via the Lüders instrument. Under this convention, the induced classical–quantum state, and hence the fragment ensembles and I_{acc} -based record predicates, is uniquely determined by $\rho_{SE}(t)$ and the specified PVM. The redundancy notions below are therefore state-level predicates determined by $\rho_{SE}(t)$ and the chosen observables.

Accessible record predicates of this type are related in spirit to environment-as-witness and quantum Darwinism analyses of redundant classical records in environmental fragments [2, 3, 4]. The present formulation differs in aim: rather than deriving objectivity from dynamical decoherence models, it isolates structural admissibility conditions under which certified redundancy predicates can be formulated and bounded.

Standing Assumptions

All Hilbert spaces in this chapter are finite-dimensional.

We assume a fixed tensor-product decomposition

$$\mathcal{H} = \mathcal{H}_S \otimes \bigotimes_{k=1}^N \mathcal{H}_{E_k}$$

which is treated as laboratory-defined modeling input and is held fixed throughout the analysis. Alternative regroupings or changes in granularity may alter redundancy counts and are not considered here.

Global evolution is unitary under a Hamiltonian of the form

$$H = H_S + H_E + \sum_{k=1}^N H_{SE_k},$$

where each H_{SE_k} acts nontrivially only on $\mathcal{H}_S \otimes \mathcal{H}_{E_k}$.

Initial states are restricted to the product manifold

$$\mathcal{M}_{\text{prod}} = \left\{ \rho_S \otimes \bigotimes_{k=1}^N \rho_{E_k} \right\}.$$

This product-state restriction is imposed solely for the parameter-space and genericity program (tested preparations within $\mathcal{P}_\Lambda$). It does not constrain the kinematic or time-local definitions at fixed time t .

Genericity statements are taken with respect to the relative topology on

$$\mathcal{P} = \mathcal{H}_{\text{int}} \times \mathcal{M}_{\text{prod}},$$

where $\mathcal{H}_{\text{int}}$ denotes a fixed finite-dimensional parameterization of the interaction coefficients of the H_{SE_k} in a chosen operator basis (Euclidean topology), and $\mathcal{M}_{\text{prod}}$ is endowed with the trace-norm topology.

Remark: In finite dimension, the map $(H, \rho_{SE}(0)) \mapsto \rho_{SE}(t)$ is continuous in trace norm for fixed t . Predicates built from continuous quantities define Borel subsets of $\mathcal{P}$. For the I_{acc} -based predicates, see §13.1.

Projective Observables and Incompatibility

All observables on $\mathcal{H}_S$ are taken to be rank-1 PVMs $X = \{P_x\}$ and $Z = \{Q_z\}$. They are *compatible* iff all projectors commute:

$$[P_x, Q_z] = 0 \quad \forall x, z.$$

Equivalently, there exists a common refinement PVM $\{R_\alpha\}$ such that each P_x and Q_z is a sum of some subcollection of the R_α .

They are *incompatible* iff

$$\exists(x, z) \text{ such that } [P_x, Q_z] \neq 0.$$

η -Transverse PVMs

A pair of PVMs $X = \{P_x\}$ and $Z = \{Q_z\}$ on $\mathcal{H}_S$ is called η -*transverse* if

$$c := \max_{x,z} \|P_x Q_z\|_\infty^2, \quad c \leq 1 - \eta, \quad (12.9)$$

for some fixed $\eta \in (0, 1)$.

This is a quantitative incompatibility condition strictly stronger than mere noncommutation. Equivalently, one may define the unsquared overlap $\max_{x,z} \|P_x Q_z\|_\infty$. We adopt the squared convention throughout.

Remark: Quantitative refinements of incompatibility and approximate joint measurability have been extensively studied in the operational measurement literature [10, 11]. The present overlap-based parameter is adopted as a structural control variable governing admissible redundancy and coexistence within the DSR framework.

Conditional Fragment States (Lüders convention)

Fix a rank-1 PVM $X = \{P_x\}$ and time t , define

$$p_x(t) = \text{Tr}[(P_x \otimes I_E)\rho_{SE}(t)]. \quad (12.10)$$

For outcomes with $p_x(t) > 0$, define the conditional fragment state

$$\rho_{E_k|x}(t) = \frac{\text{Tr}_{S,E \setminus E_k}[(P_x \otimes I_E)\rho_{SE}(t)(P_x \otimes I_E)]}{p_x(t)}. \quad (12.11)$$

Outcomes with $p_x(t) = 0$ are retained in the outcome alphabet, with the convention $0 \log 0 := 0$ in entropy expressions. When $p_x(t) = 0$, the conditional state $\rho_{E_k|x}(t)$ may be chosen arbitrarily, since it only appears via the subnormalized operator $p_x(t)\rho_{E_k|x}(t) = 0$. Under the Lüders convention adopted here, this conditioning is canonical relative to that instrument, and the induced fragment ensembles are uniquely determined by the joint state and the specified PVM.

Classical X -Law Distance

Note: All states appearing in time-indexed quantities are understood as the evolved states $\rho_{SE}(t) = U_t \rho_{SE}(0) U_t^\dagger$ obtained from their respective initial conditions.

For two initial joint states ρ_{SE} and ρ'_{SE} , we define the classical X -law distance at time t by

$$\Delta_X(\rho_{SE}, \rho'_{SE}; t) := \frac{1}{2} \sum_{x \in \mathcal{X}} |p_x(t; \rho_{SE}) - p_x(t; \rho'_{SE})|,$$

where

$$p_x(t; \rho_{SE}) := [(P_x \otimes I_E)U_t \rho_{SE} U_t^\dagger].$$

This is the total variation distance between the induced X -outcome probability laws.

Entropy and Nondegeneracy

Shannon entropy is defined on the full outcome set:

$$H(X; t) = - \sum_x p_x(t) \log p_x(t).$$

We restrict attention to regimes satisfying

$$H(X; t) \geq h_{\min} > 0$$

to exclude the deterministic case in which record thresholds become trivial.

Accessible Information and Record Criterion

For fragment E_k , define the induced ensemble

$$\mathcal{E}_X^{(k)}(t) = \{(p_x(t), \rho_{E_k|x}(t))\}_x.$$

Define accessible information by

$$I_{\text{acc}}(X : E_k; t) = \sup_{\{M_y\}} I(X : Y),$$

where the supremum is over all POVMs $\{M_y\}$ on $\mathcal{H}_{E_k}$, and Y is the classical outcome of measuring $\{M_y\}$ on the ensemble $\mathcal{E}_X^{(k)}(t)$.

Fragment E_k contains a δ -record of X at time t if

$$I_{\text{acc}}(X : E_k; t) \geq (1 - \delta)H(X; t), \quad 0 < \delta < 1. \quad (12.12)$$

Define the fragment record set at state $\omega_{SE}(t)$ by

$$\omega_\delta(X; t) := \{k : I_{\text{acc}}(X : E_k; t)_\omega \geq (1 - \delta)H(X; t)_\omega\}. \quad (12.13)$$

Define the redundancy number at state ω by

$$R_\delta^\omega(X; t) := |\omega_\delta(X; t)|. \quad (12.14)$$

When the underlying state is clear from context we suppress the superscript ω .

Capacity Sanity Condition

To avoid trivial suppression due solely to fragment dimension, we interpret redundancy claims only in regimes where candidate record fragments satisfy

$$\log \dim \mathcal{H}_{E_k} \geq (1 - \delta)H(X; t),$$

and analogously for Z when both observables are considered.

12.3.1 The Disjoint Simultaneous Redundancy (DSR) Predicate

$\text{DSR}_\delta(X, Z; t)$

Let $R_\delta(X; t)$, $R_\delta(Z; t)$ and $\delta(X; t)_\delta(Z; t)$ be as defined above.

We define the disjoint simultaneous redundancy (DSR) predicate $\text{DSR}_\delta(X, Z; t)$ (with tolerance δ) by

$$\text{DSR}_\delta(X, Z; t) := \left(R_\delta(X; t) \geq 2 \wedge R_\delta(Z; t) \geq 2 \wedge \delta(X; t) \cap_\delta (Z; t) = \emptyset \right). \quad (12.15)$$

We use $\text{DSR}_\delta(X, Z; t)$ as the canonical disjoint simultaneous redundancy predicate throughout Sections 13–16.

Remark: Related structural characterizations of redundant classical record formation appear in the spectrum broadcast structure (SBS) framework, which identifies multipartite states admitting fragment-wise classical broadcasting forms [12, 13]. The present treatment differs in emphasis: rather than characterizing exact broadcast structure, it isolates quantitative admissibility and incompatibility constraints governing certified redundancy predicates.

Classical Compatibility Set

Define the compatibility (classical) set for X and Z as

$$\mathcal{C}_{XZ} = \{\sigma \in \mathcal{D}(\mathcal{H}_S) : [\sigma, P_x] = 0 \ \forall x, \ [\sigma, Q_z] = 0 \ \forall z\}. \quad (12.16)$$

Equivalently,

$$\mathcal{C}_{XZ} = \mathcal{D}(\mathcal{H}_S) \cap \{P_x\}' \cap \{Q_z\}'.$$

In finite dimension, this equals the commutant of the unital $*$ -algebra generated by $\{P_x, Q_z\}$. Moreover, $\mathcal{C}_{XZ}$ is closed (hence compact), such that

$$\text{dist}_{\text{tr}}(\rho_S, \mathcal{C}_{XZ}) = \min_{\sigma \in \mathcal{C}_{XZ}} \frac{1}{2} \|\rho_S - \sigma\|_1.$$

For commuting PVMs this coincides with states block-diagonal in the joint refinement algebra. For noncommuting PVMs it remains nonempty and contains at least the maximally mixed state. This notion of simultaneous classicality is algebraic: states in $\mathcal{C}_{XZ}$ commute with the full PVM algebras generated by $\{P_x\}$ and $\{Q_z\}$, not merely with their outcome distributions.

Structural Admissibility Hypothesis (Typed, Time-Local)

Fix t and parameters $(H, \rho_{SE}(0)) \in \mathcal{P}$.

For uniformly transverse PVMs X, Z , there exist constants $\epsilon > 0$ and $\delta \in (0, 1)$ such that, on an open dense subset of $\mathcal{P}$ (relative topology), $\text{DSR}_\delta(X, Z; t)$ is absent whenever

$$\text{dist}_{\text{tr}}(\rho_S(t), \mathcal{C}_{XZ}) > \epsilon. \tag{12.17}$$

This is stated as a structural hypothesis, not a theorem, and its scope is restricted to the dynamical and operational conventions fixed above.

13 Hypothesis Under Stress

Section 13 subjects the Structural Admissibility Hypothesis of §12.3 to systematic stress under quantitative incompatibility. The aim is not yet to prove a rigidity theorem, but to make precise what would count as a robust violation and to clarify which ingredients are genuinely dynamical or generic versus purely kinematic.

We proceed in three moves. First (§13.1), we harden the hypothesis by fixing an explicit parameter space, topology, and continuity assumptions, and we formulate a falsification criterion in terms of open neighborhoods. Second (§13.2), we present an adversarial engineering template isolating what would be required to realize $\text{DSR}_\delta(X, Z; t)$ for incompatible observables while keeping $\rho_S(t)$ at positive trace distance from $\mathcal{C}_{XZ}$. Third (§13.3), we drop genericity assumptions entirely and analyze $\text{DSR}_\delta(X, Z; t)$ as a fixed-state predicate, introducing a multiplicity-refined variant to track how obstruction strengthens with redundancy counts.

13.1 Hardened Structural Admissibility Hypothesis

Throughout this subsection we make the Structural Admissibility Hypothesis of §12.3 precise in the sense required for genericity and robustness claims. We therefore fix an explicit parameter space for fragment-local interaction architectures and product initializations, specify the relevant topology (Euclidean on coefficients, trace norm on states), and record the continuity properties needed to treat $\text{DSR}_\delta(X, Z; t)$ as a well-defined Borel predicate. All Hilbert spaces are finite-dimensional, and we fix η -transverse rank-1 PVMs X and Z as in §12.3.

13.1.1 Parameter Space and Dynamics

Fix a time $t > 0$ and a fragment decomposition

$$\mathcal{H}_E = \bigotimes_{k=1}^N \mathcal{H}_{E_k}.$$

Fix system and environment self-Hamiltonians H_S and H_E . These are held fixed throughout.

Fix an operator-basis architecture for fragment-local system–environment interactions:

$$H_{\text{int}}(g) = \sum_{k=1}^N \sum_{j=1}^{m_k} g_{j,k} A_{j,k} \otimes B_{j,k}, \quad g = \{g_{j,k}\} \in \mathbb{R}^m, \quad m = \sum_k m_k,$$

where each $B_{j,k}$ acts nontrivially on at most one fragment E_k and trivially on $E \setminus E_k$.

Fix a bounded coefficient domain

$$\mathcal{G}_\Lambda := \{g \in \mathbb{R}^m : \|g\|_2 \leq \Lambda\} \quad (\Lambda > 0).$$

Fix the manifold of product initial states

$$\mathcal{M}_{\text{prod}} = \left\{ \rho_{SE}(0) = \rho_S(0) \otimes \bigotimes_{k=1}^N \rho_{E_k}(0) \right\} \subset \mathcal{D}(\mathcal{H}_{SE}).$$

Define the parameter space

$$\mathcal{P}_\Lambda = \mathcal{G}_\Lambda \times \mathcal{M}_{\text{prod}},$$

equipped with the product topology (Euclidean on $\mathcal{G}_\Lambda$ and the relative trace-norm topology on $\mathcal{M}_{\text{prod}}$).

Dynamics and Reduced States

For a parameter point $(g, \rho_{SE}(0)) \in P_\Lambda$, let

$$H(g) := H_S + H_E + H_{\text{int}}(g), \quad U_g(t) := e^{-itH(g)}.$$

The joint state then evolves as

$$\rho_{SE}(t) := U_g(t)\rho_{SE}(0)U_g(t)^\dagger, \quad \rho_S(t) := \text{Tr}_E[\rho_{SE}(t)].$$

Classical Compatibility Set

Throughout this subsection, $\mathcal{C}_{XZ}$ denotes the classical compatibility set defined in Eq. (12.16).

Outcome Entropy at Time t

For a rank-1 PVM $X = \{P_x\}$ define

$$p_x(t) = \text{Tr}[P_x \rho_S(t)], \quad H(X; t) := - \sum_x p_x(t) \log p_x(t),$$

with the convention that outcomes with $p_x(t) = 0$ contribute 0 to all entropy and information expressions. In particular, when $p_x(t) = 0$, the corresponding conditional state $\rho_{E_k|x}(t)$ may be chosen arbitrarily on that null event, since its weighted contribution $p_x(t)\rho_{E_k|x}(t)$ vanishes identically.

13.1.2 Record Structure and Redundancy Geometry

Outcomes with $p_x(t) = 0$ are retained in the outcome alphabet. No support excision is performed.

For a rank-1 PVM $X = \{P_x\}$, define the Lüders-conditioned fragment states

$$\rho_{E_k|x}(t) = \frac{\text{Tr}_{S, E \setminus E_k}[(P_x \otimes I_E)\rho_{SE}(t)(P_x \otimes I_E)]}{p_x(t)}, \quad p_x(t) = \text{Tr}[(P_x \otimes I_E)\rho_{SE}(t)],$$

with the convention that outcomes with $p_x(t) = 0$ contribute 0 to all entropy and information expressions.

As in §12.3, we restrict to rank-1 PVMs and fix the Lüders measurement map throughout this analysis. Under this standing convention, the conditioning is canonical relative to that instrument, and the induced cq ensembles are uniquely determined by $\rho_{SE}(t)$ and the specified PVM.

Accessible Information (Single-Copy, Single-Fragment)

For an ensemble $\mathcal{E} = \{(p_x, \rho_x)\}_x$ on a finite-dimensional Hilbert space, define

$$I_{\text{acc}}(\mathcal{E}) := \sup_{\{M_y\}} I(X : Y), \quad Y := \text{POVM outcome obtained from measuring } \{M_y\}.$$

In finite dimension and for a fixed finite alphabet X , the classical mutual information $I(X:Y)$ is continuous in the joint distribution $p(x, y)$ with the convention $0 \log 0 := 0$.

For a fixed finite-outcome POVM $\{M_y\}$, the induced distribution

$$p(x, y) = p_x \text{Tr}[M_y \rho_x]$$

depends continuously on the ensemble data (p_x, ρ_x) in the trace-norm topology and on the POVM elements in the operator-norm topology.

Hence $I(X:Y)$ is jointly continuous in the ensemble and POVM data.

Accessible-Information Compactness and Continuity

Extremality and Carathéodory-type reduction arguments imply that the accessible-information optimization may be restricted to POVMs with a uniformly bounded finite number of outcomes. Consequently, the supremum in $I_{\text{acc}}(\mathcal{E})$ may be taken over such POVMs.

For each fixed outcome number, the set

$$\{(M_y)_y : M_y \geq 0, \sum_y M_y = I\}$$

is a closed and bounded subset of a finite-dimensional real vector space of Hermitian operators (equipped with the operator-norm topology), and is therefore compact.

Since only finitely many outcome numbers need be considered, the feasible set is a finite union of compact sets and is therefore compact. The supremum is attained. By Berge's Maximum Theorem, because the feasible set is compact and independent of the ensemble parameter and $I(X:Y)$ is jointly continuous, the value function $\text{acc}(\mathcal{E})$ is continuous (hence Borel measurable) on the state manifold. Additionally, $I_{\text{acc}}(\mathcal{E}) \leq \chi(\mathcal{E})$, and $\chi(\mathcal{E})$ is continuous by continuity of von Neumann entropy [14].

δ -Record and δ -Redundancy (Recall)

We use the δ -record fragment sets $\delta(X; t)$ and redundancy counts $R_\delta(X; t)$ as defined in Eqs. (12.13) and (12.14). No new definition is introduced here.

Nontriviality Regime

Fix constants

$$h_{\min} > 0, \quad 0 < \delta_{\max} < 1, \quad 0 < \epsilon_{\max} < \infty.$$

Only tolerance pairs $(\delta, \epsilon) \in (0, \delta_{\max}] \times (0, \epsilon_{\max}]$ are considered.

We refer to the disjoint simultaneous redundancy (DSR) predicate $\text{DSR}_\delta(X, Z; t)$ defined in Eq. (12.15).

13.1.3 Hardened Hypothesis and Falsification Criterion

Fix an η -transverse pair (X, Z) .

For every $(\delta, \epsilon) \in (0, \delta_{\max}] \times (0, \epsilon_{\max}]$, there exists an open dense subset

$$\mathcal{U}_{\delta, \epsilon}(t) \subseteq \mathcal{P}_\Lambda$$

such that for all $(g, \rho_{SE}(0)) \in \mathcal{U}_{\delta, \epsilon}(t)$,

$$\left(H(X; t) \geq h_{\min} \wedge H(Z; t) \geq h_{\min} \wedge \text{dist}_{\text{tr}}(\rho_S(t), \mathcal{C}_{XZ}) > \epsilon \right) \implies \neg \text{DSR}_{\delta}(X, Z; t). \quad (13.1)$$

Equivalently, on $\mathcal{U}_{\delta, \epsilon}(t)$ and away from the ϵ -neighborhood of $\mathcal{C}_{XZ}$ at time t , the predicate $\text{DSR}_{\delta}(X, Z; t)$ does not hold.

Falsification Criterion

The central claim is falsified if there exist $(\delta, \epsilon) \in (0, \delta_{\max}] \times (0, \epsilon_{\max}]$ and a nonempty open set $\mathcal{O} \subseteq \mathcal{P}_{\Lambda}$ such that for all $(g, \rho_{SE}(0)) \in \mathcal{O}$,

$$H(X; t) \geq h_{\min} \wedge H(Z; t) \geq h_{\min} \wedge \text{dist}_{\text{tr}}(\rho_S(t), \mathcal{C}_{XZ}) > \epsilon \wedge \text{DSR}_{\delta}(X, Z; t). \quad (13.2)$$

This falsification criterion is intentionally stronger than the literal logical negation of the central claim. It demands a robust open-neighborhood counterexample set.

13.2 Adversarial Engineering Scenario: Incompatible-Basis Attainability Test

This subsection presents an adversarial engineering scenario intended to probe the attainability of strong disjoint records for incompatible observables. It should be read as a structural design template clarifying what $\text{DSR}_{\delta}(X, Z; t)$ would require of the joint state $\rho_{SE}(t)$ and of the fragment-local architecture; it is not, by itself, a counterexample to the hardened hypothesis (Eq. (13.1)).

In Eq. (13.2) we formulated a robust falsification criterion: to refute the genericity claim one would need an open neighborhood in parameter space on which $\text{DSR}_{\delta}(X, Z; t)$ holds while $\rho_S(t)$ remains bounded away from the compatibility set $\mathcal{C}_{XZ}$. The present construction isolates the structural tension at the level of a single engineered instance, thereby indicating what an open-neighborhood violation would need to stabilize under perturbation.

Engineering Template

Let the environment factor as

$$E = E_X \otimes E_Z.$$

For disjoint index sets $K_X, K_Z \subset \{1, \dots, N\}$, define

$$E_X = \bigotimes_{k \in K_X} E_k, \quad E_Z = \bigotimes_{k \in K_Z} E_k.$$

Thus two disjoint fragment subcollections indexed by K_X and K_Z are selected within E in the adversarial scenario.

Suppose one engineers dynamics such that fragments within E_X acquire robust redundant records of the X -measurement outcomes, while fragments within E_Z redundantly encode outcomes of the incompatible observable Z .

If $\text{DSR}_{\delta}(X, Z; t)$ (Eq. (12.15)) were jointly realizable on a single fixed state $\rho_{SE}(t)$ while remaining bounded away from the classical compatibility set $\mathcal{C}_{XZ}$ (Eq. (12.16)), this would contradict the hardened structural admissibility hypothesis. The present scenario isolates the structural tension

without asserting that such joint realizability is achieved in the sense of $\text{DSR}_\delta(X, Z; t)$ as defined in Eq. (12.15).

Violation of the hardened structural admissibility hypothesis would require the existence of a distance threshold $\epsilon > 0$ and a time t in the redundancy window such that

$$\text{dist}_{\text{tr}}(\rho_S(t), \mathcal{C}_{XZ}) > \epsilon,$$

while the predicate $\text{DSR}_\delta(X, Z; t)$ (Eq. (12.15)) holds.

For transverse rank-1 PVMs whose generated $\ast$ -algebra acts irreducibly on $\mathcal{H}_S$ (i.e., the joint commutant reduces to $\mathbb{C}I$), the compatibility set reduces to

$$\mathcal{C}_{XZ} = \{I/d\}.$$

In this case, maintaining $\text{DSR}_\delta(X, Z; t)$ while keeping $\rho_S(t)$ nonclassical requires sustaining a state bounded away from maximal mixing.

Thus the stress test reduces to a single structural question:

Can one export near-complete classical information about two incompatible PVMs into disjoint environmental sectors without driving $\rho_S(t)$ toward the joint compatibility set $\mathcal{C}_{XZ}$?

13.3 Disjoint Redundancy for Incompatible Observables

We now switch from the parameter-space and genericity framing associated with the DSR predicate (Eq. (12.15)) and the channel-based rigidity expectation (Eq. (14.3)) to a purely kinematic analysis. Throughout this subsection we fix a single joint state $\rho_{SE}(t)$ and study the predicate $\text{DSR}_\delta(X, Z; t)$ without invoking product-manifold assumptions, openness claims, or dynamical typicality statements.

The aim is to determine what structural constraints arise at the level of a fixed state when X and Z are quantitatively incompatible. In addition to the baseline predicate $\text{DSR}_\delta(X, Z; t)$ from Eq. (12.15), we introduce a multiplicity-refined variant $\text{DSR}_\delta^{(r_X, r_Z)}(X, Z; t)$ to track how obstruction strengthens as redundancy counts increase.

13.3.1 Fragment-Level Redundancy

Let S be a finite-dimensional quantum system with Hilbert space $\mathcal{H}_S$, and let the environment decompose as

$$\mathcal{H}_E = \bigotimes_{k=1}^N \mathcal{H}_{E_k}.$$

At a fixed time t , consider a joint state $\rho_{SE}(t)$.

All measurement statements below are purely analytic. We apply the Lüders measurement map associated with X to define induced classical–quantum (cq) states and associated information quantities. No physical intervention on $\rho_{SE}(t)$ is assumed; the construction refers only to the action of the measurement channel in analysis. Because we fix the Lüders measurement map for all PVMs in this analysis, the induced cq state is uniquely determined by $\rho_{SE}(t)$ and X . The conditioning used below is therefore canonical relative to this standing convention.

Let $X = \{P_x\}$ be a rank-1 PVM on $\mathcal{H}_S$, define

$$p_x(t) = \text{Tr}[(P_x \otimes I_E)\rho_{SE}(t)],$$

and, for $p_x(t) > 0$, the conditional environment state

$$\rho_{E|x}(t) = \frac{\text{Tr}_S[(P_x \otimes I_E)\rho_{SE}(t)(P_x \otimes I_E)]}{p_x(t)}.$$

For fragment E_k ,

$$\rho_{E_k|x}(t) = \text{Tr}_{E \setminus E_k} \rho_{E|x}(t).$$

The Shannon entropy of X at time t is

$$H(X; t) = - \sum_x p_x(t) \log p_x(t).$$

We restrict attention to a nontrivial outcome regime to avoid vacuous “records” arising from near-deterministic observables. This restriction depends on the chosen PVM, including any coarse-graining.

We assume

$$H(X; t) \geq h_{\min} > 0, \quad H(Z; t) \geq h_{\min} > 0,$$

for any observables X, Z considered below.

Accessible Information and Records

For fragment E_k , define the ensemble

$$\mathcal{E}_X^{(k)}(t) = \{(p_x(t), \rho_{E_k|x}(t))\}_x,$$

and the associated cq state

$$\rho_{XE_k}(t) = \sum_x p_x(t) |x\rangle\langle x| \otimes \rho_{E_k|x}(t).$$

Define the accessible information

$$I_{\text{acc}}(X : E_k; t) = \sup_{\{M_y\}} I(X : Y),$$

where the supremum is over POVMs $\{M_y\}$ on $\mathcal{H}_{E_k}$ applied to $\rho_{XE_k}(t)$. This captures the existence of a decoding measurement on E_k that reveals most of the classical variable X .

We will frequently upper-bound I_{acc} using the Holevo quantity (Eq. (12.7))

$$\chi(X : E_k; t) = S\left(\sum_x p_x(t) \rho_{E_k|x}(t)\right) - \sum_x p_x(t) S(\rho_{E_k|x}(t)),$$

which satisfies

$$I_{\text{acc}}(X : E_k; t) \leq \chi(X : E_k; t). \quad (13.3)$$

Definition 13.1: Single-fragment δ -record

Fragment E_k contains a δ -record of X at time t if

$$I_{\text{acc}}(X : E_k; t) \geq (1 - \delta)H(X; t), \quad 0 < \delta < 1,$$

cf. the record condition in Eq. (12.12).

Define

$$\delta(X; t) = \{k : I_{\text{acc}}(X : E_k; t) \geq (1 - \delta)H(X; t)\}, \quad R_\delta(X; t) = |\delta(X; t)|,$$

cf. Eqs. (12.13) and (12.14).

14 Entropic Tradeoff and Geometric Rigidity

Section 14 converts the qualitative obstruction of §13 into a sharper map of what “two incompatible records” can and cannot look like. We first prove a quantitative, state-level incompatibility constraint using an entropic uncertainty relation with side information, which rules out simultaneous *high predictability* of incompatible observables from disjoint environment blocks. We then present a channel-level picture motivating Hypothesis 14.1, explaining why strong fragment records should behave like approximate measurement channels and why disjoint simultaneous redundancy (DSR) is therefore expected to force the induced system state $\rho_S(t)$ close to the classical compatibility set $\mathcal{C}_{XZ}$.

Because that geometric step is not derived here, we state it explicitly as *Hypothesis 14.1 (Geometric Rigidity)* and provide a representative amplification mechanism showing why robust counterexamples are structurally difficult within fragment-local, product-initial architectures. We close by distinguishing clearly between what is proved and what is hypothesized, and by discussing where in parameter space the DSR antecedent is expected to be rare versus engineered.

14.1 Entropic Obstruction via Side Information

This subsection provides the theorem-level anchor for Section 14. We temporarily lift the rank-1 restriction imposed in §§12.3–13.3 (which was required for instrument-independence of the record predicate), because the present obstruction is purely entropic and holds for arbitrary PVMs. Our objective is to formalize a quantitative incompatibility constraint: if one disjoint environment block predicts X with high accuracy, then even the entire purifying complement cannot simultaneously render Z predictable when X and Z are incompatible. To do so, we embed the dynamics into a purified tripartite setting and apply an entropic uncertainty relation with quantum side information, yielding a bound controlled only by the overlap parameter c .

Purified Setup

Let $\rho_{SE_X E_Z}$ be an arbitrary (possibly mixed) state. Introduce purifying registers $E_R R$ such that the global state

$$\rho_{SE_X E_Z E_R R}$$

is pure.

Define

$$F := E_Z E_R R.$$

Here $\rho_{SE_X F}$ denotes the same global pure state with registers regrouped; no subsystem is traced out in forming F . By construction, the reduced state $\rho_{SE_X F}$ is pure, and F therefore constitutes the full purifying complement of SE_X .

Let $\{P_x\}_x$ and $\{Q_z\}_z$ be projective measurements (PVMs) on S , corresponding to observables X and Z . For PVMs, define the overlap constant (cf. Eq. (12.9))

$$c := \max_{x,z} \|P_x Q_z\|_\infty^2, \quad 0 < c \leq 1.$$

Measurement Incompatibility Parameter

The parameter c quantifies measurement incompatibility through maximal operator overlap. It is state-independent and determined entirely by the PVM structure. In the commuting case $c = 1$, the uncertainty bound becomes trivial. We restrict attention to the incompatible regime $c < 1$.

Define the post-measurement classical–quantum (cq) states by discarding the measured system S :

$$\begin{aligned}\rho_{XE_X} &= \sum_x |x\rangle\langle x| \otimes \text{Tr}_{SF}[(P_x \otimes I_{E_X F})\rho_{SE_X F}(P_x \otimes I_{E_X F})], \\ \rho_{ZF} &= \sum_z |z\rangle\langle z| \otimes \text{Tr}_{SE_X}[(Q_z \otimes I_{E_X F})\rho_{SE_X F}(Q_z \otimes I_{E_X F})].\end{aligned}$$

Each of these cq states is normalized by construction, since $\sum_x P_x = \sum_z Q_z = I_S$. Indeed, summing over outcomes and using the completeness relations recovers $\text{Tr}_{SE_X F}[\rho_{SE_X F}] = 1$.

We use smooth min- and max-entropies defined with respect to purified distance. For a state ρ_{AB} and smoothing parameter $\varepsilon_{\text{sm}} \geq 0$,

$$H_{\min}^{\varepsilon_{\text{sm}}}(A|B)_\rho = \sup_{\tilde{\rho}: P(\tilde{\rho}, \rho) \leq \varepsilon_{\text{sm}}} H_{\min}(A|B)_{\tilde{\rho}},$$

and similarly for $H_{\max}^{\varepsilon_{\text{sm}}}$.

Tripartite Smooth Entropic Uncertainty

Redundant encodability of X in fragment families yields ε_{sm} -smooth predictability from E_X and thus small $H_{\min}^{\varepsilon_{\text{sm}}}(X|E_X)$. To formalize the resulting constraint on incompatible observables, we invoke the smooth min-/max-entropy form of the tripartite entropic uncertainty relation with quantum side information (equal smoothing parameter) as established in [9]; see also [15, Thm. 7.6]. All logarithms are base 2, and we invoke the base-2 formulation of the entropic uncertainty relation so that the constant $\log(1/c)$ appears without additional conversion factors.

In purified-distance form and equal smoothing, for any $\varepsilon_{\text{sm}} \geq 0$,

$$H_{\min}^{\varepsilon_{\text{sm}}}(X|E_X) + H_{\max}^{\varepsilon_{\text{sm}}}(Z|F) \geq \log(1/c). \quad (14.1)$$

For classical–quantum states,

$$H_{\min}^{\varepsilon_{\text{sm}}}(X|E_X) = -\log P_{\text{guess}}^{\varepsilon_{\text{sm}}}(X|E_X),$$

where $P_{\text{guess}}^{\varepsilon_{\text{sm}}}(X|E_X)$ denotes the smoothed optimal guessing probability.

Predictability Assumption

Assume X is ε_{sm} -smoothly predictable from E_X :

$$P_{\text{guess}}^{\varepsilon_{\text{sm}}}(X|E_X) \geq 1 - \delta_X.$$

Then

$$H_{\min}^{\varepsilon_{\text{sm}}}(X|E_X) \leq -\log(1 - \delta_X).$$

Substituting into (14.1) yields

$$H_{\max}^{\varepsilon_{\text{sm}}}(Z|F) \geq \log\left(\frac{1 - \delta_X}{c}\right). \quad (14.2)$$

Remark: The lower bound is strictly positive only when $1 - \delta_X > c$. If $1 - \delta_X \leq c$, the right-hand side may be nonpositive. This does not invalidate the inequality, since conditional max-entropy can be negative in quantum settings; it simply indicates that the obstruction becomes nontrivial only in the regime $1 - \delta_X > c$.

Structural Consequence

Even when conditioning on the full purifying complement $F = E_Z E_R R$, near-perfect predictability of X from E_X forces the Z outcomes to retain substantial smooth conditional max-entropy. For classical-quantum states, $H_{\max}^{\varepsilon_{\text{sm}}}(Z|F)$ admits an operational interpretation in one-shot compression with quantum side information. The bound therefore shows that, even granting maximal side information compatible with global purity, the Z outcomes retain intrinsic conditional description length determined solely by the measurement incompatibility parameter c .

14.2 Channel Picture Motivating Hypothesis 14.1 (Heuristic)

Whereas §14.1 established an entropic incompatibility constraint using uncertainty relations with side information, we now analyze the problem at the level of the induced system-to-fragments channel. The focus shifts from entropy bounds to the operational structure required for strong fragment records to behave like effective measurement channels. This perspective clarifies the structural tension between disjoint sharp recording of incompatible PVMs and stability of the induced system state, and motivates the geometric rigidity hypothesis.

14.2.1 Operational Channel Hypotheses

Let $\Phi_t : \mathcal{D}(\mathcal{H}_S) \rightarrow \mathcal{D}\left(\bigotimes_{k=1}^N \mathcal{H}_{E_k}\right)$ denote the induced system-to-environment channel at the fixed time parameter t .

For $J \subseteq \{1, \dots, N\}$, define its complement

$$\bar{J} := \{1, \dots, N\} \setminus J.$$

Write

$$\Phi_t^{(J)} :=_{E_{\bar{J}}} \Phi_t$$

for the reduced channel to the fragment set J .

Operational Hypotheses (Programmatic)

(B1) Uniform Decoding of an X -Classical Ensemble by a Fixed POVM:

Fix a PVM $X = \{P_x\}$ on S and an associated X -classical preparation ensemble $\{\rho_x\}_x$ (e.g. $\rho_x = P_x/P_x$, or more generally any family of states commuting with all P_x). For each $k \in K_X$, there exists a single POVM $\{M_x^{(k)}\}_x$ on $\mathcal{H}_{E_k}$ (independent of x and independent of the particular ρ_x within the specified family) such that

$$\Pr[\hat{x} = x \mid \rho_x] := [M_x^{(k)} \Phi_t^{(k)}(\rho_x)] \geq 1 - \varepsilon \quad \text{uniformly for all } x.$$

Thus each fragment in K_X admits a fixed measurement that decodes the classical X -label with uniformly high success over the entire X -classical ensemble. (Analogously, fix $Z = \{Q_z\}$ and a Z -classical ensemble $\{\eta_z\}_z$, and impose the analogous condition for each $k \in K_Z$.)

(B2) Approximate Channel Factorization Across Fragment Sets (Diamond-Norm Control):

We measure channel distance in the diamond norm [16]

$$\|\Psi\|_\diamond := \sup_{\rho_{SA}} \|(\Psi \otimes \text{id}_A)(\rho_{SA})\|_1,$$

where the supremum ranges over all density operators on $\mathcal{H}_S \otimes \mathcal{H}_A$ with $\dim \mathcal{H}_A = \dim \mathcal{H}_S$.

Assume the reduced channels to the recorded fragment sets approximately factorize at the channel level:

$$\left\| \Phi_t^{(K_X)} - \bigotimes_{k \in K_X} \Phi_t^{(k)} \right\|_\diamond \leq \delta, \quad \left\| \Phi_t^{(K_Z)} - \bigotimes_{k \in K_Z} \Phi_t^{(k)} \right\|_\diamond \leq \delta.$$

Thus correlations between fragments within each recorded block are perturbatively small in the completely bounded sense.

Heuristic Channel-Stability Step

Assumption (B1) asserts uniform high-success decoding of a classical ensemble by a single fixed POVM per fragment. A natural strengthening, requiring an explicit channel-stability lemma, is that such uniform decoding forces $\Phi_t^{(k)}$ to be close, in diamond norm and under suitable uniformity conditions, to an entanglement-breaking measure-and-prepare channel of the form

$$\mathcal{M}_X^{(k)}(\rho) = \sum_x (P_x \rho) \sigma_x^{(k)},$$

for some output states $\{\sigma_x^{(k)}\}_x$ on $\mathcal{H}_{E_k}$.

Informally, each fragment behaves approximately as an implementation of an X -measurement channel. An analogous statement would hold for Z on K_Z .

Approximate Two-Output Broadcasting Configuration

If both K_X and K_Z satisfy these properties on disjoint outputs, then the overall channel

$$\Phi_t : S \rightarrow E_{K_X} \otimes E_{K_Z}$$

simultaneously approximates

- an X -measurement channel upon tracing out E_{K_Z} , and
- a Z -measurement channel upon tracing out E_{K_X} .

This constitutes a two-output broadcasting configuration for typically incompatible observables X and Z .

Structural Expectation

By analogy with no-broadcasting and joint-measurability constraints, such simultaneous approximation to incompatible measurement channels on disjoint outputs is expected to be quantitatively unstable. In particular, the channel picture motivates the expectation that two-output broadcasting behavior can persist only when the realized system state $\rho_S(t)$ lies close to the classical compatibility set $\mathcal{C}_{XZ}$. Throughout this subsection, $\mathcal{C}_{XZ}$ denotes the classical compatibility set (Eq. (12.16)). We quantify proximity using the trace norm $\|A\|_1$ and the induced trace distance $\text{dist}_{\text{tr}}(\rho, \sigma)$ together with the associated distance-to-set $\text{dist}_{\text{tr}}(\rho, \mathcal{C})$ (Eqs. (12.4) and (12.5)).

The structural expectation is that, whenever Φ_t yields simultaneous high-quality approximations to sharp X - and Z -measurement channels on disjoint outputs for the realized input $\rho_S(t)$, there exists a function $f(\varepsilon, \delta)$, vanishing as the channel-approximation errors vanish, such that

$$\text{dist}_{\text{tr}}(\rho_S(t), \mathcal{C}_{XZ}) \leq f(\varepsilon, \delta). \quad (14.3)$$

Equivalently, under the operational hypotheses above, the realized system state is expected to lie close in trace distance to the set of states simultaneously classical with respect to both observables. For generic irreducible incompatible pairs (e.g. nondegenerate mutually unbiased PVMs), the set $\mathcal{C}_{XZ}$ may be very small and may reduce to the maximally mixed state. In such cases this compatibility regime represents a strong structural restriction.

Status

The channel-structural motivation presented here is heuristic in its present form. Turning it into a theorem requires:

- (i) A channel-stability lemma upgrading uniform high-success decoding on a classical ensemble with a fixed POVM to diamond-norm closeness to an entanglement-breaking measurement channel.
- (ii) An incompatibility-to-instability bound showing that a single channel cannot simultaneously approximate sharp measurement channels for incompatible PVMs on disjoint outputs beyond quantitative limits governed by joint-measurability or no-broadcasting constraints; in particular, such simultaneous approximation must force $\rho_S(t)$ close in trace distance to $\mathcal{C}_{XZ}$.

14.3 Disjoint Predictability Tradeoff and a Geometric Rigidity Hypothesis

We state and prove a quantitative disjoint-predictability tradeoff derived from the entropic uncertainty framework. High predictability of X from one environment block forces large conditional uncertainty for Z relative to its complement whenever X and Z are incompatible ($c < 1$). The result is formulated in min-/max-entropy terms and therefore constrains one-shot predictability. Because $\text{DSR}_\delta(X, Z; t)$ is defined via accessible information rather than entropic quantities, an additional system-level step is required to connect disjoint redundancy to geometric restriction of $\rho_S(t)$. We formulate that step as Hypothesis 14.1 (Geometric Rigidity)

Standing Assumptions

Fix $t > 0$ and finite-dimensional Hilbert spaces $\mathcal{H}_S$ and $\{\mathcal{H}_{E_k}\}_{k=1}^N$ with fixed fragment decomposition. Fix rank-1 PVMs $X = \{P_x\}$ and $Z = \{Q_z\}$ on $\mathcal{H}_S$ with incompatibility parameter

$$c := \max_{x,z} \|P_x Q_z\|_\infty^2 < 1.$$

We write

$$H(X; t) := H(X; \rho_S(t)), \quad H(Z; t) := H(Z; \rho_S(t)).$$

We restrict attention to parameter points satisfying

$$H(X; t) > 0 \quad \text{and} \quad H(Z; t) > 0.$$

All entropies in this subsection are the conditional *non-smooth* entropies evaluated at time t . We invoke the non-smooth ($\varepsilon = 0$) form of the tripartite entropic uncertainty relation with quantum side information as stated in [15, Thm. 7.6].

Theorem 14.1: Disjoint Predictability Tradeoff (EUR Consequence) [9, 15]

Fix $t > 0$ and rank-1 PVMs $X = \{P_x\}$ and $Z = \{Q_z\}$ on $\mathcal{H}_S$ with incompatibility parameter

$$c := \max_{x,z} \|P_x Q_z\|_\infty^2 < 1 \quad (\text{cf. Eq. (12.9)}).$$

Let $\rho_{SE}(t)$ be any joint state on $\mathcal{H}_S \otimes \bigotimes_{k=1}^N \mathcal{H}_{E_k}$.

For any index set $K_X \subseteq \{1, \dots, N\}$ write

$$E_{K_X} = \bigotimes_{k \in K_X} E_k, \quad E_{\overline{K_X}} = \bigotimes_{k \notin K_X} E_k.$$

If

$$P_{\text{guess}}(X|E_{K_X}; t) \geq 1 - \alpha,$$

equivalently

$$H_{\min}(X|E_{K_X}; t) \leq -\log(1 - \alpha),$$

then

$$H_{\max}(Z|E_{\overline{K_X}}; t) \geq \log \frac{1}{c} - H_{\min}(X|E_{K_X}; t).$$

Proof of Theorem 14.1

Purify $\rho_{SE}(t)$ by adjoining a reference system R so that $\rho_{SER}(t)$ is pure, and apply the tripartite entropic uncertainty relation to the decomposition

$$S : E_{K_X} : E_{\overline{K_X}} R.$$

Taking the $\varepsilon = 0$ specialization of Eq. (14.1) yields

$$H_{\min}(X|E_{K_X}; t) + H_{\max}(Z|E_{\overline{K_X}}; t) \geq \log \frac{1}{c}.$$

Now let $K_Z \subseteq \overline{K_X}$. Then the subsystem E_{K_Z} is obtained from $E_{\overline{K_X}}$ by the CPTP map

$$\text{Tr}_{E_{\overline{K_X}} \setminus E_{K_Z}}.$$

Monotonicity of conditional max-entropy under CPTP maps on the conditioning system (see [15, Thm. 6.19]) implies

$$H_{\max}(Z|E_{\overline{K_X}};t) \leq H_{\max}(Z|E_{K_Z};t).$$

Combining the two inequalities gives

$$H_{\max}(Z|E_{K_Z};t) \geq \log \frac{1}{c} - H_{\min}(X|E_{K_X};t). \quad (14.4)$$

□

Consequence

If α is sufficiently small relative to c , so that $H_{\min}(X|E_{K_X};t)$ is small, then the tradeoff forces $H_{\max}(Z|E_{\overline{K_X}};t)$ to be large. This consequence is purely entropic. It does not assert that any specific disjoint block enables high predictability for Z . It establishes only that simultaneous high predictability for incompatible observables from disjoint environment blocks is constrained.

Predictability vs. Redundancy

The tradeoff theorem concerns simultaneous high predictability expressed in terms of conditional min- and max-entropies. This notion is weaker than the operational definition of disjoint simultaneous redundancy $\text{DSR}_{\delta}(X, Z; t)$ given in §12.3. DSR additionally requires accessible-information thresholds, multiplicity conditions, and fragment-set disjointness. Relating $\text{DSR}_{\delta}(X, Z; t)$ to the present tradeoff requires a bridge from accessible-information record conditions to guessing-probability or min-entropy bounds. No such bridge is assumed here.

Hypothesis 14.1: Geometric Rigidity Hypothesis (System-Level Form)

We assume that within the admissible parameter space $\mathcal{P}_{\Lambda}$ defined in §13.1, restricted to the nondegenerate subset specified above, whenever $\text{DSR}_{\delta}(X, Z; t)$ (Eq. (12.15)) is satisfied for two incompatible observables under admissible coupling, the induced system state $\rho_S(t)$ lies within a controlled trace-distance neighborhood of the compatibility set $\mathcal{C}_{XZ}$ (Eq. (12.16)).

That is,

$$\rho_S(t) \in \mathcal{N}_{\epsilon}(\mathcal{C}_{XZ}) \quad (14.5)$$

for some $\epsilon = \epsilon(\delta, \Lambda, c)$.

Optional Variant (Quantitative Rigidity)

In addition to Hypothesis 14.1, one may ask whether the proximity parameter ϵ can be chosen to obey an explicit scaling law in the redundancy multiplicities (for example, exponential decay in $|K_X|$ and $|K_Z|$ for fixed single-fragment record quality). Section §14.4 provides a representative mechanism supporting this expectation.

System Compatibility Neighborhood

Define

$$\mathcal{N}_{\epsilon}(\mathcal{C}_{XZ}) = \left\{ \sigma_S \in \mathcal{D}(\mathcal{H}_S) : \inf_{\omega_S \in \mathcal{C}_{XZ}} \frac{1}{2} \|\sigma_S - \omega_S\|_1 \leq \epsilon \right\}.$$

All neighborhoods are taken with respect to the trace-norm topology on density operators.

14.4 Hard Test: Adversarial Counterexample Template and Decoherence—Amplification Mechanism

Hypothesis 14.1 is intended as a *generic* geometric rigidity claim: within an open dense (or full-measure) subset of the admissible parameter space $\mathcal{P}_\Lambda$ of §13.1, strict disjoint simultaneous redundancy (DSR) for two incompatible rank-1 PVMs forces the induced system state $\rho_S(t)$ into a controlled trace-distance neighborhood of the compatibility set $\mathcal{C}_{XZ}$ (Eq. (12.16)).

This subsection isolates (a) what a robust counterexample would have to look like inside $\mathcal{P}_\Lambda$, and (b) the principal structural obstruction encountered under the standing assumptions of §§12–14 (finite dimension, fragment-local interactions, product initial states, fixed Luders conditioning, and nondegenerate outcome entropies).

14.4.1 Adversarial Construction Template

To falsify Hypothesis 14.1 *as a generic statement* within $\mathcal{P}_\Lambda$, one would need to exhibit constants $\delta \in (0, 1)$ and $\epsilon_0 > 0$ and a nonempty open set (or full-measure subset) $\mathcal{O} \subseteq \mathcal{P}_\Lambda$ such that for every $(g, \rho_{SE}(0)) \in \mathcal{O}$, at the fixed time t :

- (i) there exists a large fragment set K_X for which many *individual* fragments satisfy the strict single-fragment record condition for X :

$$I_{\text{acc}}(X : E_k; t) \geq (1 - \delta) H(X; t) \quad \text{for many } k \in K_X;$$

- (ii) there exists a disjoint fragment set K_Z with $K_X \cap K_Z = \emptyset$ such that many individual fragments satisfy

$$I_{\text{acc}}(Z : E_\ell; t) \geq (1 - \delta) H(Z; t) \quad \text{for many } \ell \in K_Z;$$

- (iii) the induced system state remains bounded away from $\mathcal{C}_{XZ}$:

$$\text{dist}_{\text{tr}}(\rho_S(t), \mathcal{C}_{XZ}) > \epsilon_0, \quad \text{with } \mathcal{C}_{XZ} \text{ as in (12.16).}$$

Any such open-neighborhood construction would realize strict DSR while keeping $\rho_S(t)$ macroscopically nonclassical relative to (X, Z) , and would therefore refute the intended geometric rigidity.

14.4.2 Decoherence: Amplification Mechanism (Representative Model)

A recurring structural feature of strong record formation is that high single-fragment record quality forces the *conditional fragment states* associated with distinct outcomes to become nearly perfectly distinguishable. Under fragment-local dynamics with product initial states, this distinguishability typically coincides with suppression of system coherences in the corresponding measurement basis.

We illustrate the mechanism in the qubit case. Let $S \simeq \mathbb{C}^2$, let $X = \{P_+, P_-\}$ and $Z = \{Q_0, Q_1\}$ be incompatible rank-1 PVMs (so $c < 1$), and assume product initialization $\rho_{SE}(0) = \rho_S(0) \otimes \bigotimes_k \rho_{E_k}(0)$.

In measurement-like (record-forming) regimes it is common, and analytically transparent, to approximate the interaction on the X -recording block K_X by an X -controlled unitary

$$U_X = \bigotimes_{k \in K_X} (P_+ \otimes V_+^{(k)} + P_- \otimes V_-^{(k)}). \quad (14.6)$$

We emphasize that (14.6) is *not* assumed as a global normal form on all of $\mathcal{P}_\Lambda$; it is a representative model that makes explicit the record—decoherence coupling anticipated in the channel picture of §14.2.

For a pure fragment input $\rho_{E_k}(0) = |\phi_k\rangle\langle\phi_k|$, define

$$\gamma_X^{(k)} := \langle\phi_k|(V_+^{(k)})^\dagger V_-^{(k)}|\phi_k\rangle, \quad |\gamma_X^{(k)}| \leq 1.$$

$$|\langle+|\rho_S(t)|-\rangle| = |\langle+|\rho_S(0)|-\rangle| \prod_{k \in K_X} |\gamma_X^{(k)}|. \quad (14.7)$$

14.4.3 Quantitative Scaling from Single-Fragment Record Quality (Qubit Case)

Assume $S \simeq \mathbb{C}^2$ and $X = \{P_+, P_-\}$, and consider the representative controlled-unitary record model (14.6) on a fragment block K_X with product fragment inputs. Assume for simplicity that the X -outcomes are equiprobable at time t , $p_+(t) = p_-(t) = \frac{1}{2}$, so that $H(X; t) = 1$.

Suppose that for each $k \in K_X$ the strict single-fragment record condition holds at level $\delta \in (0, 1)$ in the binary form

$$I_{\text{acc}}(X : E_k; t) \geq 1 - \delta.$$

Then, by definition of I_{acc} , there exists a POVM on E_k whose classical outcome $\hat{X}$ satisfies

$$I(X : \hat{X}) \geq 1 - \delta. \quad (14.8)$$

Let $p_{\text{err}}^{(k)} := \Pr[\hat{X} \neq X]$ denote the resulting (not necessarily optimal) decoding error probability. By binary Fano's inequality, (14.8) implies

$$h_2(p_{\text{err}}^{(k)}) \leq \delta, \quad \text{hence} \quad p_{\text{err}}^{(k)} \leq h_2^{-1}(\delta) =: p_\delta,$$

where h_2 is the binary entropy and $h_2^{-1} : [0, 1] \rightarrow [0, 1/2]$ denotes its inverse on $[0, 1/2]$.

Let $p_{\text{err, opt}}^{(k)}$ be the *optimal* (minimum) error probability for discriminating $\rho_{E_k|+}(t)$ from $\rho_{E_k|-}(t)$ with equal priors.

Since the above decoder is a feasible discriminator,

$$p_{\text{err, opt}}^{(k)} \leq p_{\text{err}}^{(k)} \leq p_\delta.$$

For binary discrimination with equal priors, the Helstrom formula gives

$$\frac{1}{2} \|\rho_{E_k|+}(t) - \rho_{E_k|-}(t)\|_1 = 1 - 2p_{\text{err, opt}}^{(k)} \geq 1 - 2p_\delta. \quad (14.9)$$

In the controlled-unitary model (14.6) the conditional fragment states are pure and satisfy $F(\rho_{E_k|+}(t), \rho_{E_k|-}(t)) = |\gamma_X^{(k)}|^2$.

By the Fuchs—van de Graaf inequality,

$$\frac{1}{2} \|\rho_{E_k|+}(t) - \rho_{E_k|-}(t)\|_1 \leq \sqrt{1 - |\gamma_X^{(k)}|^2}. \quad (14.10)$$

Combining (14.9) and (14.10) yields the quantitative overlap bound

$$|\gamma_X^{(k)}| \leq 2\sqrt{p_\delta(1-p_\delta)}, \quad p_\delta := h_2^{-1}(\delta) \in [0, 1/2]. \quad (14.11)$$

Substituting (14.11) into the overlap-multiplication formula (14.7) gives the exponential suppression estimate

$$|\langle +|\rho_S(t)|- \rangle| \leq |\langle +|\rho_S(0)|- \rangle| \left(2\sqrt{p_\delta(1-p_\delta)}\right)^{|K_X|}.$$

Equivalently, using the qubit identity $\text{dist}_{\text{tr}}(\rho_S(t), \Delta_X(\rho_S(t))) = |\langle +|\rho_S(t)|- \rangle|$,

$$\text{dist}_{\text{tr}}(\rho_S(t), \Delta_X(\rho_S(t))) \leq |\langle +|\rho_S(0)|- \rangle| \left(2\sqrt{p_\delta(1-p_\delta)}\right)^{|K_X|}.$$

The same reasoning applies to a disjoint Z -recording block K_Z under an analogous measurement-like approximation, yielding a corresponding bound with (Z, K_Z) in place of (X, K_X) .

Remark: These bounds exhibit a concrete scaling mechanism by which strict single-fragment redundancy, when replicated across many fragments in disjoint blocks for incompatible observables, generically drives $\rho_S(t)$ toward the joint commutant $\mathcal{C}_{XZ}$, thereby clarifying why open-neighborhood counterexamples to Hypothesis 14.1 are structurally difficult within $\mathcal{P}_\Lambda$.

From Record Quality to Small Overlap

Under (14.6), the conditional fragment states are pure:

$$\rho_{E_k|\pm}(t) = V_\pm^{(k)} |\phi_k\rangle\langle\phi_k| (V_\pm^{(k)})^\dagger, \quad F(\rho_{E_k|+}(t), \rho_{E_k|-}(t)) = |\gamma_X^{(k)}|^2.$$

By the Helstrom formula,

$$p_{\text{err}}^{(k)} = \frac{1}{2} \left(1 - \frac{1}{2} \|\rho_{E_k|+}(t) - \rho_{E_k|-}(t)\|_1\right),$$

and using $\|\rho - \sigma\|_1 \leq 2\sqrt{1 - F(\rho, \sigma)}$, hence

$$p_{\text{err}}^{(k)} \geq \frac{1}{2} \left(1 - \sqrt{1 - |\gamma_X^{(k)}|^2}\right).$$

Equation (14.7) therefore exhibits *exponential* suppression of X -basis coherences as the number of X -recording fragments grows.

Connection to the Trace-Distance Decoherence Criterion of §12.2 (Qubit Identity)

For a qubit, the X -dephased state is

$$\Delta_X(\rho_S(t)) := \sum_{s \in \{+, -\}} P_s \rho_S(t) P_s,$$

and

$$\text{dist}_{\text{tr}}(\rho_S(t), \Delta_X(\rho_S(t))) = |\langle +|\rho_S(t)|- \rangle|.$$

Thus $\rho_S(t)$ is ϵ -decohered in the X basis in the precise trace-distance sense of §12.2 (cf. (12.6)) whenever the off-diagonal entry is $\leq \epsilon$.

Connection to the Trace-Distance Decoherence Criterion of §12.2 (General d)

For a general rank-1 PVM $X = \{P_x\}_x$ on $d = \dim \mathcal{H}_S$, define the X -dephasing channel

$$\Delta_X(\rho) := \sum_x P_x \rho P_x.$$

Then

$$\text{dist}_{\text{tr}}(\rho, \Delta_X(\rho)) = \frac{1}{2} \|\rho - \Delta_X(\rho)\|_1 \leq \frac{1}{2} \sum_{x \neq x'} \|P_x \rho P_{x'}\|_1.$$

In particular, small off-diagonal blocks in the X basis (in trace norm) suffice to certify ϵ -decoherence in the sense of §12.2 (cf. (12.6)).

14.5 Structural Regimes and Typicality of DSR

The preceding subsections establish a quantitative predictability tradeoff and a representative amplification mechanism for incompatible observables via an entropic uncertainty relation with side information and isolates Hypothesis 14.1, which postulates a link between disjoint simultaneous redundancy (DSR) and proximity of $\rho_S(t)$ to the classical compatibility set $\mathcal{C}_{XZ}$.

We now address a logically distinct question: how large is the subset of parameter space in which the predicate $\text{DSR}_\delta(X, Z; t)$ is realized? This subsection concerns structural prevalence in parameter space and does not supply a derivation of Hypothesis 14.1.

How frequently should the predicate $\text{DSR}_\delta(X, Z; t)$ be realized?

Topological genericity and measure-typicality are distinct notions. Statements below refer explicitly to measure-theoretic typicality within coefficient space; no claim of openness or topological robustness is asserted unless explicitly stated.

Parameter Space and Measure

We assume fragment-local interactions of the form

$$H_{\text{int}}(g) = \sum_{j=1}^m g_j A_j \otimes B_j,$$

where each B_j acts nontrivially on at most one fragment $\mathcal{H}_{E_k}$.

Let

$$\mathcal{G}_\Lambda \subset \mathbb{R}^m$$

be a bounded coefficient domain equipped with the Euclidean topology.

Whenever we refer to “full measure,” we mean full measure with respect to probability measures absolutely continuous with respect to Lebesgue measure on $\mathbb{R}^m$. Typicality statements concern subsets of $\mathcal{G}_\Lambda$. Rigidity statements, by contrast, concern trace-norm neighborhoods in state space.

General Structural Mechanism

Disjoint simultaneous redundancy for two η -transverse PVMs requires that:

1. Distinct fragment subsets independently acquire high accessible information about X and Z , respectively;
2. These fragment records remain disjoint;
3. The induced joint state does not collapse into approximate joint classicality with respect to both observables.

For incompatible observables ($c < 1$), these requirements impose coordinated structural constraints on the interaction coefficients g . Heuristically, such coordination corresponds to alignment conditions among coupling directions that are not generically enforced in fragment-local architectures. The following dynamical classes clarify where such coordination is structurally suppressed or attainable.

Class I: Pointer-Selection (Pure Dephasing) Regimes

Suppose the interaction Hamiltonian is dominated by a single system observable A ,

$$H_{\text{int}} \approx \sum_{k=1}^N A \otimes B_k,$$

with each B_k acting on a distinct fragment.

Such dynamics generically induce decoherence in the eigenbasis of A , producing redundant fragment records of the associated PVM. For η -transverse PVM pairs (X, Z) not aligned with this pointer observable, disjoint simultaneous redundancy would require additional couplings approximately diagonal in a second incompatible basis. This introduces additional alignment constraints among the coefficients g .

Heuristically, these alignment conditions correspond to lower-dimensional structural constraints inside $\mathcal{G}_\Lambda$. Under mild regularity assumptions on the redundancy functionals (e.g. continuity in g of the induced reduced states and accessible-information functionals), the subset of coefficients producing $\text{DSR}_\delta(X, Z; t)$ is expected to be measure-suppressed within this dynamical class. No openness or codimension statement is asserted. In view of the quantitative amplification mechanism identified in §14.4, small misalignments between coupling directions and incompatible observables are multiplicatively amplified across fragment families, further sharpening the effective alignment requirements in this regime.

Class II: Random Fragment-Local Couplings

Let g be drawn from a probability measure on $\mathcal{G}_\Lambda$ that is absolutely continuous with respect to Lebesgue measure and imposes no special symmetry or commutation structure. In such cases, the induced system–environment channel typically selects an effective decoherence basis determined by dominant interaction directions. To obtain high-quality redundant encoding of two incompatible PVMs on disjoint fragment subsets, the coupling coefficients must satisfy correlated alignment conditions for two noncommuting operator families.

These alignment conditions represent nongeneric structural constraints in coefficient space. Accordingly, $\text{DSR}_\delta(X, Z; t)$ is heuristically expected to be measure-suppressed within $\mathcal{G}_\Lambda$ relative to smooth coefficient distributions. The exponential sensitivity to fragment multiplicity exhibited in §14.4 further suggests that generic coefficient draws amplify small incompatibilities, rendering simultaneous high-quality disjoint redundancy structurally unstable under smooth perturbations.

This statement concerns prevalence in the measure-theoretic sense only. It is not assumed as a formal premise in subsequent structural derivations, though it informs later interpretive and architectural discussions.

Class III: Engineered Multi-Channel Measurement Architectures

One may intentionally construct interactions in which distinct fragment subsets couple predominantly via operators approximately diagonal in X , while other subsets couple via operators approximately diagonal in Z . Sequential or Trotterized constructions can approximate two distinct measurement channels acting on S .

Such architectures demonstrate logical possibility: $\text{DSR}_\delta(X, Z; t)$ can be realized for carefully tuned parameters and initial states. However, their structural stability is delicate. Small perturbations in coupling strengths or timing generally degrade at least one of the redundant records unless quantitative margins are maintained. These constructions therefore occupy finely structured regions of parameter space rather than representing typical behavior.

Class IV: Nonlocal or Global Couplings

If interaction operators act nonlocally across multiple fragments, the fragment decomposition ceases to define dynamically independent record carriers. In such cases the notion of disjoint fragment redundancy may lose operational meaning. Hypothesis 14.1 is therefore intended only for fragment-local architectures preserving subsystem individuality.

Summary

Hypothesis 14.1 constrains the structure of states conditional on realization of $\text{DSR}_\delta(X, Z; t)$. The present discussion addresses the structural prevalence of that antecedent in fragment-local finite-dimensional models. In pointer-dominated and random fragment-local regimes, disjoint simultaneous redundancy for incompatible PVMs is heuristically expected to require nongeneric alignment conditions in coefficient space and thus to be measure-suppressed within $\mathcal{G}_\Lambda$ relative to smooth coefficient distributions. Engineered architectures demonstrate logical possibility but occupy structurally coordinated regions of parameter space. These regime considerations complement the geometric rigidity hypothesis. They do not derive it and do not strengthen it.

Handoff to §15

The preceding subsections separate three layers: a proved entropic obstruction, a heuristic channel-based motivation for geometric rigidity, and a parameter-space discussion of the prevalence of the DSR antecedent. What remains is to clarify attainability. Section 15 therefore provides explicit constructions illustrating how strong two-task record behavior can be realized under controlled architectures, before returning to unconditional structural constraints in §16.

15 Construction and Attainability

Section 15 clarifies the distinction between logical attainability and structural genericity. The constructions above demonstrate that strong two-task record behavior for incompatible observables can be engineered under controlled, fresh-initialization architectures. In particular, tested-channel redundancy for X and Z can be realized in separate runs without contradicting the entropic obstruction of §14.

However, such constructions do not by themselves establish disjoint simultaneous redundancy (DSR) on a single unknown joint state. They require coordinated initialization, controlled scheduling, or parameter tuning, and therefore occupy structured regions of parameter space rather than representing typical fragment-local dynamics.

Accordingly, §15 does not refute the geometric rigidity hypothesis of §14; it calibrates it. Engineered attainability is compatible with the claim that DSR, when realized under fragment-local product architectures, is structurally constrained and potentially nongeneric. The question of whether DSR forces trace-distance proximity to $\mathcal{C}_{XZ}$ remains conditional on Hypothesis 14.1 and its possible quantitative refinements.

We now turn to results that are logically independent of rigidity and typicality. Section 16 establishes unconditional stability and distributed-erasure constraints for redundancy certification, showing that record loss across many fragments requires proportionate multi-fragment disturbance.

15.1 Fully Worked Toy Model: Engineered Tested-Channel Redundancy (Two Tasks)

We give an explicit finite-dimensional construction showing that strong tested-channel redundancy for two incompatible observables can be engineered under controlled initialization and separate runs. In one experiment a disjoint fragment family reliably records X , and in another a disjoint family reliably records Z . This does not constitute the predicate $\text{DSR}_\delta(X, Z; t)$ (Eq. (12.15)) for a single unknown joint state evolving once. Rather, it provides an existence witness for two-task tested-channel redundancy under separate controlled initializations, and calibrates the degree of fine-tuning required to achieve it. The construction therefore clarifies the distinction between logical attainability and structural typicality.

15.1.1 Scope and Intent (Tested-Channel Redundancy)

The construction in this subsection analyzes tested-channel redundancy. The system is freshly initialized in product form, evolved under a fixed Hamiltonian, and redundancy is evaluated on the resulting fragment records obtained in each controlled run. This notion differs from disjoint simultaneous redundancy (DSR), which concerns a single fixed joint state and Lüders-conditioned ensembles defined on that state.

The present construction does not claim to instantiate $\text{DSR}_\delta(X, Z; t)$ for a single unknown joint input undergoing one uninterrupted evolution. Rather, it demonstrates the logical attainability of tested-channel redundancy under fresh initialization. Accordingly, no claim is made that a single unknown input state evolving once under the Hamiltonian produces simultaneous fragment-level records of incompatible observables.

Purpose

We present a concrete finite-dimensional construction that realizes tested-channel redundancy (two tasks) in an explicitly engineered, non-generic way. Specifically, two disjoint fragment families redundantly encode their respective observables under separate controlled initializations. We do not claim that a single unknown input undergoing one uninterrupted evolution produces simultaneous records of incompatible observables.

Can disjoint fragments carry near-complete records of two incompatible PVMs under separately tested record channels, without asserting rigidity or generic attainability?

The answer here is “yes” in a deliberately engineered sense. We construct a two-stage instrument and define two tested record channels, one per observable, on classical inputs. The result establishes logical attainability together with a quantitative fine-tuning estimate. It does not assert simultaneity for unknown inputs, nor any rigidity or genericity theorem.

15.1.2 Construction: System, Fragments, and Tested Record Channels

Let $S \simeq \mathbb{C}^2$. The environment consists of four qubit fragments $E_1, E_2, E_3, E_4 \simeq \mathbb{C}^2$.

Define

$$E_X := E_1 \otimes E_2, \quad E_Z := E_3 \otimes E_4.$$

Note: The labels X and Z refer to the redundancy tasks rather than the Pauli naming convention. In particular, the computational basis corresponds to the eigenbasis of σ_z .

Fix two incompatible PVMs on S :

$$W_X = \sigma_z \quad (\text{eigenstates } |0\rangle, |1\rangle), \quad W_Z = \sigma_x \quad (\text{eigenstates } |+\rangle, |-\rangle).$$

Their incompatibility parameter is

$$c = \max_{x,z} \|P_x Q_z\|_\infty^2 = \frac{1}{2}.$$

Sequential Instrument (Bookkeeping Only)

Fix durations $\tau_X, \tau_Z > 0$ and couplings $\lambda_X, \lambda_Z \in \mathbb{R}$.

Define

$$U_X := \exp(-i\tau_X H_X), \quad U_Z := \exp(-i\tau_Z H_Z), \quad V := U_{\text{Had}} \quad (\text{Hadamard on } S),$$

with

$$H_X := \lambda_X \sigma_z^{(S)} \otimes (\sigma_z^{(1)} + \sigma_z^{(2)}), \quad H_Z := \lambda_Z \sigma_z^{(S)} \otimes (\sigma_z^{(3)} + \sigma_z^{(4)}).$$

Within each stage the fragment terms commute, since they act on disjoint fragment supports with a common system operator. Thus U_X factorizes across E_1, E_2 and U_Z factorizes across E_3, E_4 .

For bookkeeping one may write

$$U_{\text{tot}} = U_Z(V \otimes I_E)U_X,$$

but the redundancy guarantees below are not derived from a single application of U_{tot} to an unknown input. They arise from separate classical-input channel tests.

Environment Initialization

All fragments are initialized in the σ_x eigenstate:

$$|\psi_E\rangle = |+\rangle^{\otimes 4}.$$

Operational Record Channels (Separate Tasks)

Redundancy is evaluated via classical-input cq channels. The record property is therefore a property of the tested map $L_W \mapsto \rho_{E_k}^{L_W}$ under fresh initialization, not a statement about a single unknown input producing simultaneous records.

(Test $\mathcal{N}_X$) W_X Record Channel

Let $L_X \in \{0, 1\}$ be equiprobable. Prepare $|x\rangle \in \{|0\rangle, |1\rangle\}$ on S .

Apply U_X .

For $k \in \{1, 2\}$ define

$$\rho_{E_k}^x =_{S, E \setminus E_k} \left[U_X(|x\rangle\langle x| \otimes |+\rangle\langle +|^{\otimes 4}) U_X^\dagger \right].$$

Fragments E_3, E_4 are uncoupled during this test and remain in product state. Tracing them out does not affect $\rho_{E_k}^x$.

(Test $\mathcal{N}_Z$) W_Z Record Channel

Let $L_Z \in \{+, -\}$ be equiprobable. Prepare $|z\rangle \in \{|+\rangle, |-\rangle\}$. As part of the W_Z record task, apply the basis-change unitary V on S , then apply U_Z .

For $k \in \{3, 4\}$ define

$$\tilde{\rho}_{E_k}^z =_{S, E \setminus E_k} \left[U_Z(V|z\rangle\langle z| V^\dagger \otimes |+\rangle\langle +|^{\otimes 4}) U_Z^\dagger \right].$$

The resulting fragment states depend on $\lambda_Z \tau_Z$. Their distinguishability is quantified explicitly in (TD-Z) below. Fragments E_1, E_2 are uncoupled during this test and remain in product state. These are distinct experiments with fresh initialization.

15.1.3 Trace-Distance and Accessible-Information Bounds

For $k \in \{1, 2\}$, the two conditional fragment states are pure and satisfy

$$\frac{1}{2} \|\rho_{E_k}^0 - \rho_{E_k}^1\|_1 = |\sin(2\lambda_X \tau_X)|. \quad (15.1)$$

For $k \in \{3, 4\}$,

$$\frac{1}{2} \|\tilde{\rho}_{E_k}^+ - \tilde{\rho}_{E_k}^-\|_1 = |\sin(2\lambda_Z \tau_Z)|. \quad (15.2)$$

From Trace Distance to Accessible Information (Binary Case)

For binary discrimination with equal priors, the Helstrom optimal POVM M^* achieves [16]

$$p_{\text{err}}^* = \frac{1}{2} \left(1 - \frac{1}{2} \|\rho_0 - \rho_1\|_1 \right). \quad (15.3)$$

Thus if

$$\frac{1}{2} \|\rho_0 - \rho_1\|_1 \geq 1 - \alpha,$$

then $p_{\text{err}}^* \leq \alpha/2$.

Let Y denote the outcome of $\mathbf{M}^*$ and let $\hat{L}$ be the induced guess. By the binary Fano inequality (in the equiprobable binary case) [17],

$$I(X; Y) \geq 1 - h_2(p_{\text{err}}).$$

Since $\hat{L}$ is a deterministic function of Y , data processing gives

$$I(L : Y) \geq I(L : \hat{L}).$$

Accessible information is defined as

$$I_{\text{acc}}(L : E_k) = \max_{\text{POVM}} I(L : Y).$$

The Helstrom POVM need not maximize mutual information. It suffices that it provides one admissible POVM. Since I_{acc} is the supremum over POVMs, this furnishes a valid lower bound.

Therefore,

$$I_{\text{acc}}(L : E_k) \geq H(L) - h_2(p_{\text{err}}^*). \quad (15.4)$$

For equiprobable binary labels, where $H(L) = 1$, if $h_2(\alpha/2) \leq \delta$ and

$$\frac{1}{2} \|\rho_0 - \rho_1\|_1 \geq 1 - \alpha,$$

then

$$I_{\text{acc}}(L : E_k) \geq 1 - \delta.$$

Tested-Channel Redundancy (Two Tasks, Per Task)

With redundancy defined operationally for a tested record channel as the existence of two disjoint fragments each satisfying

$$I_{\text{acc}}(L_W : E_{F_i}) \geq (1 - \delta)H(L_W),$$

the singleton fragments

$$F_1^X = \{1\}, \quad F_2^X = \{2\}, \quad F_1^Z = \{3\}, \quad F_2^Z = \{4\}$$

witness tested-channel redundancy (two tasks) for $\mathcal{N}_X$ and $\mathcal{N}_Z$ respectively whenever the above conditions hold. No simultaneity claim across tasks is made.

15.1.4 Fine-Tuning Windows, Non-Genericity, and Scaling

In finite dimension, the relevant entropic and distinguishability quantities are continuous in the state. Throughout this section we restrict attention to parameter regimes bounded away from degeneracy, so that small perturbations in coupling parameters produce controlled variations in the associated information measures.

On bounded Hamiltonian domains, the map

$$(H, \rho(0)) \mapsto e^{-itH} \rho(0) e^{itH}$$

is continuous in trace norm for fixed t . Hence the induced distinguishability and information functionals vary continuously in the parameters.

Consider the constraint

$$|\sin(2\lambda\tau)| \geq 1 - \alpha.$$

Fix a maximizer λ^* satisfying $2\lambda^*\tau = \pi/2 + \pi n$, and write

$$\Delta = 2\tau(\lambda - \lambda^*).$$

Near this maximizer,

$$|\sin(2\lambda\tau)| = |\cos \Delta|.$$

Restrict to the principal neighborhood $|\Delta| \leq \pi/2$. This restriction analyzes a sufficient local neighborhood around each maximizer. The full admissible set is the union of such neighborhoods over all maximizers.

Within this window,

$$|\cos \Delta| \geq 1 - \alpha \iff |\Delta| \leq \arccos(1 - \alpha).$$

As $\alpha \rightarrow 0$,

$$\arccos(1 - \alpha) = \sqrt{2\alpha} + O(\alpha^{3/2}).$$

Hence each admissible window around a maximizer has half-width

$$|\lambda - \lambda^*| = \Theta\left(\frac{\sqrt{\alpha}}{\tau}\right). \quad (15.5)$$

For sufficiently small α , the admissible windows around distinct interior maximizers remain disjoint. Overlap effects are therefore negligible in the local scaling analysis. The full admissible set is the union of such windows over all maximizers in $[-\Lambda, \Lambda]$. Since maximizers are spaced by $\pi/(2\tau)$, their number scales as $\Theta(\Lambda\tau)$.

Thus total one-dimensional admissible measure scales as

$$\Theta(\Lambda\sqrt{\alpha}) = O(\sqrt{\alpha}) \quad (\Lambda \text{ fixed}).$$

Boundary truncation and integer-rounding effects are suppressed in the $\Theta(\cdot)$ scaling. Because the constraints separate in λ_X and λ_Z , the admissible region in $[-\Lambda, \Lambda]^2$ is, up to boundary effects, $A_X \times A_Z$.

Its area therefore scales as

$$\text{Area}(\mathcal{A}_{\text{adm}}) = O(\alpha).$$

15.1.5 Structural Fragility, Regime Separation, and Conclusion

The preceding construction establishes logical attainability of tested-channel redundancy (two tasks). However, the admissible parameter region occupies vanishing measure as the discrimination threshold tightens. For each observable, the admissible one-dimensional parameter region has half-width $\Theta(\sqrt{\alpha}/\tau)$. Thus the total admissible measure in $[-\Lambda, \Lambda]$ scales as $O(\sqrt{\alpha})$ for fixed Λ . Because the X and Z constraints separate in (λ_X, λ_Z) , the admissible region in the square $[-\Lambda, \Lambda]^2$ has area

$$O(\alpha).$$

As $\alpha \rightarrow 0$, meaning near-complete distinguishability is required, the admissible region shrinks to measure zero. Small perturbations larger than $O(\sqrt{\alpha}/\tau)$ generically destroy the tested-channel redundancy condition. In this sense the construction is structurally fragile. It demonstrates existence but not robustness.

This clarifies the regime distinction:

- Tested-channel redundancy (two tasks) is logically attainable under engineered, fine-tuned parameters.
- It is not generic and is unstable under perturbation.
- Therefore the toy model does not contradict any structural obstruction to uniform or robust realization of the predicate $\text{DSR}_\delta(X, Z; t)$.

The model occupies a nongeneric, measure-suppressed regime within parameter space, while the obstruction arguments apply to the robust regime.

Conclusion

This construction establishes logical attainability of the tested-channel redundancy notion within a controlled engineered regime. It does not assert generic realization, dynamical inevitability, or disjoint simultaneous redundancy for a single unknown input. The example is explicitly non-generic and requires parameter tuning within bounded windows. Its purpose is structural: to demonstrate compatibility and scaling behavior within a precisely specified parameter regime under explicit dynamics.

15.2 Post-Toy-Model Significance and Regime Qualification

The fully worked toy model in §15.1 establishes logical attainability of near-maximal distinguishability in two incompatible measurement bases under carefully engineered coupling parameters. It is essential to clarify precisely what this construction does and does not imply for the structural program developed above.

Predicate Clarification

Section 12.3 defines the predicate $\text{DSR}_\delta(X, Z; t)$ on a single joint state $\rho_{SE}(t)$ with a fixed fragment decomposition and Lüders-induced conditional ensembles. By contrast, the toy model of §15.1 establishes logical attainability of near-complete records for incompatible observables only in

a tested-channel sense, where the X - and Z -record channels are verified under separate runs with fresh initialization.

Accordingly, §15.1 should be read as an attainability witness rather than as a counterexample or stress test of Hypothesis 14.1. Bridging these notions would require additional structural hypotheses relating separately tested channels to a single jointly realizing state $\rho_{SE}(t)$.

What the Toy Model Does *Not* Imply

- It does not invalidate the structural admissibility framework of §§12.3–14.3.
- It does not demonstrate generic tested-channel redundancy across incompatible PVMs.
- It does not eliminate the obstruction associated with incompatibility parameter $c < 1$.
- It does not show that tested-channel redundancy persists under perturbation or typical environmental couplings.

The construction is explicitly parameter-sensitive and does not constitute evidence of structural stability beyond the specified tuning regime.

What the Toy Model Does Imply

- **Logical Attainability:** There exist parameter configurations under which distinguishability measures in two incompatible bases can each be tuned arbitrarily close to unity in their respective tested tasks.
- **Fine-Tuning Scaling:** The admissible window for such configurations scales as

$$|\lambda - \lambda^*| = \Theta\left(\frac{\sqrt{\alpha}}{\tau}\right),$$

indicating shrinking parameter tolerance as $\alpha \rightarrow 0$.

- **Non-Genericity:** The allowable region in parameter space is measure-small relative to the full coupling domain.
- **Regime Qualification:** The construction depends on controlled engineered coupling structure rather than generic decoherence dynamics.

Interpretive Discipline

The toy model demonstrates existence, not prevalence. It shows that tested-channel redundancy (two tasks) is mathematically achievable under structured tuning, while leaving open questions of structural cost, detectability, and stability under perturbation. Subsequent options (A/B/C) examine whether additional constraints strengthen detectability, impose structural cost scaling, or suggest broader measure-theoretic hypotheses regarding admissible tested-channel regimes.

15.3 Detectability and Rigidity Amplification

We now analyze what redundancy certification yields independently of rigidity assumptions. The first result establishes unconditional certification stability: if a fixed finite-outcome witness certifies the record inequality with a strict margin at a reference state, then sufficiently small perturbations of the induced classical joint law preserve that certification, with controlled degradation. We then examine how, under Hypothesis 14.1, certified redundancy can serve as the input to a rigidity amplification mechanism, converting $\text{DSR}_\delta(X, Z; t)$ into a geometric restriction on $\rho_S(t)$.

15.3.1 Certification Stability

Let $X = \{|x\rangle\langle x|\}_x$ be a rank-1 PVM on $\mathcal{H}_S$ with finite outcome set, and fix time t . For a fragment E_k and a finite-outcome POVM $M = \{M_y\}_y$ on $\mathcal{H}_{E_k}$, define the induced joint law

$$p_{X,Y}^{(M)}(x, y) := p_x(t) [M_y \rho_{E_k|x}(t)],$$

and write $I_{\mathcal{E}_k}(X:Y_M)$ for the Shannon mutual information computed from this joint law.

We write $H(X; t)$ for the Shannon entropy of $p_x(t)$.

Total Variation Convention

Throughout §§15.3–16 we use the normalized total variation distance

$$\text{TV}(p, q) := \frac{1}{2} \|p - q\|_1.$$

All continuity statements are expressed in this convention.

Continuity Modulus

For an integer $D \geq 2$ and $\varepsilon \in [0, 1 - 1/D]$, define

$$f_D(\varepsilon) := \varepsilon \log(D - 1) + h_2(\varepsilon). \quad (15.6)$$

where h_2 is the binary entropy with the same log base as H .

By the classical Fannes–Audenaert inequality [14], for distributions $p, \tilde{p}$ on a D -point alphabet,

$$\frac{1}{2} \|p - \tilde{p}\|_1 \leq \varepsilon \implies |H(p) - H(\tilde{p})| \leq f_D(\varepsilon).$$

Theorem 15.1: Margin-Witnessed Robustness of Redundancy Certification

Let $\rho_{SE}(t)$ and $\tilde{\rho}_{SE}(t)$ be two joint states with the same fragment decomposition $E = E_1 \cdots E_N$.

Let $\mathcal{E}_k, \tilde{\mathcal{E}}_k$ denote the corresponding X -conditioned fragment ensembles at time t .

Assume that for some $m \geq 1$ there exist distinct fragments $k_1, \dots, k_m$ and finite-outcome POVMs $M^{(k_j)}$ such that for each j ,

$$I_{\mathcal{E}_{k_j}}(X:Y_{M^{(k_j)}}) \geq (1 - \delta) H(X; t) + \gamma. \quad (15.7)$$

for some $\delta \in (0, 1)$ and some strict margin $\gamma > 0$.

Let $|X|$ denote the number of X outcomes and $|Y_j|$ denote the number of outcomes of $M^{(k_j)}$. Define $D_j := |X| |Y_j|$.

Then there exists $\varepsilon > 0$ depending only on $\gamma, \delta, |X|, |Y_1|, \dots, |Y_m|$ such that if for all j ,

$$\frac{1}{2} \|p_{X,Y}^{(M^{(k_j)})} - \tilde{p}_{X,Y}^{(M^{(k_j)})}\|_1 \leq \varepsilon,$$

then for each j ,

$$I_{\tilde{\mathcal{E}}_{k_j}}(X:Y_{M^{(k_j)}}) \geq (1 - \delta) H_{\tilde{\rho}}(X; t),$$

and consequently

$$I_{\text{acc}}^{\tilde{P}}(X : E_{k_j}; t) \geq (1 - \delta) H_{\tilde{P}}(X; t),$$

so that $R_{\delta}^{\tilde{P}}(X; t) \geq m$.

Proof of Theorem 15.1

Fix $j \in \{1, \dots, m\}$ and let $Y := Y_{M^{(k_j)}}$ denote the outcome of the fixed finite-outcome POVM $M^{(k_j)}$.

Write

$$P := P_{XY}^{(M^{(k_j)})}, \quad \tilde{P} := \tilde{P}_{XY}^{(M^{(k_j)})},$$

with alphabet sizes $|X|$ and $|Y| := |Y_j|$.

Assume

$$\frac{1}{2} \|P - \tilde{P}\|_1 \leq \varepsilon.$$

Then the marginal distances satisfy

$$\frac{1}{2} \|P_X - \tilde{P}_X\|_1 \leq \varepsilon, \quad \frac{1}{2} \|P_Y - \tilde{P}_Y\|_1 \leq \varepsilon,$$

since total variation distance decreases under marginalization.

Without loss of generality, restrict to $\varepsilon \leq 1 - 1/(|X||Y|)$ so that the Fannes—Audenaert modulus applies.

By the Fannes—Audenaert inequality and the definition of $f_D(\varepsilon)$,

$$\begin{aligned} |H_P(X) - H_{\tilde{P}}(X)| &\leq f_{|X|}(\varepsilon), \\ |H_P(Y) - H_{\tilde{P}}(Y)| &\leq f_{|Y|}(\varepsilon), \\ |H_P(X, Y) - H_{\tilde{P}}(X, Y)| &\leq f_{|X||Y|}(\varepsilon). \end{aligned} \tag{15.8}$$

Using $I(X; Y) = H(X) + H(Y) - H(X, Y)$, we obtain

$$\begin{aligned} |I_P(X; Y) - I_{\tilde{P}}(X; Y)| &\leq |H_P(X) - H_{\tilde{P}}(X)| + |H_P(Y) - H_{\tilde{P}}(Y)| + |H_P(X, Y) - H_{\tilde{P}}(X, Y)| \\ &\leq f_{|X|}(\varepsilon) + f_{|Y|}(\varepsilon) + f_{|X||Y|}(\varepsilon). \end{aligned} \tag{15.9}$$

By hypothesis,

$$I_P(X; Y) \geq (1 - \delta) H_P(X) + \gamma.$$

Combining this with (15.9) and (15.8), we obtain

$$\begin{aligned} I_{\tilde{P}}(X; Y) &\geq I_P(X; Y) - (f_{|X|}(\varepsilon) + f_{|Y|}(\varepsilon) + f_{|X||Y|}(\varepsilon)) \\ &\geq (1 - \delta) H_P(X) + \gamma - (f_{|X|}(\varepsilon) + f_{|Y|}(\varepsilon) + f_{|X||Y|}(\varepsilon)) \\ &\geq (1 - \delta) H_{\tilde{P}}(X) + \gamma - ((2 - \delta) f_{|X|}(\varepsilon) + f_{|Y|}(\varepsilon) + f_{|X||Y|}(\varepsilon)). \end{aligned}$$

For each j this yields a threshold ε_j depending on $\gamma, \delta, |X|, |Y_j|$ such that

$$(2 - \delta) f_{|X|}(\varepsilon_j) + f_{|Y_j|}(\varepsilon_j) + f_{|X||Y_j|}(\varepsilon_j) \leq \gamma.$$

Set

$$\varepsilon := \min_{1 \leq j \leq m} \varepsilon_j,$$

which depends only on $\gamma, \delta, |X|, |Y_1|, \dots, |Y_m|$.

Then the above bound holds simultaneously for all $j = 1, \dots, m$, yielding

$$I_{\tilde{P}}(X; Y_{M^{(k_j)}}) \geq (1 - \delta) H_{\tilde{P}}(X).$$

Since the X -marginal of $\tilde{P}$ coincides with that of $\tilde{\rho}(t)$, we have $H_{\tilde{P}}(X) = H_{\tilde{\rho}}(X; t)$.

By definition of accessible information,

$$I_{\text{acc}}^{\tilde{\rho}}(X : \tilde{E}_{k_j}; t) = \sup_M I_{\tilde{P}}(X; Y_M) \geq I_{\tilde{P}}(X; Y_{M^{(k_j)}}) \geq (1 - \delta) H_{\tilde{\rho}}(X; t).$$

Since this holds for each $j = 1, \dots, m$, each fragment $\tilde{E}_{k_j}$ certifies δ -redundancy, and therefore

$$R_{\delta}^{\tilde{\rho}}(X; t) \geq m.$$

□

15.3.2 Conditional Rigidity Amplification

We work in the parameter space $\mathcal{P}_{\Lambda}$ defined in §13.1 and assume the Geometric Rigidity Hypothesis 14.1.

Proposition 15.1: Rigidity Amplification

Assume the Geometric Rigidity Hypothesis 14.1, and let X and Z be incompatible rank-1 PVMs on $\mathcal{H}_S$. If a parameter point in $\mathcal{P}_{\Lambda}$ satisfies $\text{DSR}_{\delta}(X, Z; t)$, then the induced system marginal $\rho_S(t)$ satisfies

$$\text{dist}_{\text{tr}}(\rho_S(t), \mathcal{C}_{XZ}) \leq \varepsilon(t, c, \delta, d_{E_1}, \dots, d_{E_N}).$$

where $\varepsilon(\cdot)$ is the function postulated in the Hypothesis 14.1 (Geometric Rigidity).

Proof of Proposition 15.1

This follows directly from the Geometric Rigidity Hypothesis 14.1.

□

Corollary 15.1: Conditional Rigidity-to-Robustness Transfer

Under the assumptions of Proposition 15.1, suppose in addition that the induced classical joint laws associated with the certified fragments satisfy the total-variation hypotheses of Theorem 15.1. Then certified δ -redundancy is stable under the rigidity-induced perturbation.

Proof of Corollary 15.1

Proposition 15.1 yields trace-distance proximity of the system marginal $\rho_S(t)$ to the compatibility set $\mathcal{C}_{XZ}$ defined in Eq. (12.16), which restricts admissible joint classical records of the observables under the Geometric Rigidity Hypothesis 14.1. This statement concerns the system marginal alone.

The conclusion regarding redundancy stability requires, in addition, control of the induced classical joint laws associated with the witness POVMs on the fragments. When these joint laws satisfy the total-variation bounds required in Theorem 15.1, the robustness conclusion follows directly from that theorem.

No implication from system-marginal proximity to fragment-level joint-law proximity is asserted here; such control must be supplied independently (see §16).

□

15.3.3 Interpretive Discipline

The certification stability result shows that margin-witnessed redundancy is robust under sufficiently small perturbations of the induced classical law. The conditional rigidity amplification establishes that, assuming Hypothesis 14.1, realization of $\text{DSR}_\delta(X, Z; t)$ propagates certified detectability into a geometric constraint on the system marginal.

The amplification remains explicitly conditional. No independent derivation of geometric rigidity is asserted in this section. The subsequent chapter examines whether additional structural hypotheses can internalize or strengthen this amplification mechanism beyond the conditional framework adopted here.

15.4 Exploratory Hypothesis: Redundancy Cost Scaling

The previous subsection established certification stability under perturbations of the induced classical law and, conditional on Hypothesis 14.1, rigidity amplification into proximity of the compatibility set $\mathcal{C}_{XZ}$. These results remain local and conditional. We now isolate a possible quantitative strengthening: displacement of the system marginal away from $\mathcal{C}_{XZ}$ forces loss of *certified* redundancy at a rate that scales with the displacement. No derivation is claimed in this section.

Remark: Scaling intuition is related to no-broadcasting constraints for noncommuting quantum states [18], which prohibit exact broadcasting of incompatible state families. The present program differs in scope: rather than establishing universal impossibility, it isolates regime-dependent suppression and quantitative loss under admissible coupling.

15.4.1 Certified Redundancy Count

Fix a time t and incompatible rank-1 PVMs

$$X = \{P_x\}_x, \quad Z = \{Q_z\}_z$$

on $\mathcal{H}_S$, together with tolerances $\delta \in (0, 1)$ and a strict margin $\gamma > 0$.

Fix witnessing POVM families

$$\mathbf{M}_X := \{M_X^{(k)}\}_{k=1}^N, \quad \mathbf{M}_Z := \{M_Z^{(k)}\}_{k=1}^N,$$

where each $M_X^{(k)} = \{M_{X,y}^{(k)}\}_y$ and $M_Z^{(k)} = \{M_{Z,y}^{(k)}\}_y$ is a finite-outcome POVM on $\mathcal{H}_{E_k}$.

For a joint state $\omega_{SE}(t)$ on the fixed decomposition $S : E_1 \cdots E_N$, let $\mathcal{E}_X^{\omega, (k)}(t)$ and $\mathcal{E}_Z^{\omega, (k)}(t)$ denote the corresponding canonically conditioned fragment ensembles induced by X and Z . Write $H_\omega(X; t)$ and $H_\omega(Z; t)$ for the Shannon entropies of the induced outcome laws.

For each fragment E_k , define the $(X, \mathbf{M}_X)$ -*certificate* predicate on $\omega_{SE}(t)$ by

$$\text{Cert}_{\delta, \gamma}^\omega(X; k) : \Longleftrightarrow I_{\mathcal{E}_X^{\omega, (k)}}(X : Y_{M_X^{(k)}}) \geq (1 - \delta) H_\omega(X; t) + \gamma,$$

and analogously $\text{Cert}_{\delta, \gamma}^\omega(Z; k)$.

Define the associated certified record sets and certified redundancy counts:

$$\text{cert}_{\delta, \gamma}^{\omega}(X; t; \mathbf{M}_X) := \{k : \text{Cert}_{\delta, \gamma}^\omega(X; k)\}, \quad R_{\delta, \gamma}^{\text{cert}, \omega}(X; t; \mathbf{M}_X) := \left| \text{cert}_{\delta, \gamma}^{\omega}(X; t; \mathbf{M}_X) \right|,$$

and similarly for Z .

15.4.2 Scaling Hypothesis

Fix a baseline joint state $\rho_{SE}(t)$.

Assume that $\rho_{SE}(t)$ satisfies certified redundancy for both observables:

$$R_{\delta,\gamma}^{\text{cert},\rho}(X;t;\mathbf{M}_X) \geq m_X, \quad R_{\delta,\gamma}^{\text{cert},\rho}(Z;t;\mathbf{M}_Z) \geq m_Z.$$

The corresponding certified record sets are disjoint:

$$R_{\delta,\gamma}^{\text{cert},\rho}(X;t;\mathbf{M}_X) \cap R_{\delta,\gamma}^{\text{cert},\rho}(Z;t;\mathbf{M}_Z) = \emptyset.$$

Scaling Hypothesis (Protective Form)

There exists a nondecreasing function

$$g : [0, 1] \rightarrow \mathbb{N}, \quad g(0) = 0, \quad g(\varepsilon) \leq \min\{m_X, m_Z\},$$

such that for any perturbed joint state $\tilde{\rho}_{SE}(t)$ with system marginal $\tilde{\rho}_S(t) :=_E \tilde{\rho}_{SE}(t)$ and displacement

$$\Delta(\tilde{\rho}) := \text{dist}_{\text{tr}}(\tilde{\rho}_S(t), \mathcal{C}_{XZ}),$$

the implication

$$\Delta(\tilde{\rho}) \geq \varepsilon \implies \left(R_{\delta,\gamma}^{\text{cert},\tilde{\rho}}(X;t;\mathbf{M}_X) \leq m_X - g(\varepsilon) \text{ or } R_{\delta,\gamma}^{\text{cert},\tilde{\rho}}(Z;t;\mathbf{M}_Z) \leq m_Z - g(\varepsilon) \right)$$

holds for all $\varepsilon \in [0, 1]$.

In words: displacement of the system marginal away from the compatibility set forces loss of at least $g(\varepsilon)$ certified fragments for at least one of the incompatible observables.

15.4.3 Linear–Geometric Strengthening (Variant)

Assume the fragment family $\{E_k\}_{k=1}^N$ is equipped with a fixed architectural profile

$$\Lambda_{\text{geom}} \in \mathfrak{G},$$

where $\mathcal{G}$ denotes a class of admissible fragment geometries such as interaction graphs with bounded coupling range, maximum degree, or correlation length.

Linear–Geometric Scaling Variant

There exist constants $c > 0$ and $\varepsilon_0 \in (0, 1]$, depending only on $\delta, \gamma, \Lambda_{\text{geom}}$ and fragment dimensions, such that the base scaling hypothesis holds with g satisfying, for all $\varepsilon \in [0, \varepsilon_0]$,

$$g(\varepsilon) \geq \max\{1, \lfloor c\varepsilon \min\{m_X, m_Z\} \rfloor\}.$$

Equivalently, for every perturbed joint state $\tilde{\rho}_{SE}(t)$,

$$\Delta(\tilde{\rho}) \geq \varepsilon \implies \text{at least } \max\{1, \lfloor c\varepsilon \min\{m_X, m_Z\} \rfloor\} \text{ certified fragments fail}$$

for at least one of X or Z .

Thus for sufficiently small $\varepsilon > 0$, at least one certified fragment fails. For larger displacements, the number of failures grows proportionally to $\varepsilon \min\{m_X, m_Z\}$ up to saturation.

15.4.4 Interpretation and Boundaries

Because trace distance is bounded by 1, no unbounded growth is claimed. The base scaling hypothesis asserts monotonic protection: larger displacement away from $\mathcal{C}_{XZ}$ forces loss of more fixed-witness certificates for at least one observable. The linear-geometric variant is substantially stronger. It posits that redundancy behaves as a distributed stabilizer: displacement of the system marginal propagates across certified fragments at a rate proportional, for small ε , to both displacement magnitude and baseline redundancy scale.

This scaling hypothesis remains exploratory. It does not follow from §15.3 or §14.3. Its validity may depend on additional structural assumptions such as fragment independence, uniform continuity of witnessing POVMs, locality constraints, or restrictions on admissible perturbations induced by small parameter motion in $\mathcal{P}_\Lambda$. The purpose of this section is to isolate precise scaling laws under which redundancy ceases to be merely detectable and instead imposes quantitative propagation cost.

15.5 Illustrative Broadcast Model (Conditional, Non-Derivational)

We sketch a restricted architecture in which a margin-controlled stabilization mechanism arises. This example is conditional and does not establish the general scaling hypothesis of §15.4.

Fragmentwise Product Measurement Channel

Assume that the joint state at time t is generated by a fragmentwise product measurement channel in the X basis:

$$\rho_{SE}(t) = \left(\text{id}_S \otimes \bigotimes_{k=1}^N \mathcal{M}_X^{(k)} \right) (\rho_S(t)),$$

where

$$\mathcal{M}_X^{(k)}(\omega) = \sum_x \text{Tr}(P_x \omega) \sigma_x^{(k)},$$

with $\{\sigma_x^{(k)}\}_x$ states on $\mathcal{H}_{E_k}$.

Let perturbations of the system marginal be propagated through the same channel:

$$\tilde{\rho}_{SE}(t) := \left(\text{id}_S \otimes \bigotimes_{k=1}^N \mathcal{M}_X^{(k)} \right) (\tilde{\rho}_S(t)).$$

Assume that for each k there exists a fixed POVM $M_X^{(k)}$ (certification is with respect to this fixed witness family $\mathbf{M}_X = \{M_X^{(k)}\}$, not re-optimized over all POVMs) satisfying the baseline certification condition

$$I_{\mathcal{E}_X^{\rho, (k)}}(X : Y_{M_X^{(k)}}) \geq (1 - \delta) H_\rho(X; t) + \gamma,$$

where certification is benchmarked relative to the baseline entropy $H_\rho(X; t)$.

In particular,

$$R_{\delta, \gamma}^{\text{cert}, \rho}(X; t; \mathbf{M}_X) = N.$$

Assumption 15.1: Uniform Continuity Modulus

We assume a uniform continuity bound for the fixed broadcast architecture described above. This assumption is structural and is not derived here.

Specifically, there exists a constant $C > 0$ such that

$$\left| I_{\mathcal{E}_X^{\tilde{\rho},(k)}}(X : Y_{M_X^{(k)}}) - I_{\mathcal{E}_X^{\rho,(k)}}(X : Y_{M_X^{(k)}}) \right| \leq C \|\tilde{\rho}_S(t) - \rho_S(t)\|_1,$$

with C independent of k .

The constant C may depend on $|X|$, the POVM outcome alphabets, and the fixed fragment channels $\mathcal{M}_X^{(k)}$, but is uniform in k .

Remark: A continuity estimate of this type is compatible with finite-alphabet Fannes—Audenaert continuity bounds for the von Neumann entropy (see Audenaert [14]), together with contractivity of trace distance under CPTP maps. The present subsection assumes such a uniform bound rather than deriving a sharp modulus.

Robustness Scale

Suppose

$$\|\tilde{\rho}_S(t) - \rho_S(t)\|_1 \leq \varepsilon.$$

Then for each fragment k ,

$$I_{\mathcal{E}_X^{\tilde{\rho},(k)}}(X : Y_{M_X^{(k)}}) \geq I_{\mathcal{E}_X^{\rho,(k)}}(X : Y_{M_X^{(k)}}) - C\varepsilon.$$

Consequently, if $C\varepsilon \leq \gamma$, the certification inequality holds for all k , and hence

$$R_{\delta,\gamma}^{\text{cert},\tilde{\rho}}(X; t; \mathbf{M}_X) = N.$$

Relation to $\mathcal{C}_{XZ}$

Let $\sigma^* \in \mathcal{C}_{XZ}$ be a minimizer of $\text{dist}_{\text{tr}}(\tilde{\rho}_S(t), \mathcal{C}_{XZ})$, which exists by compactness in finite dimension.

Then

$$\text{dist}_{\text{tr}}(\tilde{\rho}_S(t), \mathcal{C}_{XZ}) = \text{dist}_{\text{tr}}(\tilde{\rho}_S(t), \sigma^*).$$

In the 1/2-normalized trace-distance convention,

$$\text{dist}_{\text{tr}}(\tilde{\rho}_S(t), \mathcal{C}_{XZ}) \leq \varepsilon \implies \|\tilde{\rho}_S(t) - \sigma^*\|_1 \leq 2\varepsilon.$$

In particular, if the baseline state satisfies $\rho_S(t) = \sigma^* \in \mathcal{C}_{XZ}$, certified redundancy persists whenever

$$2C\varepsilon \leq \gamma.$$

Interpretation

In this broadcast architecture, certified redundancy behaves as a margin-controlled stabilizer. Perturbations of the system marginal smaller than a scale of order γ/C preserve all certified witnesses, up to the factor-of-two trace-distance normalization. Beyond this scale, Lipschitz continuity alone no longer guarantees preservation, and certified redundancy need not persist. This example illustrates how locality and uniform continuity can produce protective behavior consistent with §15.4, without asserting that such threshold behavior holds in general fragment geometries.

Handoff to §16

Section 15 has clarified the distinction between tested-channel attainability and state-level redundancy, and has isolated where additional assumptions, such as rigidity-based amplification, enter the analysis. While engineered architectures can realize strong two-task record behavior in controlled settings, such constructions do not by themselves settle the stability or erasure behavior of certified fragment records under arbitrary state transformations. The next section therefore removes all rigidity assumptions and studies record certification under perturbation directly.

16 Stability, Witness Approximation, and Distributed Erasure

Section 16 is deliberately unconditional: it does not assume rigidity, typicality, or any special dynamics, but instead studies how redundancy certification behaves under arbitrary transformations between joint states.

The first result is a forward stability statement. If a fragment certifies the δ -record predicate with strict margin at a reference state, then sufficiently small subnormalized disturbance of the conditioned fragment ensemble guarantees that the same finite-outcome witness continues to certify a δ -record after perturbation, with controlled margin degradation. The second result shows that, for the purposes of this stability analysis, finite-outcome near-optimal witnesses suffice; no infinite-outcome optimization is required. The third result is a distributed erasure principle. If many previously certified fragments lose record status under a state change, then many fragments must have experienced super-threshold disturbance. Redundancy therefore converts large-scale record loss into a multi-fragment disturbance requirement.

These statements are structural constraints on state-to-state transformations. They are compatible with, but logically independent of, the conjectural rigidity narrative of §§14–15.

16.1 Witness Stability Under Subnormalized Perturbations

This subsection establishes a local stability lemma used in subsequent distributed erasure arguments. Working under the fixed Lüders-conditioning convention (no support excision; subnormalized operators as primitives), we prove a forward stability statement: if a fragment’s δ -record status is certified at a joint state ρ_{SE} by a finite witness with strict margin, then any nearby state ρ'_{SE} whose subnormalized conditioned ensemble is disturbed below a computable threshold continues to satisfy the record inequality, possibly with reduced margin. Equivalently, if a fragment ceases to be a δ -record after a transformation, then that fragment must have undergone super-threshold subnormalized disturbance. The lemma is kinematic and does not assume particular dynamics; it provides the local disturbance–record link that will later support global counting arguments.

Scope of Stability Results

All stability and distributed-erasure results in this section apply only to fragments that satisfy the δ -record predicate with strict margin. That is, we restrict attention to interior certification in the sense defined below. These results do not apply to fragments that merely meet the δ -threshold without slack.

Definition 16.1: Margin-Certified δ -Record Fragment

For $\gamma > 0$, we say that fragment E_k is (δ, γ) -certified for X at state $\rho_{SE}(t)$ if there exists a finite-outcome POVM $M = \{M_y\}_{y \in Y}$ on $\mathcal{H}_{E_k}$ such that

$$I(X:Y; t)_M(\rho_{SE}) \geq (1 - \delta)H(X; t)_{\rho_S} + \gamma.$$

Entropy Continuity Modulus

For a probability distribution on an alphabet of size d , define

$$\phi_d(\varepsilon) := \varepsilon \log(d-1) + h_2(\varepsilon), \quad \varepsilon \in [0, 1 - 1/d]. \quad (16.1)$$

where h_2 is the binary entropy.

Whenever $\frac{1}{2}\|p - q\|_1 \leq \varepsilon$, we have

$$|H(p) - H(q)| \leq \phi_d(\varepsilon).$$

This is the classical Audenaert continuity bound.

Lemma 16.1: Witness Stability; Margin-Certified Form

Fix $k \in \{1, \dots, N\}$ and $\delta \in (0, 1)$. Suppose fragment E_k is (δ, γ) -certified for X at ρ_{SE} for some $\gamma > 0$.

Fix $\eta \in (0, \gamma/2]$. Let $M^* = \{M_y\}_{y \in Y}$ be a finite-outcome POVM certifying this condition (assumed to exist; see Proposition 16.1 for finite-outcome sufficiency), i.e.,

$$I(X; Y; t)_{M^*}(\rho_{SE}) \geq (1 - \delta)H(X; t)_{\rho_S} + \gamma.$$

Let $D := |\mathcal{X}||Y|$. Then there exists

$$\varepsilon_*(\eta, \delta; |\mathcal{X}|, |Y|) > 0$$

depending only on $\eta, \delta, |\mathcal{X}|, |Y|$ such that for any second joint state ρ'_{SE} , if

$$D_{\text{sub}}^{(k)}(\rho_{SE}, \rho'_{SE}) \leq \varepsilon_*,$$

then

$$I_{\text{acc}}(X; E_k; t)_{\rho'_{SE}} \geq (1 - \delta)H(X; t)_{\rho'_S} + \gamma - \eta.$$

Proof of Lemma 16.1

Define subnormalized conditionals

$$A_x := p_x(\rho_{SE}) \rho_{E_k|x}(\rho_{SE}), \quad A'_x := p_x(\rho'_{SE}) \rho_{E_k|x}(\rho'_{SE}),$$

so that

$$D_{\text{sub}}^{(k)}(\rho_{SE}, \rho'_{SE}) = \frac{1}{2} \sum_x \|A_x - A'_x\|_1. \quad (16.2)$$

Under M^* define

$$p_{XY}(x, y) = [A_x M_y], \quad p'_{XY}(x, y) = [A'_x M_y].$$

Since $A_x - A'_x$ is Hermitian and $0 \leq M_y \leq I$,

$$|[A_x - A'_x] M_y| \leq [A_x - A'_x | M_y].$$

Summing over x, y and using $\sum_y M_y = I$ gives

$$\sum_{x,y} |p_{XY}(x, y) - p'_{XY}(x, y)| \leq \sum_x \|A_x - A'_x\|_1.$$

Hence

$$\frac{1}{2}\|p_{XY} - p'_{XY}\|_1 \leq D_{\text{sub}}^{(k)}(\rho_{SE}, \rho'_{SE}). \quad (16.3)$$

matching the normalized total variation convention of §15.3.

Marginalization does not increase total variation, so

$$\frac{1}{2}\|p_X - p'_X\|_1 \leq \varepsilon, \quad \frac{1}{2}\|p_Y - p'_Y\|_1 \leq \varepsilon.$$

Applying the entropy continuity bound to the marginals p_X, p'_X, p_Y, p'_Y , and the joint p_{XY}, p'_{XY} , and using

$$I(X:Y; t) = H(X; t) + H(Y; t) - H(X, Y; t),$$

yields

$$|H(X; t)_{\rho_S} - H(X; t)_{\rho'_S}| \leq \phi_{|\mathcal{X}|}(\varepsilon),$$

and

$$|I(X:Y; t)_{M^*}(\rho_{SE}) - I(X:Y; t)_{M^*}(\rho'_{SE})| \leq \phi_{|\mathcal{X}|}(\varepsilon) + \phi_{|Y|}(\varepsilon) + \phi_D(\varepsilon).$$

Choose $\varepsilon_* > 0$ such that

$$\varepsilon_* \leq 1 - \frac{1}{D},$$

$$\phi_{|\mathcal{X}|}(\varepsilon_*) \leq \frac{\eta}{4(1 - \delta)},$$

$$\phi_{|\mathcal{X}|}(\varepsilon_*) + \phi_{|Y|}(\varepsilon_*) + \phi_D(\varepsilon_*) \leq \frac{\eta}{2}.$$

Combining the continuity bounds gives

$$I(X:Y; t)_{M^*}(\rho'_{SE}) \geq I(X:Y; t)_{M^*}(\rho_{SE}) - \left(\phi_{|\mathcal{X}|}(\varepsilon) + \phi_{|Y|}(\varepsilon) + \phi_D(\varepsilon) \right),$$

and also

$$(1 - \delta) H(X; t)_{\rho_S} \geq (1 - \delta) H(X; t)_{\rho'_S} - (1 - \delta) \phi_{|\mathcal{X}|}(\varepsilon).$$

Therefore,

$$I(X:Y; t)_{M^*}(\rho'_{SE}) \geq (1 - \delta) H(X; t)_{\rho'_S} + \gamma - \left(\phi_{|\mathcal{X}|}(\varepsilon) + \phi_{|Y|}(\varepsilon) + \phi_D(\varepsilon) + (1 - \delta) \phi_{|\mathcal{X}|}(\varepsilon) \right).$$

By the choice of ε_* (the two displayed smallness conditions), the bracketed term is $\leq \eta$.

Hence

$$I(X:Y; t)_{M^*}(\rho'_{SE}) \geq (1 - \delta) H(X; t)_{\rho'_S} + \gamma - \eta.$$

If

$$D_{\text{sub}}^{(k)}(\rho_{SE}, \rho'_{SE}) \leq \varepsilon_*,$$

then

$$I(X:Y; t)_{M^*}(\rho'_{SE}) \geq (1 - \delta) H(X; t)_{\rho'_S} + \gamma - \eta,$$

and hence

$$I_{\text{acc}}(X:E_k; t)_{\rho'_{SE}} \geq (1 - \delta) H(X; t)_{\rho'_S} + \gamma - \eta.$$

□

Corollary 16.1: Subnormalized Disturbance Implies Joint-Law Stability

Fix a fragment E_k and a finite-outcome POVM $M = \{M_y\}_y$. Let p_{XY} and p'_{XY} denote the induced classical joint laws under M at ρ_{SE} and ρ'_{SE} respectively.

If

$$D_{\text{sub}}^{(k)}(\rho_{SE}, \rho'_{SE}) \leq \varepsilon,$$

then

$$\frac{1}{2} \|p_{XY} - p'_{XY}\|_1 \leq \varepsilon.$$

In particular, the total-variation hypotheses required in Theorem 15.1 are satisfied whenever the corresponding subnormalized disturbance bound holds.

The finite-partition reduction above yields the following distributed certification consequence.

Proof of Corollary 16.1 This is immediate from the estimate derived in the proof of Lemma 16.1:

$$\frac{1}{2} \|p_{XY} - p'_{XY}\|_1 \leq D_{\text{sub}}^{(k)}(\rho_{SE}, \rho'_{SE}).$$

□

16.2 Finite-Outcome Witnesses for Accessible Information

This subsection provides a technical closure step required by the stability analysis of §16.1. Since accessible information is defined as a supremum over all POVMs, a priori including those with infinite outcome spaces, we justify that in finite-dimensional settings it suffices to optimize over finite-outcome POVMs. The argument proceeds by approximating relative entropy via finite measurable partitions and invoking data processing. This ensures that margin-certifying witnesses may be chosen with finite outcome sets, allowing the stability thresholds of §16.1 to depend only on finite parameters.

We justify that for finite $\mathcal{X}$ and finite-dimensional E_k , it suffices to optimize I_{acc} over finite-outcome POVMs.

Relative Entropy Convention

Throughout this subsection, relative entropy $D(P\|Q)$ is measured in bits (all logarithms are base 2). We use the following variational representation of relative entropy:

$$D(P\|Q) = \sup_f \left\{ \mathbb{E}_P[f] - \log_2 \mathbb{E}_Q[2^f] \right\},$$

where the supremum is taken over bounded measurable functions f . This is the base-2 form of the standard variational characterization of Kullback–Leibler divergence.

Setup

Fix $|\mathcal{X}| < \infty$ and a cq state

$$\omega_{XE_k}^{(k)} = \sum_{x \in \mathcal{X}} p_x |x\rangle\langle x| \otimes \rho_{E_k|x}, \quad \dim \mathcal{H}_{E_k} < \infty.$$

All informations below are evaluated on $\omega^{(k)}$.

For a POVM M on $\mathcal{H}_{E_k}$ with outcome space $(\mathcal{Y}, \Sigma)$, define

$$\mu_x(B) := [\rho_{E_k|x} M(B)], \quad P^{(M)}(x, B) = p_x \mu_x(B), \quad P_Y := \sum_x p_x \mu_x.$$

Lemma 16.2: Relative Entropy Via Finite Partitions

Let P, Q be probability measures on $(\Omega, \mathcal{F})$ with $P \ll Q$.

Then

$$D(P\|Q) = \sup_{\pi} D(P_{\pi}\|Q_{\pi}),$$

where $\pi = \{A_1, \dots, A_m\}$ ranges over finite measurable partitions and $P_{\pi}(i) = P(A_i)$, $Q_{\pi}(i) = Q(A_i)$.

Proof of Lemma 16.2

For any partition π , coarse-graining is a channel, hence $D(P_{\pi}\|Q_{\pi}) \leq D(P\|Q)$.

Using the variational representation of relative entropy introduced above,

$$D(P\|Q) = \sup_f \left\{ \int f dP - \log \int 2^f dQ \right\},$$

restrict to simple $f = \sum_{i=1}^m a_i \mathbf{1}_{A_i}$. Then

$$\int f dP - \log \int 2^f dQ = \sum_i a_i P(A_i) - \log \left(\sum_i 2^{a_i} Q(A_i) \right).$$

Optimizing over $a \in \mathbb{R}^m$ yields $D(P_{\pi}\|Q_{\pi})$; for $Q(A_i) > 0$ the maximizer is $a_i = \log \frac{P(A_i)}{Q(A_i)}$, and if $Q(A_i) = 0$ then $P(A_i) = 0$ since $P \ll Q$. Thus $D(P\|Q) \geq D(P_{\pi}\|Q_{\pi})$.

Taking the supremum over π and combining with the first inequality proves the claim. $\square$

Proposition 16.1: Infinite-Outcome Approximation of I_{acc}

For $|\mathcal{X}| < \infty$ and $\dim \mathcal{H}_{E_k} < \infty$,

$$I_{\text{acc}}(X:E_k)_{\omega^{(k)}} = \sup_{M \in \text{POVM}_{\text{fin}}} I(X:Y)_M(\omega^{(k)}).$$

Equivalently: for every $\xi > 0$ there exists a finite-outcome POVM M_{ξ} such that

$$I(X:Y)_{M_{\xi}}(\omega^{(k)}) \geq I_{\text{acc}}(X:E_k)_{\omega^{(k)}} - \xi.$$

Proof of Proposition 16.1

Fix a POVM M with induced conditional laws $\{\mu_x\}$ and $P_Y = \sum_x p_x \mu_x$. For each x with $p_x > 0$, $\mu_x \ll P_Y$ since $P_Y \geq p_x \mu_x$.

Using

$$I(X:Y)_M = \sum_x p_x D(\mu_x\|P_Y),$$

apply Lemma 16.2 to each pair (μ_x, P_Y) . For any $\xi > 0$, there exists a finite partition $\pi_x = \{B_j^{(x)}\}$ such that

$$D((\mu_x)_{\pi_x} \| (P_Y)_{\pi_x}) \geq D(\mu_x \| P_Y) - \xi.$$

Let π be a common finite refinement of all $\{\pi_x\}$. Since π_x is a coarse-graining of π , applying data processing to the coarse-graining map $\pi \rightarrow \pi_x$ gives

$$D((\mu_x)_\pi \| (P_Y)_\pi) \geq D((\mu_x)_{\pi_x} \| (P_Y)_{\pi_x}) \geq D(\mu_x \| P_Y) - \xi.$$

Equivalently: refining the partition cannot decrease the induced KL divergence.

Summing over x yields

$$\sum_x p_x D((\mu_x)_\pi \| (P_Y)_\pi) \geq I(X:Y)_M - \xi.$$

Define $M_j^\pi := M(B_j)$ for $\pi = \{B_j\}_{j=1}^m$. Then the finite-outcome POVM M^π satisfies

$$I(X:\tilde{Y})_{M^\pi}(\omega^{(k)}) \geq I(X:Y)_M(\omega^{(k)}) - \xi.$$

Since for every M and every $\xi > 0$ there exists finite M^π with $I(M^\pi) \geq I(M) - \xi$, we have $\sup_{\text{fin}} I \geq \sup I$. The reverse inequality is immediate, hence

$$\sup_{\text{fin}} I = \sup I = I_{\text{acc}}.$$

Remark: Used in §16.1 only to select M^* with finite outcome set Y and $I(X:Y)_{M^*} \geq I_{\text{acc}} - \eta/4$. $\square$

16.3 Redundancy and Distributed Record Failure

We now lift the fragmentwise stability result to a distributed redundancy setting. The question is no longer whether a single margin-certified fragment remains stable, but how redundancy failure across a collection of fragments must manifest under perturbation. The theorem below shows that distributed record loss cannot occur silently: any fragment that ceases to satisfy the δ -record predicate must cross its individual stability threshold. Thus redundancy failure is necessarily fragmentwise and threshold-detectable.

Fix a rank-1 PVM $X = \{P_x\}_x$ as in §§12.3–13.3, a parameter $\delta > 0$, and time t .

Excess Disturbance

Let ρ_{SE} and ρ'_{SE} be two joint states (at fixed time t) on the same decomposition $S : E_1 \cdots E_N$.

For each fragment E_k , recall the subnormalized disturbance

$$D_{\text{sub}}^{(k)}(\rho_{SE}, \rho'_{SE}) = \frac{1}{2} \sum_x \|p_x(\rho_{SE}) \rho_{E_k|x}(\rho_{SE}) - p_x(\rho'_{SE}) \rho_{E_k|x}(\rho'_{SE})\|_1.$$

As defined in §12.3, the classical X -law distance is

$$\Delta_X(\rho_{SE}, \rho'_{SE}) := \frac{1}{2} \sum_x |p_x(\rho_{SE}) - p_x(\rho'_{SE})|.$$

Writing

$$A_x := p_x(\rho_{SE}) \rho_{E_k|x}(\rho_{SE}), \quad A'_x := p_x(\rho'_{SE}) \rho_{E_k|x}(\rho'_{SE}),$$

we have

$$\Delta_X(\rho_{SE}, \rho'_{SE}) = \frac{1}{2} \sum_x |(A_x - A'_x)| \leq \frac{1}{2} \sum_x \|A_x - A'_x\|_1 = D_{\text{sub}}^{(k)}(\rho_{SE}, \rho'_{SE}).$$

Define the excess disturbance

$$D_{\text{excess}}^{(k)}(\rho_{SE}, \rho'_{SE}) := D_{\text{sub}}^{(k)}(\rho_{SE}, \rho'_{SE}) - \Delta_X(\rho_{SE}, \rho'_{SE}). \quad (16.4)$$

Theorem 16.2: Distributed Failure for Margin-Certified Records

Fix X , $\delta \in (0, 1)$, and time t .

Suppose there exists a subset $\mathcal{K} \subseteq \{1, \dots, N\}$ such that every $k \in \mathcal{K}$ is (δ, γ_k) -certified for X at ρ_{SE} in the sense of Definition 16.1.

That is, for every $k \in \mathcal{K}$ there exists a finite-outcome POVM

$$M^{*(k)} = \{M_y^{*(k)}\}_{y \in Y_k}$$

such that

$$I(X:Y_k; t)_{M^{*(k)}}(\rho_{SE}) \geq (1 - \delta)H(X; t)_{\rho_S} + \gamma_k.$$

Fix $\eta_k \in (0, \gamma_k/2]$ for each $k \in \mathcal{K}$, and let

$$\varepsilon_{*,k} = \varepsilon_*(\eta_k, \delta; |\mathcal{X}|, |Y_k|)$$

be the stability threshold supplied by Lemma 16.1.

For ρ'_{SE} define

$$(\rho_{SE}, \rho'_{SE}) := \left\{ k \in \mathcal{K} : D_{\text{sub}}^{(k)}(\rho_{SE}, \rho'_{SE}) > \varepsilon_{*,k} \right\}.$$

Then every fragment

$$k \in \mathcal{K} \setminus (\rho_{SE}, \rho'_{SE})$$

remains a δ -record for X at ρ'_{SE} .

In particular,

$$\left| \mathcal{K} \setminus_{\delta}^{\rho'}(X; t) \right| \leq |(\rho_{SE}, \rho'_{SE})|.$$

Equivalently, if at least m fragments in $\mathcal{K}$ cease to satisfy the δ -record predicate at ρ'_{SE} , then at least m of them must satisfy

$$D_{\text{sub}}^{(k)}(\rho_{SE}, \rho'_{SE}) > \varepsilon_{*,k}.$$

Proof of Theorem 16.2

Fix $k \in \mathcal{K}$ and suppose that

$$D_{\text{sub}}^{(k)}(\rho_{SE}, \rho'_{SE}) \leq \varepsilon_{*,k}.$$

Apply Lemma 16.1 to the certified witness $M^{*(k)}$. It yields

$$I(X:Y_k; t)_{M^{*(k)}}(\rho'_{SE}) \geq (1 - \delta)H(X; t)_{\rho'_S} + (\gamma_k - \eta_k).$$

Since $\gamma_k - \eta_k > 0$,

$$I_{\text{acc}}(X:E_k; t)_{\rho'_{SE}} \geq I(X:Y_k; t)_{M^{*(k)}}(\rho'_{SE}) \geq (1 - \delta)H(X; t)_{\rho'_S}.$$

Hence $k \in_{\delta}^{\rho'}(X; t)$.

Thus any $k \in \mathcal{K}$ that fails to remain a δ -record must satisfy

$$D_{\text{sub}}^{(k)}(\rho_{SE}, \rho'_{SE}) > \varepsilon_{*,k}.$$

Counting such fragments yields the claim. $\square$

Corollary 16.2 (Excess Disturbance Under Record Failure)

If $k \in \mathcal{K}$ fails to remain a δ -record at ρ'_{SE} , then

$$D_{\text{excess}}^{(k)}(\rho_{SE}, \rho'_{SE}) \geq \max\{0, \varepsilon_{*,k} - \Delta_X(\rho_{SE}, \rho'_{SE})\}.$$

Proof of Corollary 16.2

Record failure implies

$$D_{\text{sub}}^{(k)}(\rho_{SE}, \rho'_{SE}) > \varepsilon_{*,k}.$$

Using

$$D_{\text{excess}}^{(k)} = D_{\text{sub}}^{(k)} - \Delta_X$$

we obtain

$$D_{\text{excess}}^{(k)}(\rho_{SE}, \rho'_{SE}) > \varepsilon_{*,k} - \Delta_X(\rho_{SE}, \rho'_{SE}).$$

Since $D_{\text{excess}}^{(k)} \geq 0$ by definition, it follows that

$$D_{\text{excess}}^{(k)}(\rho_{SE}, \rho'_{SE}) \geq \max\{0, \varepsilon_{*,k} - \Delta_X(\rho_{SE}, \rho'_{SE})\}.$$

$\square$

16.4 Structural Interpretation and Scope

Throughout this subsection, “certified” means (δ, γ) -certified per Definition 16.1. All stability and distributed-failure statements below apply only to fragments that satisfy this interior certification condition at the reference state. No stability claim is made for boundary δ -records without strict margin.

Summary of Section 16

Sections 16.1–16.3 isolate two logically distinct structural phenomena governing fragment-level accessibility certification within the interior, margin-certified regime. Section 16.2 supplies the finite-outcome closure required to parameterize these statements in terms of finite alphabets.

First, Lemma 16.1 establishes local stability of margin-certified accessibility. If a fragment certifies X with strict margin, then sufficiently small subnormalized perturbations of the conditioned fragment ensemble preserve certification with degraded margin. Margins degrade continuously. This is a finite-dimensional continuity statement based on classical entropy bounds together with the certification assumption that a finite-outcome witness exists. Second, Theorem 16.2 establishes a margin-conditioned distributed failure constraint. For fragments that are (δ, γ_k) -certified, removal of δ -record status from m such fragments requires exceeding the corresponding stability threshold on at least m fragments. Widespread certified-record loss therefore requires fragment-wise threshold crossing across many fragments.

Taken together, these results describe local continuity properties of the margin-certified region (for the fixed certification witnesses) in joint-state space. They identify both its interior stability and the fragment-wise mechanism by which it can be exited. No global distance lower bound is implied.

Local Structure of the Certified Regime

Certification with strict margin defines a neighborhood in which the δ -record predicate remains valid with reduced margin. Small subnormalized perturbations of the fragment-conditioned ensembles do not immediately destroy accessibility for certified fragments. In this sense, the interior certification regime is locally stable.

The logical direction is one-sided. The excess disturbance quantity $D_{\text{excess}}^{(k)}$ admits a lower bound only when δ -record status fails at ρ'_{SE} for fragments that were (δ, γ) -certified at the reference state. No lower bound follows merely from record preservation. Stability and erasure are asymmetric.

These statements concern transformations between joint states within finite-dimensional regimes. They do not assume collapse, irreversibility, preferred-basis selection, or any geometric rigidity hypothesis. They establish continuity of certification under controlled perturbation, not static compatibility geometry.

Relation to Incompatibility Constraints

Section 14 established that incompatibility geometry constrains strong simultaneous accessibility for incompatible observables. Section 16 establishes stability properties of margin-certified fragment accessibility.

The present results do not prove that the Section 14 incompatibility obstruction itself is stable under perturbation. They show only that when certified accessibility is present with strict margin, small fragment-level perturbations cannot immediately eliminate those certificates. Whether incompatibility-based obstruction to disjoint strong accessibility inherits perturbative stability requires additional structural bridges between entropic incompatibility bounds and accessible-information certification. Such bridges are not established here.

Rigidity Program (Open Direction)

The results of §§15–16 establish stability of certification and fragmentwise distributed failure constraints. They do not establish geometric rigidity of the joint state. The only rigidity hypothesis adopted in this work appears in §14.3 and is stated as Hypothesis 14.1; it concerns the reduced system state $\rho_S(t)$.

In particular, Hypothesis 14.1 posits trace-distance proximity of $\rho_S(t)$ to the compatibility set $\mathcal{C}_{XZ}$ under $\text{DSR}_\delta(X, Z; t)$. No proximity of the full joint state $\rho_{SE}(t)$ to any structural class is assumed. No argument in §§15–16 depends on joint-state geometric rigidity.

A strictly stronger statement would instead assert proximity of the full joint state to a joint structural class. Such a statement would take the schematic form

$$\text{Strong redundancy with strict margins} \implies \text{dist}_{\text{tr}}(\rho_{SE}, \mathfrak{C}_X^{\text{joint}}) \leq \varepsilon.$$

Here $\mathfrak{C}_X^{\text{joint}}$ denotes a prospective joint-state structural class defined on $\mathcal{H}_{SE}$ and distinct from the system-level compatibility set $\mathcal{C}_{XZ}$. Establishing such a result would require additional structural bridges.

Possible approaches include:

- Quantitative bridges from accessible information to approximate orthogonality of conditioned fragment states.
- Reconstruction lemmas deriving approximate broadcast structure from multi-fragment redundancy.
- Entropic or guessing-probability inequalities yielding trace-distance bounds either for the reduced system state relative to $\mathcal{C}_{XZ}$ or for the full joint state relative to a structural class.

No such bridge is derived here. Section 16 isolates the margin-certified stability layer upon which any future rigidity theorem must build.

Noise Scaling and Quantitative Dependence

The stability thresholds $\varepsilon_{*,k}$ depend explicitly on the alphabet size $|\mathcal{X}|$, the witness outcome size $|Y_k|$, and the chosen certification margins. Because certification assumes existence of finite-outcome witnesses, these thresholds are explicit and finite-dimensional.

This makes the interior certification regime quantitatively analyzable under fragment growth, increasing alphabet size, structured noise models, and redundancy scaling. Precise scaling laws are not derived here, but the dependence is explicit.

Conceptual Boundary

Section 16 establishes:

- local stability of margin-certified accessibility,
- a fragment-wise distributed failure constraint,
- continuity properties of the certified interior regime.

The distributed constraint established in Theorem 16.2 is fragment-wise. It does not yield a single global trace-distance lower bound between joint states without additional structural assumptions. Full geometric rigidity, collapse claims, global compatibility structure, or stability of incompatibility obstruction are not established here. The rigidity program remains open. What has been established is precise and limited: margin-certified redundancy is locally stable under controlled perturbations, and loss of certified redundancy at scale requires fragment-wise threshold exceedance.

Handoff to §17

Sections 14–15 framed the global tension between incompatibility-based obstruction, conditional rigidity narratives, and engineered attainability. Section 16 has added an unconditional stability layer: within the margin-certified regime, certificates persist under small conditioned-ensemble disturbance, and large-scale certificate loss requires many fragments to cross explicit disturbance thresholds.

Section 17 consolidates these elements into a single regime-theoretic interpretation, clarifying what is proved unconditionally, what is conditional (Hypothesis 14.1), and what is merely exhibited by construction.

17 Global Meaning and Boundaries

The preceding sections developed the formal core of the program. Sections 12–16 established admissibility structure, incompatibility bounds, regime separation, and constructive attainability under controlled coordination. With those elements in place, the present section consolidates the structural meaning of the results, delineates their scope, and clarifies their limits.

The aim is not to extend the formal machinery, but to make explicit what has been established, under what assumptions it holds, and how the interpretive framework of the earlier sections stands in relation to the structural analysis. The program therefore closes in two senses: it resolves the internal tension between entropic obstruction and distributed attainability, and it marks the precise boundary within which its claims apply.

17.1 Interpretive Synthesis and Regime Closure

Sections 12–16 establish a precise structural separation between (i) incompatibility-driven obstruction under admissibility constraints and (ii) engineered coexistence achieved through non-generic coordination of fragment interactions. We recall that §12 distinguishes the state-level predicate $\text{DSR}_\delta(X, Z; t)$, defined on a single joint state $\rho_{SE}(t)$ through Lüders-conditioned fragment ensembles, from tested-channel redundancy verified across controlled runs with fresh initialization. The constructive coexistence results developed in §15 concern this tested-channel notion and therefore do not by themselves witness $\text{DSR}_\delta(X, Z; t)$. Once this predicate distinction is kept explicit, the apparent tension between rigidity and attainability dissolves as a separation of regimes within a single formal architecture.

The entropic obstruction of §14.1 establishes a quantitative disjoint-predictability constraint controlled by incompatibility. The step from the operational DSR predicate to trace-distance proximity of the system marginal to the compatibility set $\mathcal{C}_{XZ}$ is treated separately through the rigidity strengthening (Hypothesis 14.1). The channel-structural analysis of §14.2 motivates that, absent such strengthening, logical coexistence cannot be excluded in principle. The constructive toy model of §15 and the finite-operational closure of §16 confirm that tested-channel redundancy (two tasks) is attainable, but only within fine-tuned parameter windows whose total admissible measure scales as $O(\alpha)$ as the discrimination threshold tightens. Moreover, the distributed-erasure principle of §16 shows that eliminating certified records across many fragments requires correspondingly many fragments to undergo super-threshold disturbance.

The resulting picture is explicitly regime-theoretic. Under generic admissible interactions, simultaneous stabilization of incompatible observables is structurally suppressed. Coexistence is not logically impossible, but it requires coordination beyond typical admissible coupling and becomes detectable through the distributed disturbance accounting of §16 (Theorem 16.2). Rigidity is therefore not a universal prohibition; it is a structural feature of the generic regime. This regime separation resolves the interpretive concern motivating the earlier development. These results therefore support a regime-theoretic view of classical persistence: in the generic admissible regime, incompatible observables cannot simultaneously stabilize redundant records, while coexistence remains possible only through finely coordinated interaction structures. Classical persistence does not arise from ad hoc collapse, nor from unrestricted dynamical possibility. It emerges instead as a structural consequence of admissibility, constraint propagation, and incompatibility geometry.

Accordingly, the formal program developed in Sections 12–16 does not overturn the interpretive framework of Sections 1–11; it refines and sharpens it. Constraint, record stabilization, and shared

invariants are not treated as metaphysical postulates, but as organizing principles whose structural content can be isolated, bounded, and stress-tested within explicitly defined regimes. The theory thus stands as a constraint program. It identifies the admissible structures under which persistence, classical stability, incompatibility geometry, and distributed redundancy cohere, and it delineates the regimes in which those structures fail.

Results Ledger

For clarity, we summarize the logical status of the principal results developed in Sections 12–16.

- **Proved Structural Results:** The entropic disjoint-predictability obstruction (Theorem 14.1), the robustness of redundancy certification under perturbation (Theorem 15.1), local stability of certified records (Lemma 16.1), finite-outcome witness reduction for accessible information (Proposition 16.1), and the distributed erasure/counting principle (Theorem 16.2).
- **Constructive Attainability:** Section 15 provides explicit engineered constructions demonstrating coexistence of two redundancy tasks in the tested-channel sense together with quantitative scaling of the associated fine-tuning window.
- **Hypotheses and Strengthening Directions:** The Geometric Rigidity Hypothesis (Hypothesis 14.1) and the upgrade paths discussed in §17.5 identify structural conditions under which the present obstruction results may be strengthened into full rigidity theorems.

17.2 Structural Constraint Statement

Let X and Z be quantitatively incompatible rank-1 PVMs on a finite-dimensional system $\mathcal{H}_S$. Consider a fixed fragment decomposition $S : E_1 \cdots E_N$ and an admissible interaction structure satisfying the hypotheses articulated in Sections 13–14. Then the structural alternatives reduce to the following dichotomy.

Generic Regime: Under admissible, non-fine-tuned fragment coupling (generic in the sense of §13: open-dense and measure-typical parameter choices), realization of certified redundancy for both observables X and Z produces strong disjoint predictability (Theorem 14.1) and, under the rigidity strengthening of Hypothesis 14.1, implies the system marginal $\rho_S(t)$ to lie within a controlled trace-distance neighborhood of the compatibility set $\mathcal{C}_{XZ}$.

Engineered Regime: Realization of certified redundancy for both observables X and Z can occur only through non-generic coordination of fragment parameters, occupying a parameter region whose admissible measure scales as $O(\alpha)$ as the discrimination threshold tightens (see §15).

In particular, quantitative incompatibility does not render coexistence logically impossible. Rather, it induces structural rigidity. Generic admissible dynamics suppress simultaneous stabilization, while coexistence requires coordinated fine structure beyond typical admissible interaction. This dichotomy constitutes the core structural content of the present program.

17.3 Scope Conditions

The structural results of Sections 13—16 are established under the following controlled assumptions.

Finite Dimensionality: All systems and fragments are finite dimensional. No claims are made regarding infinite-dimensional Hilbert spaces, continuous spectra, or field-theoretic limits.

Fixed Fragment Decomposition: A definite fragment partition $S : E_1 \cdots E_N$ is assumed. The analysis does not address dynamically emergent or coarse-grained decompositions.

Rank-1 PVM Structure: Incompatibility bounds are derived for rank-1 PVMs. Extensions to more general POVMs or degenerate observables require separate analysis.

Admissible Interaction Class: The rigidity and suppression results apply within the admissible fragment-local interaction class defined in Sections 13—14. No universality is claimed beyond this structural regime.

Trace-Norm Proximity: All closeness statements are formulated in trace distance. No stronger norm equivalences or operator-topology claims are assumed.

Parameter-Space Typicality: Statements concerning genericity or measure suppression refer to parameter space under bounded, absolutely continuous coefficient distributions. No global topological inevitability is asserted.

Within these bounds, the regime separation and structural dichotomy stated above hold. Beyond them, the program remains open.

17.4 Non-Claims

The present work does not assert the following.

No Universal No-Go Theorem: The Geometric Rigidity Hypothesis does not establish logical impossibility of realizing the predicate $\text{DSR}_\delta(X, Z; t)$ for incompatible observables. It postulates suppression within a specified admissible generic regime.

No Dynamical Inevitability: The analysis does not claim that classicality emerges unavoidably under all open-system dynamics. It identifies structural conditions under which stabilization is enforced or suppressed.

No Collapse Mechanism: No dynamical collapse process, hidden-variable model, or modification of unitary quantum theory is introduced or required.

No Universality Beyond the Admissible Class: Results are confined to the admissible fragment-local interaction framework defined earlier. No claims are made regarding arbitrary architectures or unconstrained couplings.

No Ontological Commitment: The framework is structural. It does not prescribe a metaphysical interpretation of the quantum state beyond the constraint relations explicitly analyzed.

The program is therefore regime-theoretic and structural. Its results concern admissibility, constraint propagation, and incompatibility geometry, not universal dynamical proclamation.

17.5 From Diagnostic Framing to Structural Necessity

The preceding sections establish regime separation, admissibility constraints, quantitative incompatibility bounds, constructive attainability, and distributed threshold structure within a controlled open-system setting. These results stand independently of any further strengthening. However, the present framework also exposes several natural upgrade paths by which diagnostic principles may be strengthened into structural theorems.

Record Authority as Structural Requirement: Record stabilization has functioned as a persistence condition within the admissible framework. A structural strengthening would require demonstrating that, under generic admissible coupling, failure of stabilized correlations to constrain successive interactions leads to breakdown of global consistency across multi-step admissible evolution. Such a result would elevate record authority from interpretive organizing principle to necessary condition for sustained admissibility. Establishing this would require a multi-time extension of the present single-time redundancy analysis together with quantitative propagation bounds.

Shared Invariant Necessity: Shared invariants have been introduced diagnostically as channels enabling mutual constraint participation. A tractable strengthening would prove that, in the absence of at least one shared invariant channel, constraint propagation between subsystems generically fails to remain stable under composition. This direction appears feasible within the current finite-dimensional admissible architecture, as it depends primarily on compatibility geometry and fragment-channel structure rather than new dynamical assumptions.

Constraint-Enforced Classical Stability: The present results demonstrate suppression of realization of the predicate $\text{DSR}_\delta(X, Z; t)$ in the generic regime. A deeper strengthening would show that, within admissible open-system interaction, classical pointer structures form the only dynamically stable attractor class compatible with persistent constraint propagation. This extension is substantially more demanding, as it requires explicit dynamical stability analysis rather than static incompatibility bounds.

Global Constraint and Nonlocal Correlation: Nonlocal correlations have been treated interpretively as compatibility relations within globally constrained admissibility. A structural reformulation would require embedding Bell-type correlation structure directly into the constraint-compatibility framework. This represents a longer-range objective, as it demands integration of admissibility geometry with explicit probabilistic and causal structure.

Closing Words

These directions are not required for the validity of the present structural program. They delineate a modular path from diagnostic framing to structural necessity. Some extensions, particularly those concerning shared invariants, appear accessible within the current formal apparatus. Others

require additional dynamical machinery or multi-step constraint analysis. The program therefore closes in its present structural form while remaining open in explicitly bounded and formally articulable ways.

18 Unified Structural Posture of QCT

The manuscript has proceeded in two complementary movements. Sections 1—11 developed an interpretive framework centered on constraint, record stabilization, admissibility, and shared invariants. Sections 12—16 translated those organizing principles into explicit structural analysis, isolating quantitative incompatibility bounds, regime separation, and constructive attainability within a controlled interaction class.

These two movements are not independent layers. The interpretive framework anticipated structural tension; the formal development reframed it as a regime-theoretic problem and bounded its admissible realizations. What appeared at the outset as a conceptual concern regarding classical persistence and incompatibility has been recast as a statement about admissible interaction geometry.

The interpretive principles of constraint propagation, record authority, and invariant participation are therefore not introduced as metaphysical postulates. Within the formal arc, they function as organizing hypotheses whose structural content can be bounded, stress-tested, and delimited. Conversely, the mathematical results do not stand as isolated technical statements; they sharpen and discipline the interpretive posture by identifying precisely where generic suppression, engineered coexistence, and structural rigidity arise.

The structural analysis developed in Sections 12—16 does not by itself constitute a catalogue of empirical predictions. Rather, it establishes the regime constraints within which any predictive extension of QCT must operate. The formal core functions as a constraint architecture: it delineates what configurations are generically suppressed, what require fine coordination, and what remain admissible under controlled interaction. In this sense, QCT is neither a collapse theory, nor a hidden-variable proposal, nor a modification of unitary quantum dynamics.

It does not legislate ontology, nor does it claim dynamical inevitability. Rather, it is a constraint program: a regime-theoretic analysis of the structural conditions under which persistence, stabilization, and incompatibility geometry cohere. The central contribution of the present work is therefore not the proclamation of impossibility, but the articulation of structure. Quantitative incompatibility induces rigidity within admissible generic regimes, while engineered coordination occupies measure-suppressed regions of parameter space. Classical stability appears, within the admissible framework analyzed here, as a structural feature of constraint propagation rather than as an imposed collapse mechanism.

The structural constraints articulated here also isolate a natural rigidity question. In particular, one may ask whether sufficiently strong margin-certified redundancy forces proximity to classical broadcast structure at the joint-state level, and whether simultaneous redundancy for incompatible observables enforces compatibility-set geometry beyond the reduced state. These rigidity directions are not pursued in the present work. They represent a strengthening of the structural program rather than a prerequisite for its validity.

At the architectural level, independent of future strengthening directions, the two-part organization of this manuscript reflects that posture. Interpretation motivates structure; structure disciplines interpretation. Neither layer is sufficient in isolation. Together, they define the scope, the limits, and the identity of the program. The theory thus closes as a bounded structural analysis. Its claims are precise, its regime distinctions explicit, and its horizons identified. Beyond those bounds, further strengthening may proceed, but the present architecture stands independently: a coherent structural account of admissibility, constraint, and incompatibility within finite-dimensional open-system quantum structure.

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